University of Puerto Rico

Mayagüez Campus

Department of Mechanical Engineering

ADAS

ASSISTIVE DOOR ACTUATION SYSTEM

Opening doors. Expanding independence.

Capstone Project: Assistive Door Actuation System

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1 Project Overview

1.1 Introduction

Designing assistive mechanisms for everyday environments requires a careful balance of functionality, safety, usability, and cost. Particularly, automated door systems must integrate mechanical, electrical, and control considerations while remaining practical for residential use. This project focuses on developing a compact, low-cost interior door opener/closer that can be operated remotely and reliably under typical home conditions. The Product Design Specification (PDS) outlines the performance requirements, constraints, and evaluation metrics that will guide the engineering process—from concept development to prototyping and testing. By establishing clear customer needs, measurable metrics, and realistic marginal and ideal targets, the PDS ensures that the final design is both technically feasible and aligned with user expectations.

1.2 Problem Statement

The user is quadriplegic and can't physically open the door. This project aims to design a simple, affordable device that allows the user to remotely open and close a hinged door from approximately 12 ft away using their mobile device. The system should operate reliably without requiring Wi-Fi, be safe to use around people, quiet, compact, and easy to install in a home setting. It must also include a disengaging mechanism that allows the door to be operated manually. The goal is to increase independence and reduce the need for assistance while keeping the design practical and low-cost.

1.3 Product Design Specifications (PDS)

The Product Design Specifications for the low-cost interior door opener outlines the main requirements that the system must satisfy, along with the corresponding criteria used to evaluate the design. These specifications serve as a guideline for future design considerations throughout the development process. Each requirement is described with both ideal and marginal values to guarantee that the system will remain functional and affordable. Table 1 highlights the main customer needs, and each is assigned a level of importance (percentage) to indicate how critical it is for the final design.

Table 1: Customer Needs

#	Need	Importance
1	Mechanism to open door	7
2	Method to open doors from a distance	10
3	Compact size	5
4	Affordable	7
5	Safe operating velocity	5
6	Obstruction Detection	3
7	Easy Installation	3
8	Disengaging Mechanism	7
9	Water Resistant	7
10	Long-lasting/Reliable	10
11	Easy to Maintain	3
12	Easy to Replace Components	3
13	Smooth Movement	5
14	Workable without Wi-Fi	10
15	Can be maintained with readily available tools	3
16	Quiet	2
17	Light Weight	4
18	Workable without power	6

The following table shows the metrics and units associated with each of the customer needs presented in the previous table; this way, the system's overall performance can be measured and evaluated. The importance of values assigned to each customer's needs was established based on the specific requirements of the target user and the overall objectives of the project. Criteria directly related to user independence, reliability, and functionality, such as remote operation, long-term reliability, and operation without Wi-Fi, were assigned the highest importance values because failure in these areas would prevent the system from fulfilling its intended purpose. Medium importance values were assigned to factors such as compactness, weight, and smooth operation, while lower importance values were given to secondary considerations such as maintenance convenience and noise level. The importance rankings were determined through team discussion, user needs analysis, and comparison with the Product Design Specifications (PDS).

For example, Metric 1 was derived from Customer Needs 5 and 13, which emphasized safe operating velocity and smooth movement, respectively. Based on user expectations and

similar assistive systems, it was determined that a response time of 6 seconds represents the ideal target value, since this delay is short enough for the user to perceive the system as immediate and reliable during operation. A response time of 15 seconds was established as the marginal acceptable value because delays beyond this point could interrupt the user experience, reduce confidence in the system, and limit practical usability. (These ideal and marginal values can be seen in table 4)

Table 2: Metrics and Units

#	Needs	Metric	Imp.	Units
1	5, 13	Time to fully open/close door	7	s
2	7	Allowable Door Modifications (Screw holes added)	7	Count
3	17	Total system weight	5	kg
4	4	Component & Manufacturing Costs	10	USD
5	1	Push/Pull Capacity (ADA Standard)	7	N
6	3	Device Volume	5	cm^3
7	18	Battery Life (after power outage)	5	hr
8	7, 11	Installation & Maintenance time (governed by # of components)	3	min
9	2, 14	Can open from a distance	5	m
10	6	Obstruction Detection & Response Levels	3	Levels
11	9	Aimed IP Rating	3	IP
12	8	# of Appendages to Disengage Mechanism	5	Count
13	10	Expected Lifetime	5	Years
14	12, 15	# of Tools/parts required for installation and maintenance	3	Count
15	15	Time for Scheduled maintenance	3	Months
16	10	Maximum Ambient Temperature Allowed	5	°C
17	10	Time to first critical failure	3	Months
18	18	Rechargeability (Requires Human Involvement)	10	Yes/No
19	5,6	Safety Factor	3	-
20	1,5,6,13	Door Limits	3	deg

In the previous table, metric #10 is measured in levels; to clarify what these levels represent, Table 3 is created, which details the different obstruction detection levels.

Table 3: Obstruction Detection Levels

Level 1 – Passive detection (motor current spike), stop only
Level 2 – Detection (current spike) + timed retry or reset

Level 3 – Sensor-based detection + timed retry or reset

Table 4 takes each metric and assigns it both its ideal and marginal values, then classifies them as either demands or wishes. The demands represent the minimum requirements the system must meet to be acceptable, while the wishes represent desirable targets that may improve performance, but are not strictly required.

Table 4: Marginal and Ideal Values

#	Metric	Units	Marginal	Ideal	Imp.	W/D
1	Time to fully open/close door	s	>15	>6	7	D
2	Allowable Door Modifications (Screw holes added)	Count	<4	<2	7	D
3	Total system weight	kg	<9	<4.5	5	D
4	Component & Manufacturing Costs	USD	<170	<100	10	D
5	Push/Pull Capacity (ADA Standard)	N	>22.2	>44.48	7	D
6	Device Volume	cm^3	<14,160	<4,720	5	D
7	Battery Life (after power outage)	hr	>8	>24	5	D
8	Installation & Maintenance time (governed by # of components)	min	<15	<5	3	D
9	Can open from a distance	m	<9	>3.5	5	D
10	Obstruction Detection & Response Levels	Levels	1	3	3	D
11	Aimed IP Rating	IP	IP51	IP54	3	D
12	# of Appendages to Disengage Mechanism	Count	<4	<2	5	D
13	Expected Lifetime	Years	>2	>5	5	D
14	# of Tools/parts required for installation and maintenance	Count	<5	<3	3	D
15	Time for Scheduled maintenance	Months	>3	>6	3	D
16	Maximum Ambient Temperature Allowed	°C	<40	25-30	5	D
17	Time to first critical failure	Months	>3	>6	3	D
18	Rechargeability (Requires Human Involvement)	Yes/No	Yes	No	10	D
19	Safety Factor	-	>2	>5	3	D
20	Door Limits	deg	>80	>90	3	D

1.4 Schedule

The following table presents the project schedule, outlining the main tasks required for the development of the low-cost interior door opener and their planned order of execution. It provides a clear overview of how the work is structured over time, including the duration of each activity and how tasks are organized in sequence.

Table 5: Schedule

Task	Process	Task Description	Start	End	Duration (Days)
1	Project Initiation	Introduction to the course, Design Methodology, Forming work groups.	23/1/2026	23/1/2026	1
2	Problem Definition	Meeting #1 with PRATP. Project presentations. Project selection. Design methodology (cont.)	30/1/2026	30/1/2026	1
3	Problem Description	Design Methodology (cont.) Problem Description (PDS) (O).	30/1/2026	6/2/2026	7
4		Meeting #2 with PRATP. PDS and Gantt Chart (O + E).	13/2/2026	13/2/2026	1
5		PDS and Revised Gantt Chart (E), Idea Bank. Presentation (O))	13/2/2026	20/2/2026	7
6	Conceptual Design	Revised Idea Bank (E). +/- Method to reduce to 15, Weighted rated method to reduce from 15 to 5 (O).	20/2/2026	27/2/2026	7
7	Preliminary Design	Detailed Preliminary Design top 5, repeat Weighted rated method to identify top 3 (O+E).	27/2/2026	6/3/2026	7
8		Meeting #3 with PRATP (O) Selection of the concept to be developed	13/3/2026	13/3/2026	1
9		Progress Report (O+P).	13/3/2026	20/3/2026	7

10	Design Refinement	Progress Report (O+P). Presentation of the Complete v0 Prototype (functional) (O+P).	20/3/2026	27/3/2026	7
11	Design Refinement	Semana Santa Break	29/3/2026	5/4/2026	7
12		Meeting #4 with PRATP, Presentation of the Complete v1 Prototype (functional) (O+P).	6/4/2026	10/4/2026	4
13	Detailed Design and Manufacturing	Progress report (O+P). Design and manufacturing calculations. Revised prototype.	10/4/2026	17/4/2026	7
14		Progress report including Manufacturing Manual and User Manual (O+P).	17/4/2026	24/4/2026	7
15	Final Report Draft	Progress Report (O+P). Final Report v.0 (E) Delivery Date.	24/4/2026	1/5/2026	7
16	Final Presentation	Final Course Presentation to PRAPT (O+P).	1/5/2026	8/5/2026	7
17	Final Report	Corrected Final Report (E) plus compilation of all documentation in Google Drive.	8/5/2026	15/5/2026	7

The Gantt chart was developed to organize the project into sequential stages that aligned with the engineering design process. Early weeks focused on problem definition, PDS development, and concept generation, while later stages emphasized prototyping, manufacturing, testing, and documentation. The timeline allowed the team to distribute the workload evenly throughout the semester and establish clear milestones for each deliverable.

Overall, the schedule functioned effectively and helped maintain project progress. However, some activities required additional time due to prototyping issues, component procurement delays, and redesign iterations. In particular, the transition from Prototype V1 to V2 required modifications to the enclosure and transmission system, which shifted some manufacturing

and testing activities. Despite these adjustments, the project remained aligned with the major course deadlines, and all required deliverables were completed on time.

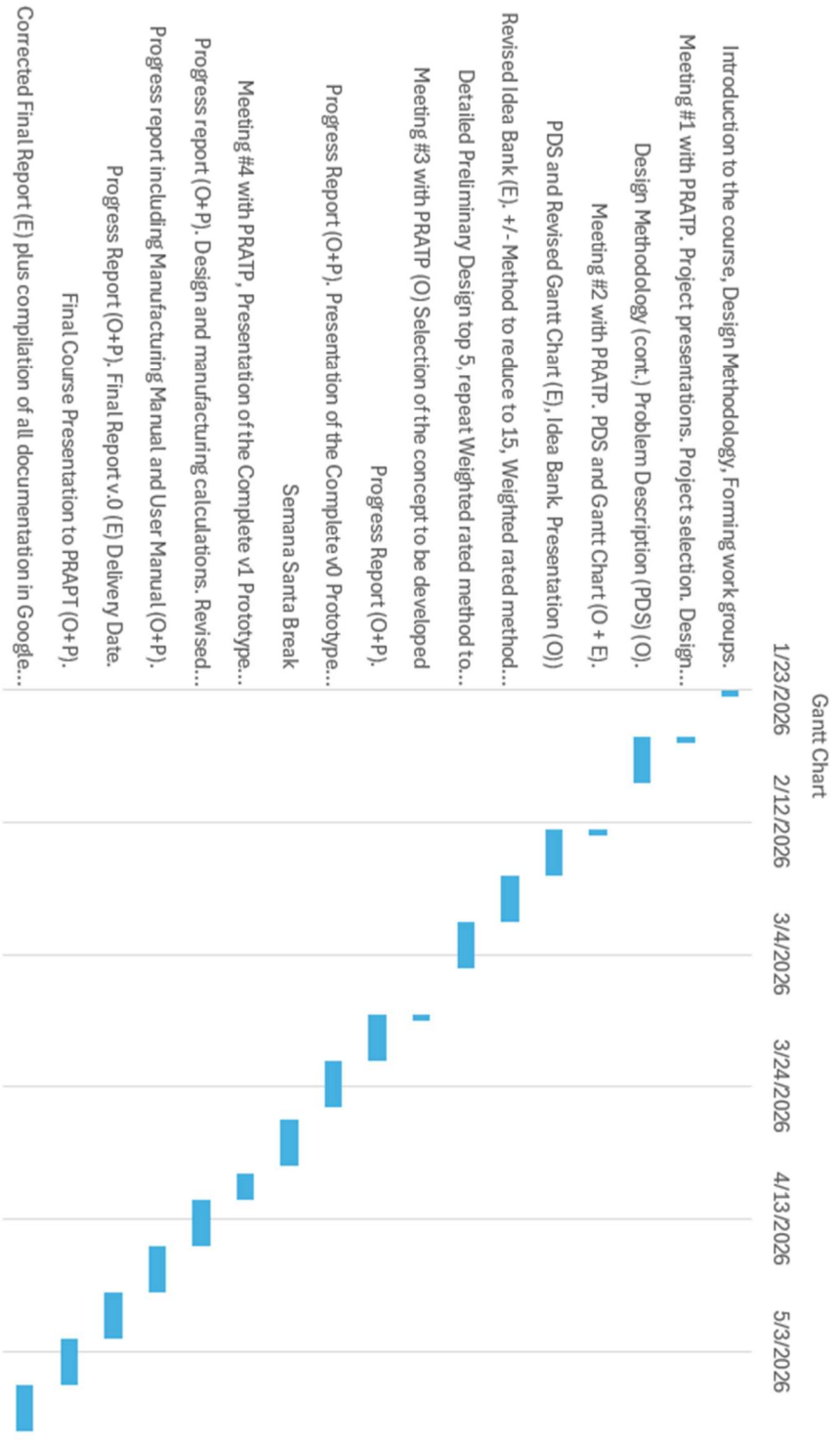


Figure 1: Gantt Chart

2 Idea Generation

2.1 Introduction

The development of the automatic door system began with a structured concept generation phase, referred to as the *bank of ideas*. This stage aimed to explore a wide range of possible solutions without early constraints, allowing creativity and functionality to guide the design process. Multiple concepts were proposed to address the core objective of enabling reliable and user-friendly door operation while meeting the defined Product Design Specifications (PDS), such as compactness, affordability, ease of installation, and mechanical efficiency.

Fifty-two different ideas were generated using brainstorming techniques and by considering different actuation methods, transmission mechanisms, mounting configurations, and control approaches. These concepts varied in complexity, cost, and feasibility, including alternatives such as pulley systems, gear-driven mechanisms, linear actuators, and friction-based wheel designs. Each idea was evaluated using the plus/minus method and the weighted rating method based on criteria such as performance, manufacturability, safety, and integration with available components. The plus/minus method was used to reduce the total number of concepts from 52 to 15. The weighted rating method was then applied to further evaluate the remaining concepts, first reducing the selection from 15 to 5 concepts and then from 5 to the 3 final concepts considered for development.

The bank of ideas served as a foundation for the subsequent concept selection process, where the most promising solutions were systematically analyzed and compared. This ensured that the final design was not chosen arbitrarily, but rather as the result of a comprehensive and methodical engineering decision-making process.

2.2 Bank of Ideas (52 ideas)

The concept generation process followed principles of engineering innovation by exploring multiple design alternatives before selecting and refining the most feasible solution (Engineering Innovation Seminar). To develop initial concepts for the system, a combinatory

method was used based on key design features that were defined and grouped into categories. The design features and options were the following:

a) Actuation Method
   - Crank Linkage
   - Wheel
   - Rack and Pinion
   - Cam Action
   - Pulley System
   - Piston/Spring

b) Disengaging Mechanism
   - Lever/4-bar
   - Clutch
   - Rail
   - Latch
   - Servomotor
   - Pin

c) Obstruction Detection
   - Current Spikes
   - Ultrasonic/Infrared sensor
   - Camera

d) Electrical Power
   - Power Bank/ Rechargeable Batteries
   - Uninterruptible Power Supply (UPS)
   - Induction Charging

e) Power Transmission
   - Motors
   - Pneumatic/Hydraulic
   - Counterweights
   - Spring

By systematically matching and merging these design options, 52 different configurations were explored and are listed in Table 6. For example, Idea #1 combines a crank-linkage as the actuation method, a clutch as the disengaging mechanism, a current spike sensor to detect obstructions, a power bank or rechargeable batteries as the electrical power source, and a motor for power transmission.

Table 6: Bank Of Ideas

Ideas	Actuation Method	Disengaging Mechanism	Obstruction Detection	Electrical Power	Power Transmission
1	Crank-Linkage	Clutch	Current Spike	Power Bank/Rechargeable Batteries	Motor
2	Crank-Linkage	Clutch	Motion Sensor	Power Bank/Rechargeable Batteries	Motor
3	Crank-Linkage	Clutch	Current Spike	UPS	Motor
4	Crank-Linkage	Clutch	Motion Sensor	UPS	Motor
5	Wheel	4bar/Lever	Current Spike	Power Bank/Rechargeable Batteries	Motor
6	Wheel	4bar/Lever	Motion Sensor	Power Bank/Rechargeable Batteries	Motor
7	Wheel	4bar/Lever	Current Spike	Induction Charging	Motor
8	Wheel	4bar/Lever	Motion Sensor	Induction Charging	Motor
9	Wheel	4bar/Lever	Current Spike	UPS	Motor
10	Wheel	4bar/Lever	Motion Sensor	UPS	Motor
11	Wheel	Rail	Current Spike	Power Bank/Rechargeable Batteries	Motor
12	Wheel	Rail	Motion Sensor	Power Bank/Rechargeable Batteries	Motor
13	Wheel	Rail	Current Spike	Induction Charging	Motor
14	Wheel	Rail	Motion Sensor	Induction Charging	Motor
15	Wheel	Rail	Current Spike	UPS	Motor
16	Wheel	Rail	Motion Sensor	UPS	Motor
17	Wheel	Clutch	Current Spike	Power Bank/Rechargeable Batteries	Motor
18	Wheel	Clutch	Motion Sensor	Power Bank/Rechargeable Batteries	Motor
19	Wheel	Clutch	Current Spike	Induction Charging	Motor
20	Wheel	Clutch	Motion Sensor	Induction Charging	Motor
21	Wheel	Clutch	Current Spike	UPS	Motor
22	Wheel	Clutch	Motion Sensor	UPS	Motor
23	Wheel	Servomotor	Current Spike	Power Bank/Rechargeable Batteries	Motor

24	Wheel	Servomotor	Motion Sensor	Power Bank/Rechargeable Batteries	Motor
25	Wheel	Servomotor	Current Spike	Induction Charging	Motor
26	Wheel	Servomotor	Motion Sensor	Induction Charging	Motor
27	Wheel	Servomotor	Current Spike	UPS	Motor
28	Wheel	Servomotor	Motion Sensor	UPS	Motor
29	Rack and Pinion	Clutch	Current Spikes	Power Bank/Rechargeable Batteries	Motor
30	Rack and Pinion	Clutch	Ultrasonic/Radar/Infrared	Power Bank/Rechargeable Batteries	Motor
31	Rack and Pinion	Clutch	Current Spikes	UPS	Motor
32	Rack and Pinion	Clutch	Ultrasonic/Radar/Infrared	UPS	Motor
33	Cam Action	Pin	Current Spikes	Power Bank/Rechargeable Batteries	Motor
34	Cam Action	Pin	Ultrasonic/Radar/Infrared	Power Bank/Rechargeable Batteries	Motor
35	Cam Action	Pin	Current Spikes	UPS	Motor
36	Cam Action	Pin	Ultrasonic/Radar/Infrared	UPS	Motor
37	Pulley	Latch	Current Spikes	Power Bank/Rechargeable Batteries	Motor
38	Pulley	Latch	Current Spikes	Power Bank/Rechargeable Batteries	Counterweights
39	Pulley	Latch	Current Spikes	UPS	Motor
40	Pulley	Latch	Current Spikes	UPS	Counterweights
41	Pulley	Clutch	Current Spikes	Power Bank/Rechargeable Batteries	Motor
42	Pulley	Clutch	Current Spikes	Power Bank/Rechargeable Batteries	Counterweights
43	Pulley	Clutch	Current Spikes	UPS	Motor
44	Pulley	Clutch	Current Spikes	UPS	Counterweights
45	Piston/Spring	Latch	Current Spikes	Power Bank/Rechargeable Batteries	Pneumatic/ Hydraulic
46	Piston/Spring	Latch	Ultrasonic/Radar/Infrared	Power Bank/Rechargeable Batteries	Pneumatic/ Hydraulic
47	Piston/Spring	Latch	Current Spikes	UPS	Pneumatic/ Hydraulic
48	Piston/Spring	Latch	Ultrasonic/Radar/Infrared	UPS	Pneumatic/ Hydraulic

49	Piston/Spring	Pin	Current Spikes	Power Bank/Rechargeable Batteries	Pneumatic/ Hydraulic
50	Piston/Spring	Pin	Ultrasonic/Radar/Infrared	Power Bank/Rechargeable Batteries	Pneumatic/ Hydraulic
51	Piston/Spring	Pin	Current Spikes	UPS	Pneumatic/ Hydraulic
52	Piston/Spring	Pin	Ultrasonic/Radar/Infrared	UPS	Pneumatic/ Hydraulic

2.3 Idea Culling (15 Ideas)

The Plus/minus method was used to filter the 52 initial design concepts, resulting in a reduced list of 15 ideas. In this method, a datum is chosen as the reference idea that will be compared against all other concepts. When an idea is assessed against a design criterion and performs better than the datum, it receives a +1 value; if it performs worse, it receives a -1 value, and if it is equivalent, it receives 0. The 15 selected ideas resulting from this evaluation can be seen in Table 7, ranked from best to worst. (A detailed example for the Plus/Minus method (Table 11), including how some of the ideas scored against the datum, is provided in the appendix.)

Table 7: Top 15 Ideas

#	Idea	Actuation Method	Disengaging Mechanism	Obstruction Detection	Electrical Power	Power Transmission
1	28	Wheel	Servomotor	Motion Sensor	UPS	Motor
2	22	Wheel	Clutch	Motion Sensor	UPS	Motor
3	16	Wheel	4bar/Rail	Motion Sensor	UPS	Motor
4	32	Rack and pinion	Clutch	Motion Sensor	UPS	Motor
5	48	Piston/Spring	Latch	Motion Sensor	UPS	Pneumatic/Hydraulic
6	36	Cam Action	Pin	Motion Sensor	UPS	Motor
7	52	Piston/Spring	Pin	Motion Sensor	UPS	Pneumatic/Hydraulic
8	4	Crank-Linkage	Clutch	Motion Sensor	UPS	Motor
9	15	Wheel	4bar/Rail	Current Spike	UPS	Motor

10	23	Wheel	Servomotor	Current Spike	Power Bank/Rechargeable Batteries	Motor
11	31	Rack and pinion	Clutch	Current Spikes	UPS	Motor
12	46	Piston/Spring	Latch	Motion Sensor	Power Bank/Rechargeable Batteries	Pneumatic/Hydraulic
13	50	Piston/Spring	Pin	Motion Sensor	Power Bank/Rechargeable Batteries	Pneumatic/Hydraulic
14	12	Wheel	Rail	Motion Sensor	Power Bank/Rechargeable Batteries	Motor
15	30	Rack and pinion	Clutch	Motion Sensor	Power Bank/Rechargeable Batteries	Motor

The application of the Plus/Minus method led to the elimination of certain concepts from the design options. Specifically, ideas involving the pulley system in the actuation method and induction charging in the electrical power parameters did not perform well against the PDS criteria. Additionally, concepts that used counterweights in the power transmission parameter were also discarded. Therefore, helping the design process to focus on the most viable solutions while maintaining alignment with the project's requirements.

2.4 Idea Culling (5 Ideas)

After narrowing the 52 initial concepts to 15 ideas, a more detailed evaluation was performed using the weighted rated method. This method consists of comparing multiple design ideas based on how well they meet the PDS criteria. Each criterion is assigned a percentage weight to reflect its level of importance in the evaluation process. Each idea is then rated on a scale from 1 to 10 based on how well it satisfies a specific criterion, where higher values represent stronger performance and lower values represent weaker performance. The rating is multiplied by the corresponding criterion weight to calculate a weighted score, allowing both the importance of the criterion and the performance of the idea to be considered in the overall evaluation.

This method allowed for a more detailed and comprehensive evaluation of each concept as it placed greater emphasis on key factors such as manufacturing cost, maintenance cost, volume, weight, etc. The following table is created to present the reduction of the previous 15 ideas to the top 5 concepts based on their weighted analysis, ranked from the highest scoring concept to the lower scoring concept. (The detailed calculations for the Weighted/Rated method are provided in the appendix.)

Table 8: Top 5 Ideas

#	Idea	Actuation Method	Disengaging Mechanism	Obstruction Detection	Electrical Power	Power Transmission
1	28	Wheel	Servomotor	Motion Sensor	UPS	Motor
2	22	Wheel	Pin	Motion Sensor	UPS	Motor
3	16	Wheel	Rail	Motion Sensor	UPS	Motor
4	36	Cam Action	Pin	Motion Sensor	UPS	Motor
5	32	Rack and pinion	Clutch	Motion Sensor	UPS	Motor

This reduction phase led to the elimination of several design concepts. In the actuation method category, crank linkage, piston/spring, and pulley systems were removed completely from consideration. For the disengaging mechanism, both the latch and 4-bar/lever configurations were eliminated. Within the electrical power section, power banks and rechargeable battery options were excluded. Finally, in the power transmission parameter, the pneumatic and hydraulic systems were also discarded.

Preliminary CAD designs were developed for the Top 5 concepts obtained from the weighted/rated method. At this stage, the models focus on the overall geometry, layout, and basic component integration rather than detailed specifications. These initial designs allow for visual comparison and early detection of design challenges. The Top 5 concepts are presented below.

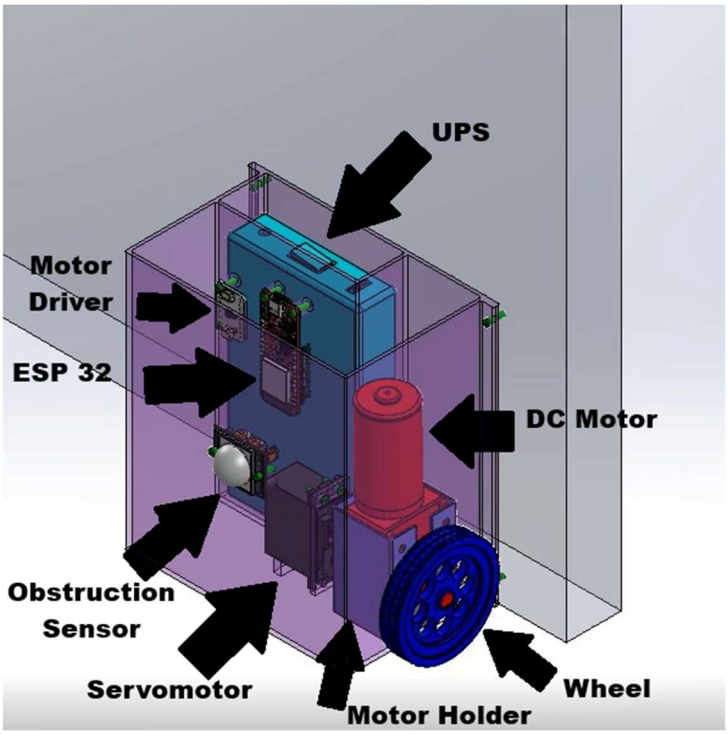


Figure 2: Idea #28 (Preliminary Design)

This design consists of a wheel actuator integrated into the system, with an uninterruptible power supply (UPS) housed inside the component box. The disengaging mechanism is controlled by a servomotor that lifts the wheel away from the floor, as each cycle ends. This design allows the door to move freely once the operation has finished, making sure that people can enter or exit the room manually without resistance or interference from the system.

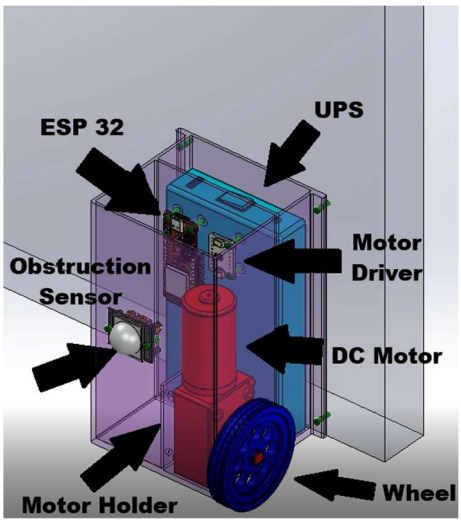


Figure 3: Idea #22 (Preliminary Design)

Idea 22 is similar to idea 28, except that the disengaging mechanism consists of a pin. In this design, the system remains engaged as long as the pin is in place, maintaining the connection between the motor and the wheel. To disengage it, the user simply needs to remove the pin, which disconnects the wheel from the motor and allows the wheel to separate from the ground quickly.

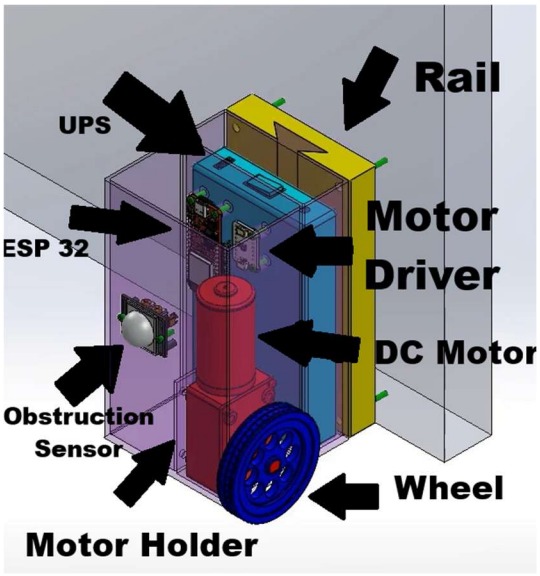


Figure 4: Idea #16 (Preliminary Design)

In Idea # 16, the system is also driven by a wheel mechanism, and the disengaging mechanism is based on a rail mounted directly onto the door. To disengage the user simply needs to lift the component box upward along the rail (manually), separating it from the ground and disconnecting it from its engaged position.

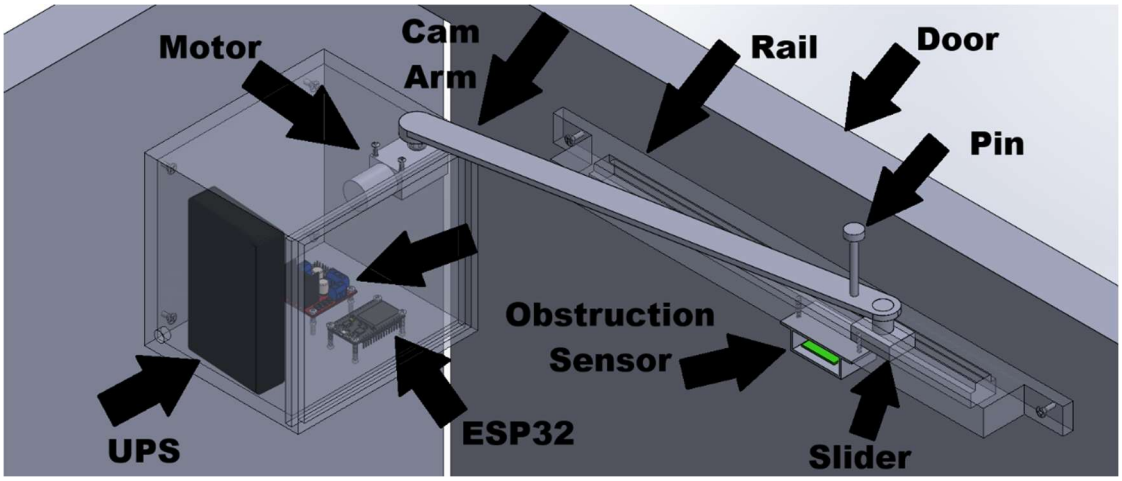


Figure 5: Idea #36 (Preliminary Design)

Idea #36 is based on a cam action mechanism like those commonly used in conventional door closers. However, unlike traditional systems that are primarily designed only to close the door automatically, this concept integrates a motorized system capable of both opening and closing the door. By combining the cam-action mechanism with a motorized actuator and the Bluetooth-based control, the design provides both automated accessibility and controlled door movement.

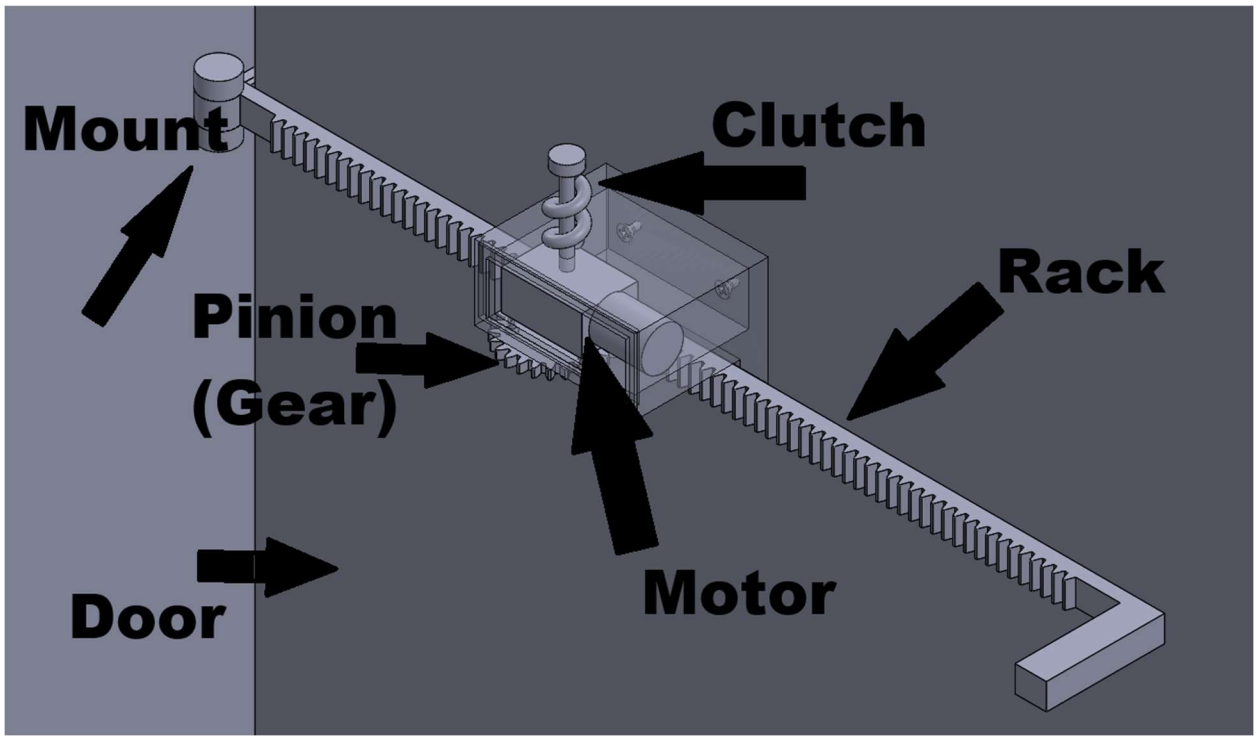


Figure 6: Idea #32 (Preliminary Design)

Idea #32 uses a rack and pinion mechanism in which the motor, pinion gear, and supporting components are mounted inside a housing attached directly to the door. A fixed gear rack is mounted on the door frame near the hinge side of the door. When the motor rotates the pinion gear, the gear travels along the stationary rack. Because the motor assembly is attached to the door itself, the movement of the pinion along the fixed rack forces the door to rotate, causing the door to open or close. In this way, the rotational motion of the motor is converted into controlled movement of the door while keeping the mechanism compact and integrated within the door structure.

2.5 Preliminary Ideas (Top 3 Ideas)

After refining the Top 5 ideas, a second weighted rated method was performed to further evaluate the concepts and to identify the top three final designs. The results are shown in table 9.

Table 9: Weighted Rated Method (Top 3)

Criterion	Weight (%)	1 (Idea# 28)	2 (Idea# 22)	3 (Idea# 16)	4 (Idea# 36)	5 (Idea# 32)
Manufacturing Cost	10.00%	0.8	1	0.9	0.7	0.7
Maintenance Cost	5.00%	0.4	0.4	0.4	0.25	0.35
Volume	5.00%	0.5	0.5	0.5	0.25	0.25
Weight	5.00%	0.5	0.5	0.5	0.25	0.25
Battery Life	5.00%	0.5	0.5	0.5	0.5	0.5
Rechargeability	15.00%	1.5	1.5	1.5	1.5	1.5
IP Rating	5.00%	0.35	0.35	0.35	0.35	0.4
Ergonomic/Safe Design	3.00%	0.21	0.15	0.15	0.21	0.24
Lifetime	8.00%	0.56	0.56	0.56	0.56	0.56
Serviceability	15.00%	1.05	1.05	1.05	1.05	1.2
Time to Open	7.00%	0.7	0.7	0.7	0.56	0.56
Response Level	7.00%	0.7	0.7	0.7	0.7	0.7
Positional Accuracy	3.00%	0.24	0.24	0.24	0.21	0.21
Disengaging Simplicity	7.00%	0.7	0.49	0.49	0.49	0.49
SUM	100%	8.71	8.64	8.54	7.58	7.91

A clear advantage of the wheel-based door opener was observed after implementing the second weighted rated method. Further evaluation of the rack and pinion and cam action ideas demonstrated limitations in terms of complexity and reliability. When compared to wheel designs. As a result, these alternatives were eliminated in favor of the wheel-based designs that offered smoother operation and implementation.

2.6 Final Selection

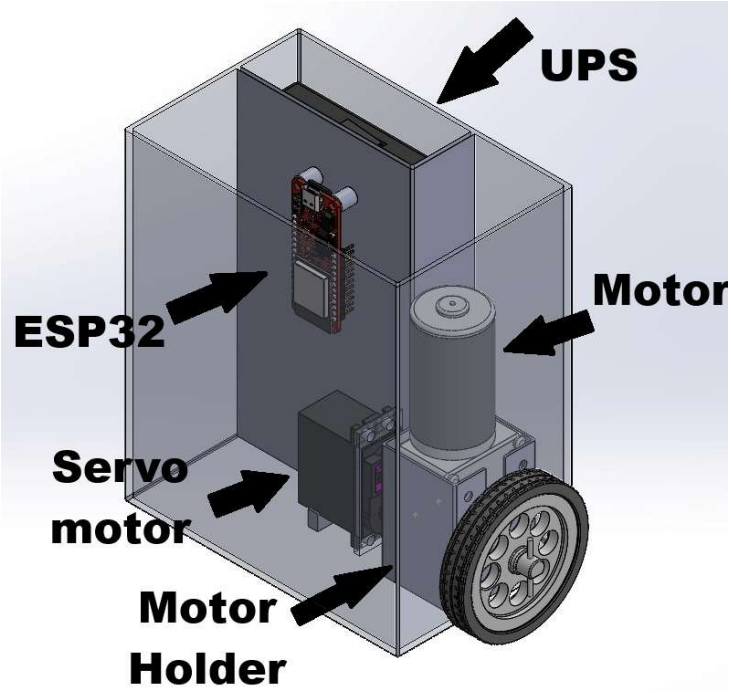


Figure 7: Final Selection (Idea #28)

The three final wheel-based designs were presented to Dr. Lizama for review, providing a clear comparison of their features. During the presentation, the team shared their assessment, highlighting that idea #28 was considered the strongest based on criteria and design analysis. After discussing the options and taking into consideration the team's recommendation, Idea #28 was selected as the preferred design to move forward with, concluding the idea generation process.

2.7 Conclusion

A structured design process was used to gradually reduce an initial set of 52 ideas into a final, well-supported solution. A plus-minus method was first implemented to eliminate ideas that deviated significantly from the PDS. These were then reviewed with a greater focus on costs, size, and weight to identify the five most promising designs using a weight-rated method. A second weighted evaluation narrowed the five preliminary designs, resulting in three final designs, all based on a wheel actuation method. After presenting these options, Idea #28 was selected as the preferred solution for further development, as it best met the requirements set out in the PDS.

3 Calculations

3.1 Introduction

This chapter presents the engineering calculations used to support the design and validate the performance of the automatic door system. The purpose of these analyses is to ensure that all selected components, including the motor, battery, and transmission system, satisfy the functional and safety requirements established in the Product Design Specifications (PDS).

The calculations include electrical power consumption, battery capacity, door kinematics, torque requirements, and energy consumption. Additionally, structural analyses were performed to evaluate the enclosure's ability to withstand accidental impacts and operational loads. Each analysis includes the parameters used, governing equations, calculation procedure, and engineering implications of the obtained results.

Together, these analyses provide quantitative validation that the proposed system is mechanically feasible, energy efficient, and capable of reliable operation under expected working conditions.

3.2 Motor Power Requirements

3.2.1 Objective

This analysis was performed to determine the electrical power requirements of the selected DC motor under both normal and worst-case operating conditions. The results are used to size the power supply and evaluate battery capacity requirements.

3.2.2 Parameters Used

Parameter	Symbol	Value
Rated Voltage	V	12 V
Rated Current	I	0.6 A
Stall Current	I_{stall}	1.7 A

The motor specifications were obtained from the technical information provided for the BRINGSMART JGY-370 12 V, 30 rpm DC worm gear motor product listing [33].

A 12 V DC motor was selected based on the torque and power requirements of the system, as well as its compatibility with commonly available power sources and motor drivers. The selected voltage provides sufficient power to generate the torque required to open the door while maintaining manageable current levels, reducing excessive power losses and component stress. Additionally, 12 V systems are widely available, cost-effective, and compatible with rechargeable battery solutions, making them suitable for portable and reliable operation.

3.2.3 Governing Equations

Electrical power is calculated using:

$$P = VI \tag{1}$$

Worst-case power occurs during stall conditions:

$$P_{max} = VI_{stall} \tag{2}$$

3.2.4 Calculations

Rated operating power:

$$P = (12)(0.6) = 7.2\ W$$

Worst-case power:

$$P_{max} = (12)(1.7) = 20.4\ W$$

3.2.5 Results and Implications

The motor requires approximately 7.2 W during normal operation and up to 20.4 W under stall conditions. These values were used to ensure that the selected battery and motor driver are capable of supporting transient peak loads without overheating or voltage instability.

3.3 Servo Power Requirements

3.3.1 Objective

This analysis was performed to estimate the electrical power required by the servo motor responsible for engaging and disengaging the transmission mechanism.

3.3.2 Parameters Used

Parameter	Symbol	Value
Servo Voltage	V	6 V
Current Range	I	0.5–0.9 A

The servo motor specifications were obtained from the technical information provided for the Hosyond MG996R digital servo motor product listing.

A 6 V servo motor was selected because it provided the torque and positional control required to reliably engage and disengage the transmission mechanism. The selected operating voltage is within the recommended range of the MG996R servo and ensures stable performance without overstressing the component. Additionally, 6 V servos are widely supported by commercially available power regulation and control systems, making integration into the overall design practical and reliable.

3.3.3 Governing Equation

Servo power consumption is calculated using Equation (1):

$$P = VI$$

3.3.4 Calculations

Worst-case servo power:

$$P = (6)(0.9) = 5.4\ W$$

Total system power:

$$P_{total} = 7.2 + 5.4 = 12.6\ W \approx 13\ W$$

3.3.5 Results and Implications

The servo contributes a relatively small portion of the total system power consumption; however, its peak current demand must still be considered during power supply selection to avoid voltage drops during engagement and disengagement operations.

3.4 Battery Capacity and Runtime

3.4.1 Objective

This analysis was performed to verify that the selected battery provides sufficient backup runtime during power outages, as required by the Product Design Specifications (PDS).

3.4.2 Parameters Used

Parameter	Symbol	Value
Battery Voltage	V	12 V
Battery Capacity	Ah	6 Ah
Total Power Consumption	P	13 W

A 12 V, 6000 mAh lithium-ion battery [35] was selected based on its voltage compatibility with the system requirements, sufficient energy capacity to provide backup runtime during power outages, and portability. Additionally, the selected battery is capable of charging while

powering the system, allowing continuous operation without interruption and improving overall system reliability.

3.4.3 Governing Equations

Battery energy capacity:

$$E = V(Ah) \tag{3}$$

Battery runtime:

$$t = \frac{E}{P} \tag{4}$$

3.4.4 Calculations

Battery energy:

$$E = (12)(6) = 72\ Wh$$

Manufacturer-rated energy:

$$E = 66.6\ Wh$$

Runtime:

$$t = \frac{66.6}{13} = 5.12\ hours$$

3.4.5 Results and Implications

Although the calculated continuous full-load runtime is approximately 5.12 hours, the expected real-world operating duration is significantly longer because the system operates intermittently rather than continuously. Since the motor and servo are only activated during door opening and closing events, the battery is expected to provide backup operation for multiple days under typical usage conditions.

3.5 Door Geometry and Travel Distance

3.5.1 Objective

This analysis determines the travel distance required to fully open the door and estimates the wheel motion necessary to achieve the desired angular displacement.

3.5.2 Parameters Used

Parameter	Symbol	Value
Distance from Hinge	r	1.016 m
Door Angle	θ	$90° = 1.57\,rad$
Wheel Diameter	D	4in = 0.1016 m

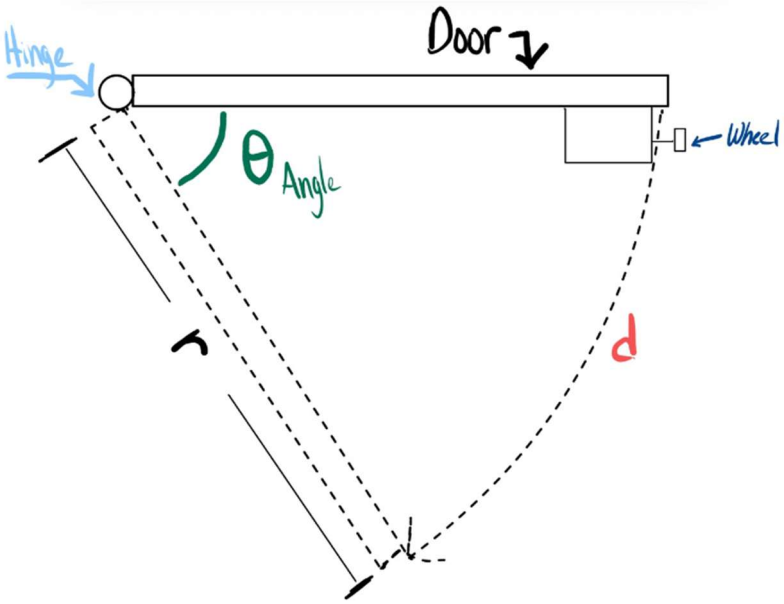


Figure 8: Door FBD

Wheel selection was based on findings from previous prototype iterations. Initially, a 65 mm wheel diameter was paired with a 30-rpm motor. Although this configuration met the Product Design Specification (PDS) time requirements for opening the door, testing revealed inadequate wheel performance, particularly in traction. To address this issue, the wheel diameter was increased to 85 mm; however, traction problems persisted. As a result, a larger and more durable wheel was tested. The final selection was a 4-inch wheel, which successfully resolved the traction issues while continuing to satisfy the PDS requirements.

3.5.3 Governing Equations

Arc distance traveled by the door:

$$d = r\theta \tag{5}$$

Wheel circumference:

$$C = \pi D \tag{6}$$

3.5.4 Calculations

Arc distance:

$$d = (1.016)(1.57) \approx 1.6\ m$$

Wheel circumference:

$$C = \pi(0.1016) = 0.319\ m$$

3.5.5 Results and Implications

The analysis shows that the door requires approximately 1.6 m of travel distance to achieve a 90° opening angle. The selected wheel diameter (4in) reduces the number of revolutions required while improving traction and motion efficiency.

3.6 Required Revolutions and Time

3.6.1 Objective

This analysis estimates the number of wheel revolutions and operating time required to fully open or close the door.

3.6.2 Parameters Used

Parameter	Symbol	Value
Travel Distance	d	1.6 m
Wheel Circumference	C	0.319 m
Motor Speed	RPM	30 RPM

3.6.3 Governing Equations

Number of revolutions:

$$N = \frac{d}{C} \tag{7}$$

Motor speed conversion:

$$30\ RPM = \frac{30}{60} = 0.5\ rev/s$$

Operating time:

$$t = \frac{N}{rev/s} \tag{8}$$

3.6.4 Calculations

Required revolutions:

$$N = \frac{1.6}{0.319} = 5.02 \approx 5\ rev$$

Time to open/close:

$$t = \frac{5}{0.5} = 10\ s$$

3.6.5 Results and Implications

The calculated opening and closing time of approximately 10 seconds satisfies the PDS requirement for smooth and safe operation while avoiding excessively rapid door movement that could compromise user safety.

3.7 Torque Analysis

3.7.1 Objective

This analysis determines the minimum torque required to actuate the door and validates the motor selection.

3.7.2 Parameters Used

Parameter	Symbol	Value
Hinge Torque	T_{hinge}	3 N·m
Distance from Hinge	r	1.016 m

Parameter	Symbol	Value
Wheel Radius	r_w	0.0508 m

An estimated hinge torque of 3 N·m was used based on prototype testing and iterative validation. Since the prototype successfully opened the door under normal operating conditions, the selected torque value was considered representative of the required opening resistance for the tested door.

3.7.3 Governing Equations

Force required to open the door

$$F = \frac{T}{r} \tag{9}$$

Wheel torque:

$$T = Fr \tag{10}$$

Required torque including safety factor:

$$T_{required} = SF(T_{theoretica}) \tag{11}$$

3.7.4 Calculations

Force required to open the door

$$F = \frac{3}{1.016} \approx 3\ N$$

Wheel radius:

$$r_w = \frac{0.1016}{2} = 0.0508\ m$$

Wheel torque:

$$T = (3)(0.0508) = 0.1524\ N \cdot m$$

Conversion to kg·cm:

$$T = 0.1524(10.2) = 1.55\ kg \cdot cm$$

Applying safety factor:

$$T_{required} = (2)(1.55) = 3.10\ kg \cdot cm$$

A safety factor of 2 was applied to the calculated torque requirement to account for uncertainties and real operating conditions not captured in the theoretical analysis. These factors include variations in door resistance, wheel slippage, friction between the wheel and floor, misalignment during operation, changes in battery voltage, and transient loads during startup. Applying a safety factor ensures that the motor can reliably generate sufficient torque to open the door under varying conditions without operating continuously near its performance limit.

3.7.5 Results and Implications

The analysis determined that a minimum torque of approximately 3.10 kg·cm is required to actuate the door reliably. A 6 kg·cm motor was selected to provide additional margin for friction losses, mechanical inefficiencies, and long-term reliability.

As explained before, a safety factor of 2 was implemented on motor selection to account for:

- Friction losses
- Door variability
- Dynamic loading effects
- Manufacturing tolerances
- Motor performance variability

3.7.6 Final Motor Selection

A motor with at least **3 kg·cm torque** is sufficient. However, selecting a **6 kg·cm motor** is recommended to provide additional margin for friction, inefficiencies, and long-term reliability.

3.8 Energy Per Cycle

3.8.1 Objective

This analysis estimates the energy consumption per operating cycle and evaluates long-term battery endurance under typical daily usage conditions.

3.8.2 Parameters Used

Parameter	Symbol	Value
Motor Power	(P_m)	7.2 W
Servo Power	(P_s)	5.4 W
Motor Operating Time per Cycle	(t_m)	20 s
Servo Operating Time per Cycle	(t_s)	4 s
Number of Daily Cycles	(n)	20
Battery Energy Capacity	(E_b)	66.6 Wh

The motor power value was obtained from the motor power requirement analysis (Section 3.2), while the servo power value was obtained from the servo power requirement analysis (Section 3.3). Operating times per cycle and daily cycle count were estimated based on expected system usage during normal operation.

3.8.3 Governing Equations

Energy consumption:

$$E = Pt \tag{12}$$

Daily energy usage:

$$E_{daily} = n(E_{cycle}) \tag{13}$$

Battery life estimate:

$$Battery\ Life = \frac{Battery\ Capacity}{Daily\ Consumption} \tag{14}$$

3.8.4 Calculations

Motor energy:

$$E = \frac{(7.2)(20)}{3600} = 0.040\ Wh$$

Servo energy:

$$E = \frac{(5.4)(4)}{3600} = 0.006\ Wh$$

Total cycle energy:

$$E_{cycle} = 0.040 + 0.006 = 0.046\ Wh$$

Daily energy consumption:

$$E_{daily} = 20(0.046) = 0.92\ Wh$$

Battery life:

$$Battery\ Life = \frac{66.6}{0.92} = 72.4\ days$$

3.8.5 Results and Implications

The low daily energy consumption demonstrates that the system is highly energy efficient and capable of extended operation without frequent charging. The estimated battery life, of 72.4 days, significantly exceeds the minimum PDS backup power requirement of 24 hours in case of power outages (metric 7).

3.9 Conclusions

The engineering analysis validates the feasibility of the proposed automatic door opener system and supports the selection of the motor, battery, and wheel.

The torque analysis determined that a minimum torque of approximately **3.10 kg · cm** is required after applying a safety factor of 2. Based on this result, a **6 kg·cm motor** was selected to provide additional margin for frictional losses, system inefficiencies, and long-term reliability. This selection ensures consistent operation without overloading the motor.

The kinematic analysis showed that using a **1.016 mm wheel diameter** reduces the required number of revolutions and achieves a door opening/closing time of approximately **10 seconds**, which satisfies the Product Design Specifications (PDS).

Energy consumption analysis demonstrated that the system requires approximately **0.046 Wh per cycle**, resulting in a daily consumption of **0.92 Wh** for typical usage. With a battery capacity of **66.6 Wh**, the system can operate for approximately **72 days per charge**, indicating high energy efficiency.

Overall, the selected motor provides sufficient torque and speed for reliable mechanical performance, while the chosen battery offers more than adequate energy capacity for

extended operation. These results confirm that the design meets performance requirements and is both practical and efficient for real-world use.

4 CAD Designs and Analysis

4.1 Introduction

This section presents the prototypes and the final design's structural evaluation of the component box developed for the automatic door system. The purpose of this section is to demonstrate how the enclosure was refined to meet functional requirements while ensuring adequate strength, durability, and efficient use of space. The design process incorporated modifications based on component availability and prototype performance, leading to a more compact and reinforced enclosure.

Following the presentation of the final CAD design, analytical methods are applied to evaluate the structural integrity of the box under realistic loading conditions. These include a drop test analysis to simulate accidental impacts during handling and an external impact analysis representing forces applied while the box is mounted on the door. Key parameters such as stress, acceleration, and factor of safety are calculated and compared against material properties to assess performance.

Together, the design improvements and engineering analyses provide validation that the component box can withstand expected operational and accidental loads, while also identifying potential areas for further enhancement.

4.2 Prototypes

4.2.1 Prototype V0

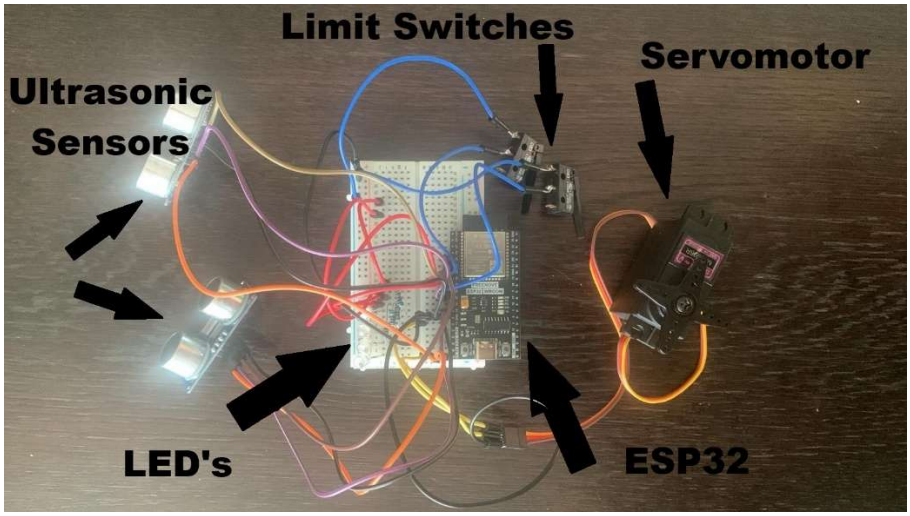


Figure 9: Prototype V0

The first prototype of the system was developed purely as an electronic model, with no 3D printed components or mechanical structures included at this stage. Its purpose was to establish and verify the core wiring and functional relationships between the different elements of the design. The setup consisted of an ESP32 microcontroller, two ultrasonic sensors, two limit switches, a servo motor, and LED lights as visual indicators to represent the behavior of key outputs, such as opening, closing, and motor engagement, allowing the system's logic to be tested before the addition of any mechanical component.

4.2.2 Prototype V1

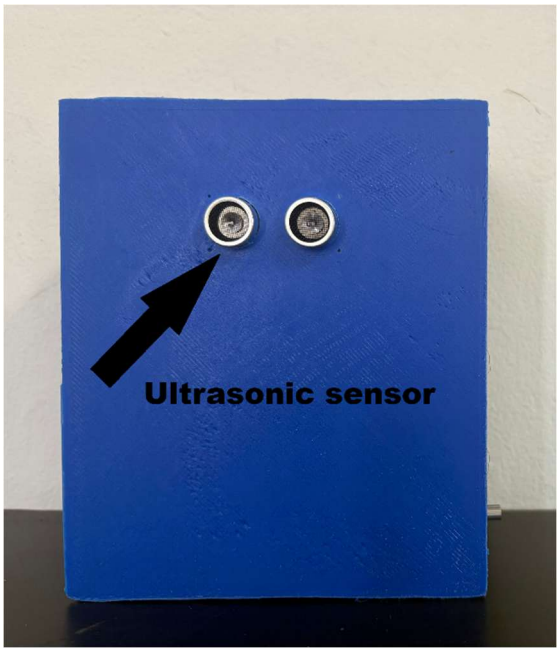


Figure 10: Prototype V1 (Front View)

Figure 9 illustrates the front view of a 3D printed enclosure featuring two precisely positioned openings used to mount an ultrasonic sensor. The idea was to determine an effective method of integrating the sensor within the box while still allowing it to accurately measure distance. The two openings help the sensor's transmitter and receiver to fully operate without interference while its internal components remain protected.

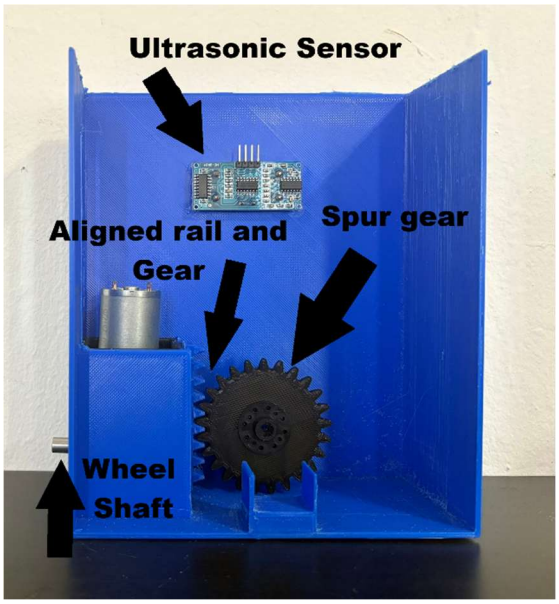


Figure 11: Prototype V1 (Section View)

A sectional view can be observed in Figure 10, to allow visualization of the internal arrangement and interaction of its components and to demonstrate that the manufactured elements fit and work as expected. In particular, the spur gear is shown to align perfectly with the rack, and the motor holder is placed precisely within its designated section, granting freedom of vertical movement as the gear rotates.

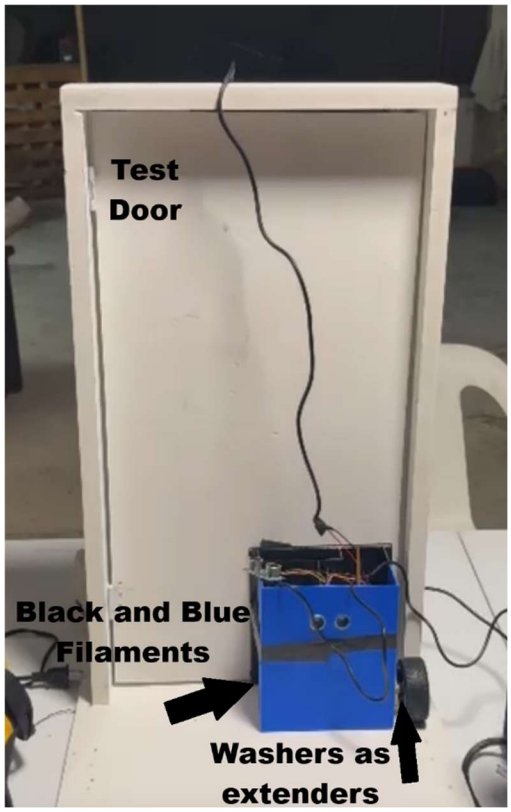


Figure 12: Prototype V1 (Mounted)

Although the components appeared to fit correctly in the assembled enclosure, several issues became evident during testing. First, the component box was not fully completed during the initial 3D printing process and stopped midway. As a result, a different filament had to be used to complete the print. Second, screws were not accounted for in the CAD design, which meant that once the screws were introduced, the motor holder no longer fit properly in the box. Furthermore, the spur gear did not couple as expected in the servo motor and required the assistance of a servo motor arm to connect properly. Finally, it was identified that the motor shaft was not long enough to reach the wheel, which prevented

transmission of motion from the motor to the wheel. These highlighted design oversights and the need for further refinement of the system.

4.2.3 Prototype V2

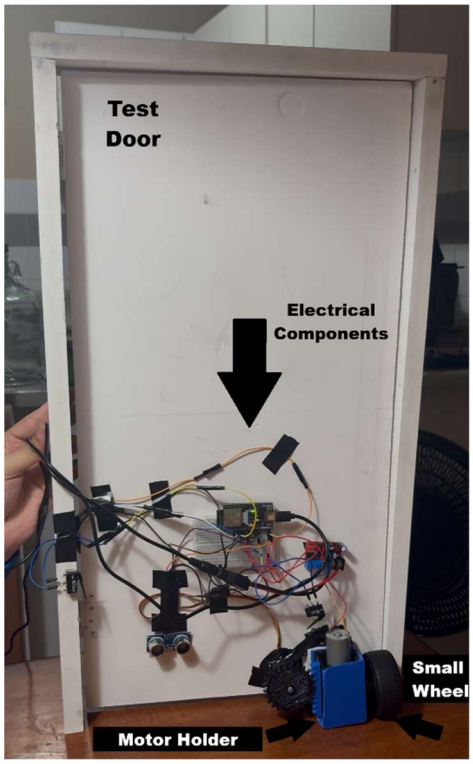


Figure 13: Prototype V2

For this prototype, the component box was completely broken down, leaving only the section housing the motor holder intact. The primary purpose of this iteration was to observe whether the system itself was functioning as expected and to address any uncertainties before moving on to the new CAD designs and 3D printing. To extend the motor's shaft transmission to the wheel, multiple washers were stacked together until the shaft reached the necessary length. During testing, it was confirmed that the door could indeed open, but it became evident that the wheel needed to be larger. This adjustment was necessary to establish better traction with the ground, as the initial wheel size of 65mm resulted in inadequate grip. To address this issue, alternative wheel sizes were evaluated. An 85 mm diameter wheel was first tested, but traction problems persisted. Consequently, a 4-inch diameter wheel was selected, as it successfully resolved the traction issue and provided reliable operation.

4.3 Final Design

The final design of the component box reflects a series of improvements made to enhance structural integrity, optimize space utilization, and adapt to changes in component availability. Initially, the design accommodated an internal uninterruptible power supply (UPS); however, due to procurement delays, a wall-mounted UPS was selected as an alternative. This modification allowed for a significant reduction in the overall volume of the enclosure (from $149.5\ in^3$ to $92.24\ in^3$), resulting in a more compact and efficient design. Additionally, the wall thickness was increased from 3 mm to 5 mm, substantially improving the stiffness and durability of the box, which addressed the structural weaknesses observed in the initial prototype. Internal space was reorganized to ensure proper fitment of all remaining components while minimizing unused volume and material waste. Furthermore, design refinements were made to the gear and mounting features to improve mechanical performance and reliability. Overall, the final enclosure achieves a balance between strength, functionality, and material efficiency, making it better suited for real-world operation.

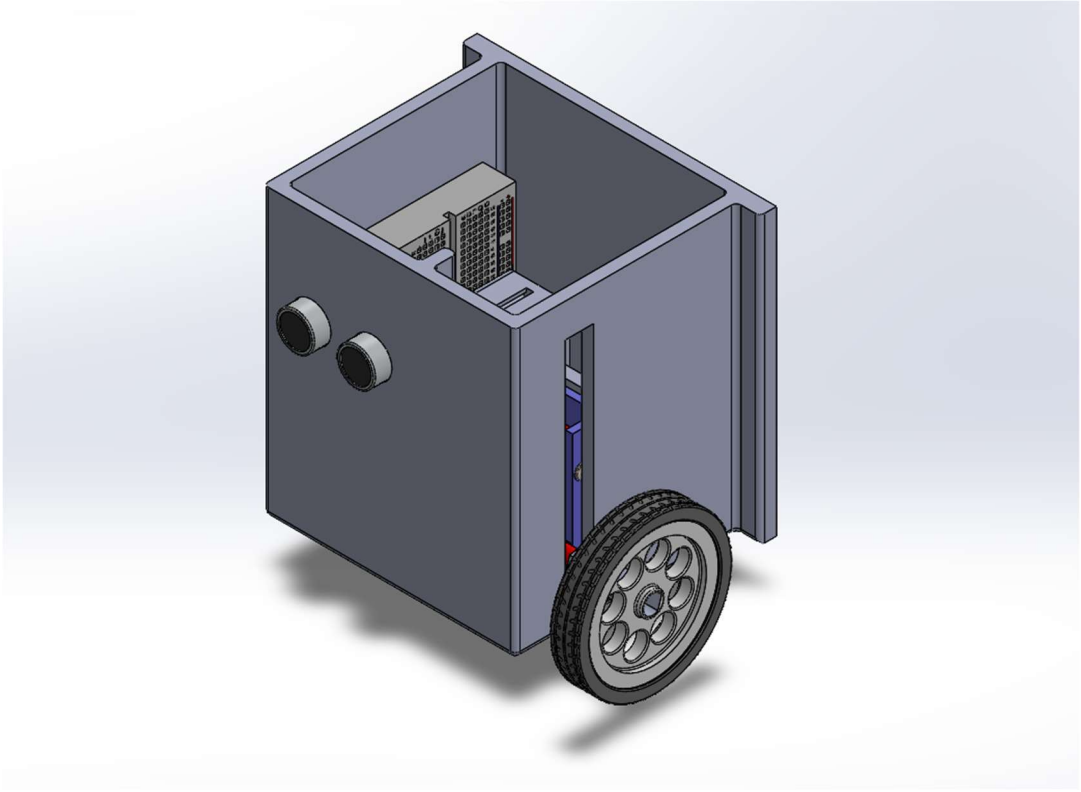


Figure 14: Final Design

4.4 Drop Test Analysis

4.4.1 Objective

This analysis was performed to evaluate the structural integrity of the enclosure under accidental drop conditions. The objective is to estimate the impact forces and stresses generated during a worst-case drop scenario and verify that the PLA enclosure can withstand these loads without structural failure.

The evaluation follows a failure-based design approach, where potential weaknesses are identified and addressed early in the development process, as emphasized in the Engineering Errors Seminar.

4.4.2 Parameters Used

Parameter	Symbol	Value
Material	—	PLA
Mass of enclosure	m	0.36 kg
Wall thickness	t	0.005 m
Drop height	h	0.65 m
Gravity	g	9.81 m/s^2
Impact duration	Δt	0.001 s
Contact area	A	$1 \times 10^{-4}\ m^2$

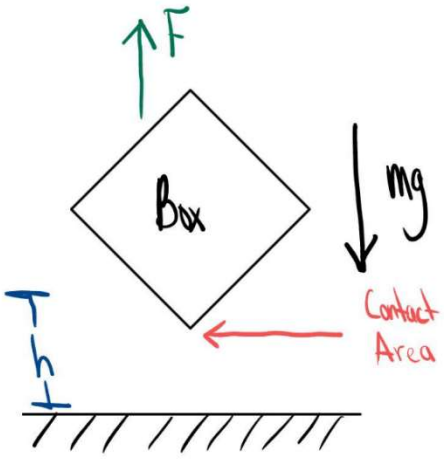


Figure 15: Drops Test FBD

The parameters used in the drop test analysis were selected based on the enclosure geometry, prototype characteristics, and assumptions commonly used in impact and drop-test literature. The enclosure mass was obtained from the CAD model and prototype measurements, while the wall thickness corresponds to the nominal enclosure design thickness. A drop height of 0.65 m was selected to represent a realistic accidental drop during handling or installation, consistent with free-fall testing approaches described in ASTM D5276 and IEC 60068-2-31 [4], [5]. The gravitational acceleration was taken as the standard value of 9.81 m/s^2. An impact duration of 0.001 s was assumed to represent the short contact times commonly observed in rigid-body drop events and simplified impact analyses [1]–[3]. The contact area was estimated as a localized impact region to represent a conservative loading condition during surface contact.

4.4.3 Governing Equations

Impact velocity:

$$V = \sqrt{2gh} \tag{15}$$

Impact energy:

$$E = \frac{1}{2}mv^2 \tag{16}$$

Impact force:

$$F = \frac{mv}{\Delta t} \tag{17}$$

Acceleration:

$$a = \frac{F}{m} \tag{18}$$

Stress:

$$\sigma = \frac{F}{A} \tag{19}$$

Factor of safety:

$$SF = \frac{\sigma_y}{\sigma} \tag{20}$$

4.4.4 Calculations

4.4.4.1 Impact Velocity

$$V = \sqrt{2(9.81)(0.65)} \qquad V = 3.57\ m/s$$

4.4.4.2 Impact Energy

$$E = \frac{1}{2}(0.36)(3.57)^2 \qquad E = 2.29\ J$$

The enclosure must absorb approximately 2.3 J of energy during impact.

4.4.4.3 Impact Force

Assuming an impact duration of:

$$\Delta t = 0.001\ s$$
$$F = \frac{(0.36)(3.57)}{0.001}$$
$$F = 1285\ N$$

4.4.4.4 Acceleration (Shock Load)

$$a = \frac{1285}{0.36}$$
$$a = 3569\ m/s^2$$

Converting to gravitational acceleration:

$$a = \frac{3569}{9.81}$$
$$a = 364\ g$$

The enclosure experiences approximately 360 g during impact.

4.4.4.5 Stress Calculation

Assuming a worst-case corner impact area:

$$A = 1\ cm^2 = 1 \times 10^{-4}\ m^2$$
$$\sigma = \frac{1285}{1 \times 10^{-4}}$$
$$\sigma = 12.85\ MPa$$

4.4.4.6 Material Strength Comparison

Typical PLA yield strength:

$$\sigma_y \approx 50\ MPa$$

Factor of safety:

$$SF = \frac{50}{12.85}$$
$$SF = 3.89$$

4.4.5 Results and Implications

The drop test analysis demonstrates that the enclosure is capable of withstanding accidental drops under the evaluated conditions. The calculated stress of 12.85 MPa remains significantly below the approximate PLA yield strength of 50 MPa, resulting in a factor of safety of approximately 3.89.

These results indicate that the selected wall thickness and enclosure geometry provide sufficient structural resistance against accidental handling impacts. The analysis also confirms that increasing the wall thickness from 3 mm to 5 mm substantially improved the enclosure's durability and rigidity.

Although the enclosure satisfies the required structural performance, the analysis assumes uniform stress distribution and ideal material behavior. In practice, localized stress concentrations, layer adhesion quality, and print orientation may influence the actual impact resistance of the 3D printed structure.

4.5 Impact Analysis

4.5.1 Objective

This analysis evaluates the enclosure's ability to withstand external impact forces while mounted on the door. The scenario represents accidental misuse conditions, such as a person kicking the enclosure during operation.

4.5.2 Parameters Used

Parameter	Symbol	Value
Mass of system	m	0.36 kg
Impact velocity	v	3 m/s
Contact deformation distance	s	0.005 m
Contact area	A	$2 \times 10^{-4}\ m^2$
PLA yield strength	σ_y	50 MPa

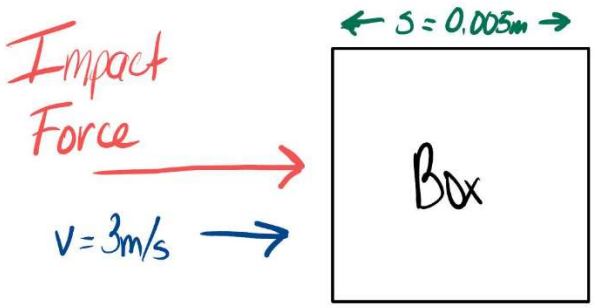


Figure 16: Impact Analysis FBD

The parameters used for the impact analysis were selected based on the physical properties of the enclosure, expected misuse conditions, and material data from referenced sources. The system mass was obtained from the CAD model and prototype measurements. An impact velocity of 3 m/s was selected to represent a moderate accidental kick or collision during operation, corresponding to realistic human interaction scenarios. The contact deformation distance was assumed to be 0.005 m to represent localized deformation during impact and energy absorption between the enclosure and contacting surface, consistent with simplified impact modeling approaches [2], [3]. The contact area was estimated as a localized region of impact to conservatively represent force concentration during collision. The PLA yield strength value of 50 MPa was obtained from published material property data for PLA [11].

4.5.3 Governing Equations

Impact energy:

$$E = \frac{1}{2}mv^2 \tag{21}$$

Average impact force using the work-energy principle:

$$F_{avg}s = \frac{1}{2}mv^2 \tag{22}$$

Average impact force:

$$F_{avg} = \frac{0.5mv^2}{s} \tag{23}$$

Peak impact force approximation:

$$F_{max} \approx 2F_{avg} \tag{24}$$

Stress calculation:

$$\sigma = \frac{F}{A} \tag{25}$$

Factor of safety:

$$SF = \frac{\sigma_y}{\sigma} \tag{26}$$

4.5.4 Calculations

4.5.4.1 Impact Energy

$$E = \frac{1}{2}(0.36)(3)^2$$
$$E = 1.62\ J$$

4.5.4.2 Average Impact Force

$$F_{avg} = \frac{0.5(0.36)(3)^2}{0.005}$$
$$F_{avg} = 324\ N$$

4.5.4.3 Peak Force Estimation

$$F_{max} \approx 2(324)$$
$$F_{max} = 648\ N$$

4.5.4.4 Stress on the Enclosure

Assuming a contact area of:

$$A = 2\ cm^2 = 2 \times 10^{-4}\ m^2$$

$$\sigma = \frac{648}{2 \times 10^{-4}}$$

$$\sigma = 3.24\ MPa$$

4.5.4.5 Factor of Safety

$$SF = \frac{50}{3.24}$$

$$SF = 15.4$$

4.5.5 Results and Implications

The impact analysis demonstrates that the enclosure can safely withstand moderate external impacts while mounted on the door. The calculated stress of 3.24 MPa is significantly lower than the yield strength of PLA, resulting in a factor of safety of approximately 15.4.

These results indicate that the enclosure possesses adequate structural strength for realistic accidental loading scenarios. Additionally, the relatively high factor of safety suggests that the enclosure design contains a significant structural margin against fracture or permanent deformation. To further improve long-term durability and impact resistance, future design iterations may consider increasing fillet radii at sharp corners to reduce stress concentrations, optimizing print orientation to improve interlayer strength, and incorporating localized reinforcement in high-stress regions. Additionally, the use of elastomeric mounting materials or protective padding may further reduce stresses transmitted to the enclosure during accidental impacts.

However, the analysis assumes simplified loading conditions and ideal material properties. Real-world performance may vary due to stress concentrations, print layer orientation, and repeated impact loading over time.

4.6 Conclusion

The CAD design and structural analyses demonstrate that the final enclosure satisfies the required mechanical and functional performance criteria for the intended application.

The redesign process, which included reducing the enclosure volume from $149.5\ in^3$ to $92.24\ in^3$ and increasing the wall thickness from 3 mm to 5 mm, significantly improved the rigidity and durability of the structure compared to earlier prototypes. The performed drop test and impact analyses confirmed that the enclosure can withstand both accidental drops and moderate external impacts while maintaining acceptable stress levels and factors of safety relative to the material properties of PLA.

The analyses also validated that the enclosure design provides sufficient structural margin for real-world operation while identifying critical areas, such as corners and mounting interfaces, that may require additional reinforcement in future iterations.

Additionally, the enclosure development followed an iterative engineering design methodology aligned with agile hardware development principles. Multiple prototypes were manufactured and evaluated to progressively improve component integration, stiffness, manufacturability, and transmission performance. This iterative approach enabled design limitations to be identified and corrected efficiently, resulting in a more optimized and reliable final design.

Overall, the final enclosure achieves an effective balance between compactness, structural strength, manufacturability, and material efficiency, making it suitable for the intended assistive door actuation application.

5 Design Specifications

5.1 Introduction

This section presents the detailed compliance of design with specifications of the automatic door system, providing all the necessary information required for manufacturing, assembly, and system integration. The objective of this section is to clearly document the components, materials, and technical elements that define the final design, ensuring that the system can be accurately reproduced and evaluated.

The section begins with the Bill of Materials (B.O.M.), which outlines all components used in the system, including mechanical, electrical, and purchased parts. This is followed by a description of the manufactured components, highlighting custom-designed parts and their fabrication requirements. The electrical schematic is then presented to illustrate the system's wiring and control architecture, ensuring proper integration of all electronic elements. Additionally, the software code and flowchart are included to describe the control logic and operational sequence of the system.

Together, these specifications provide a comprehensive overview of the final design, bridging the gap between conceptual development and practical implementation while ensuring clarity, consistency, and reproducibility.

5.2 Bill Of Materials

The Bill of Materials (BOM) table provides a comprehensive list of all the components used in the development of the door opener/closer system, detailing both purchased and custom-made parts. It includes items such as the ESP 32 microcontroller, motor driver, limit switches, and ultrasonic sensors, among others, which were sourced from various suppliers. Additionally, the table outlines the custom-created components, such as the motor holder, which were fabricated using a 3D printer. Each entry in the BOM specifies the part number, part name, description, quantity, cost, supplier, and link providing an overview of the resources used in the project.

Table 10: Bill Of Materials

Part Num.	Part Name and Number	Description	Qty	Price	Supplier	Link
1	BRINGSMART DC Motor	12V Worm Gear DC Motor (30 RPM, 6 kg·cm) – High torque, low speed, self-locking motor for door actuation	1	$ 15.87	Amazon	DC motor
2	MG 996R Servo	MG996R Servo Motor – High-torque (≈11 kg·cm), metal gear, PWM-controlled positioning actuator	1	$ 13.99	Amazon	Servomotor
3	ESP32	ESP32 Dev Board – Dual-core 240 MHz microcontroller with Wi-Fi/Bluetooth for system control	1	$ 17.95	Amazon	Esp32
4	Ultrasonic Sensor	HC-SR04 Ultrasonic Sensor – Non-contact distance measurement (2–500 cm) using 40 kHz sound waves	2	$ 9.99	Amazon	URsensor
5	H Bridge	Dual H-Bridge Motor Driver (3–14V, 5A) – Bidirectional DC motor control with PWM speed regulation	1	$ 9.88	Amazon	HBridge
6	Wheel	65mm rubber wheel	1	$ 3.24	Control Hobby	ControlHobby
7	Perfboard	ELEGOO 32 Pcs Double sided PCB Board Prototype Kit	1	$ 9.99	Amazon	PerfBoard
8	M3 Screws (35mm)	M3-0.5x35mm Zinc Pan Head Phillips	1	$ 1.95	Home Depot	

		Drive Machine Screw				M3 Screws (35 mm)
9	M3 Screws (6mm)	M3-0.5 x 6 mm Zinc Pan Head Phillips Drive Machine Screw	5	$ 1.95	Home Depot	M3 Screws (6mm)
10	M3 Washer	M3 Zinc-Plated Steel Flat Washers	2	$ 1.06	Home Depot	M3 Washer
11	Screws (6-32 X 1/2)	Round Machine Screws (6-32 X 1/2)	4	$0.32	National	National
12	Hex Nuts (6-32 ZP)	Hex Nuts (6-32 ZP)	4	$0.24	National	National
13	Component Box	Structural enclosure designed to contain and protect electronic components, providing mounting features and cable management for the system.	1	$ 4.53	In-House Fabrication	
14	Lid	Removable cover designed to secure and protect internal components while allowing access for maintenance and assembly.	1	$ 0.83	In-House Fabrication	
15	Extender	Custom extension used to secure the wheel to the motor and transfer torque effectively.	1	$0.05	In-House Fabrication	
16	Servo Holder	Custom bracket designed to securely mount the servo motor, ensuring proper alignment and stable operation during actuation.	1	$ 0.70	In-House Fabrication	
17	Motor Holder	Support structure designed to hold	1	$ 1.90	In-House Fabrication	

		and align the DC worm gear motor, providing stability and proper positioning for torque transmission.				
18	Gear	Custom-designed gear used to transfer rotational motion from the motor to the door mechanism, enabling controlled movement and torque transmission.	1	$ 0.13	In-House Fabrication	
19	Power Supply	12VDC Switching Adaptor 2A	1	$3.50	Alpha3DLab	Alpha3D
20	Female Jack	Female jack for power supply	1	$0.50	Alpha3DLab	Alpha3D
21	UPS	Talentcell Rechargeable 12 V 6000mAh/5V	1	$32.99	Amazon	UPS
22	Jumper Wires	ELEGOO 120 pcs Multicolored Dupont Wire (Male to male, male to female, female to female)	1	$6.98	Amazon	JumperWires
23	Screws (6 X 1-1/4)	Flat Wood screws 6 X 1-1/4	4	$0.40	National	National

Note: Technical Drawings if in-house fabricated parts are found in the appendix and the manufacturing and assembly manual.

5.3 Manufactured Components

The manufactured components section focuses on the parts required for the development of the system, which will be produced using 3D printing. These components are designed to form the mechanical structure and functional elements of the mechanism, making sure that

the system can open and close a door efficiently. This manufacturing method allows for precise customization of parts such as mounts, gears, housings, all configured to fit the specific dimensions and requirements of the door. Additionally, it facilitates adjustments during development and guarantees that each of the components is integrated properly.

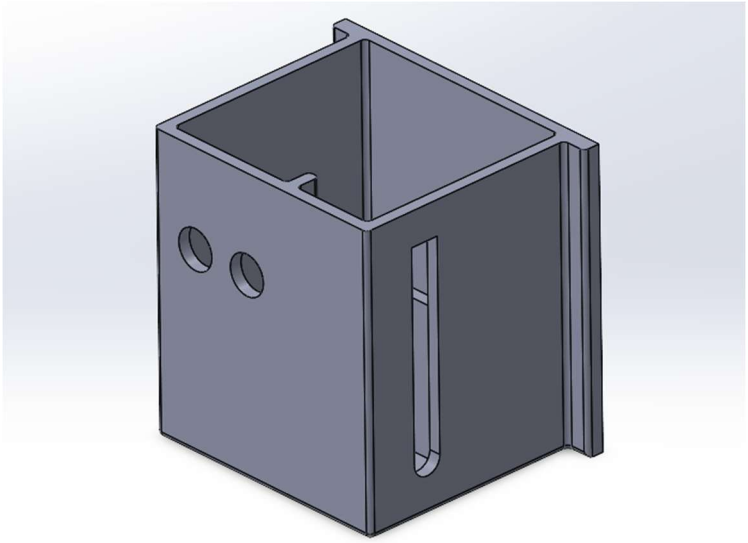


Figure 17: Component Box

Figure 14 shows the component box designed to securely store and organize the various components that will be used in the project. Each of the parts will be placed carefully in its designated section, helping maintain order. In addition to organization, it also provides protection to the components, shielding them from dust, moisture, or potential damage from its surroundings.

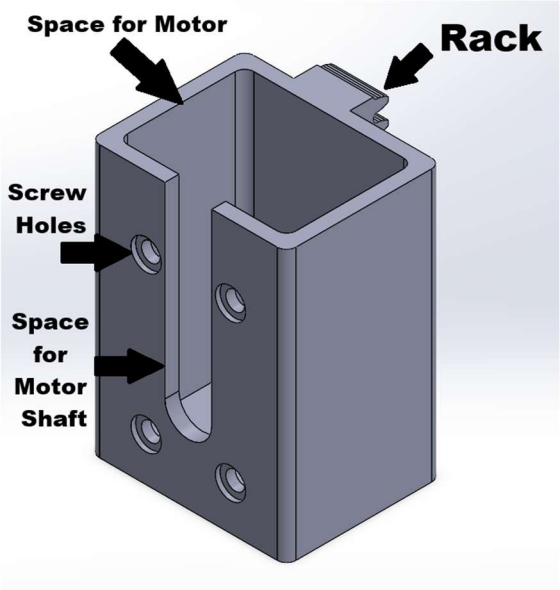


Figure 18: DC Motor Holder

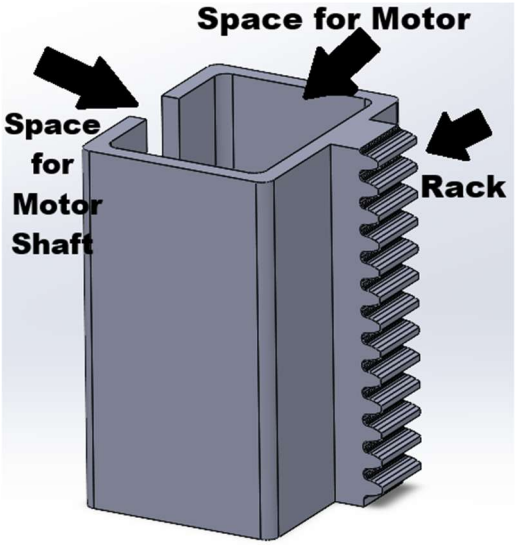


Figure 19: DC Motor Holder (Rack)

The figure depicts a motor holder engineered to firmly secure the motor in an upright orientation while keeping its shaft aligned horizontally. The positioning is critical for the transmission of motion to the driving wheel responsible for opening and closing the door. This component helps reduce any movement or vibrations that the motor may experience as it is turned on. On the rear side of the motor holder, a rack is implemented, allowing the system to be raised or lowered in a linear motion using a spur gear mechanism. This feature is added to help the door opener disengage when required, therefore permitting manual use of the door by individuals other than the client.

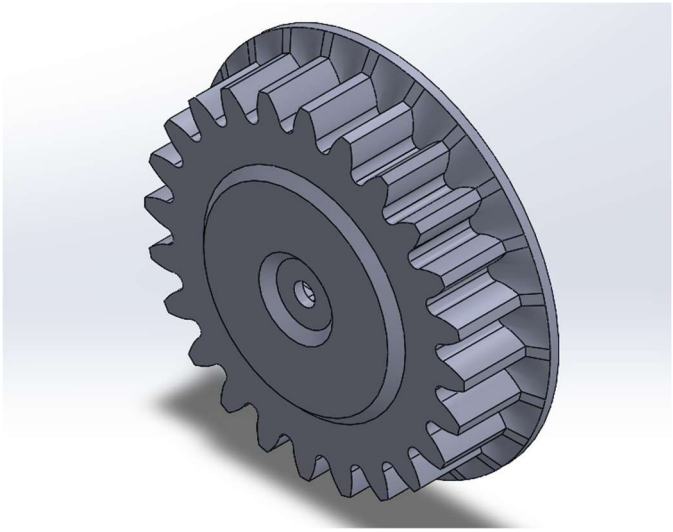


Figure 20: Spur Gear

As mentioned previously, this gear is positioned adjacent to the rack integrated in the motor holder. Its main function is to engage with the rack and convert rotational motion into linear movement, allowing the motor assembly to be raised and lowered, lifting the driving wheel away from the ground.

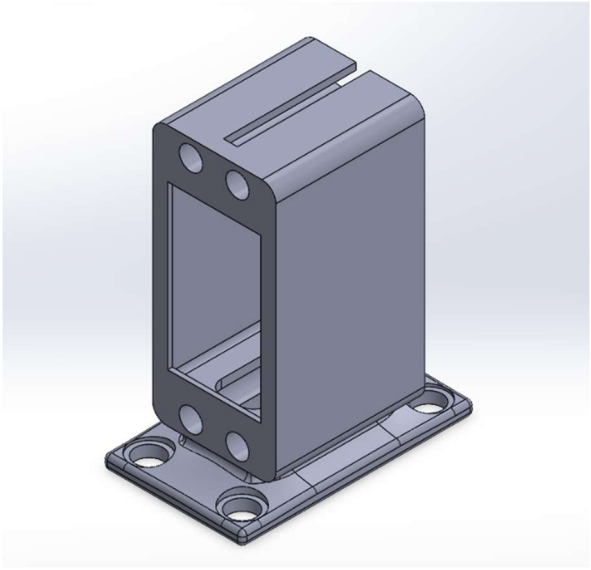


Figure 21: Servo Motor Mount

The servomotor mount is designed to securely hold the servomotor in a fixed and stable position, similar to the DC motor. The servo motor is an electromechanical device that converts electrical control signals into angular motion, rotating into specific and controlled

positions rather than continuous rotation. In this design, the servo motor is responsible for driving the spur gear, which in turn activates the rack and pinion system used to lift or lower the motor assembly.

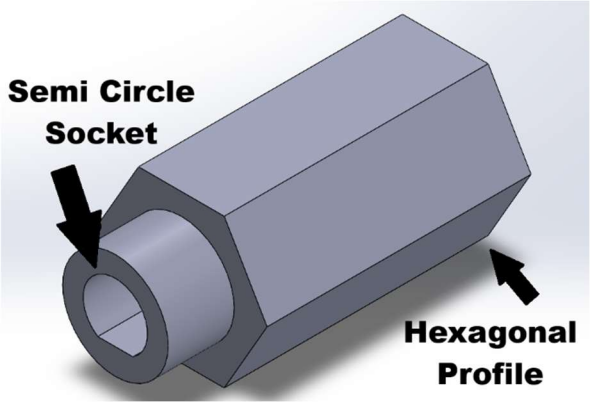


Figure 22: Extender

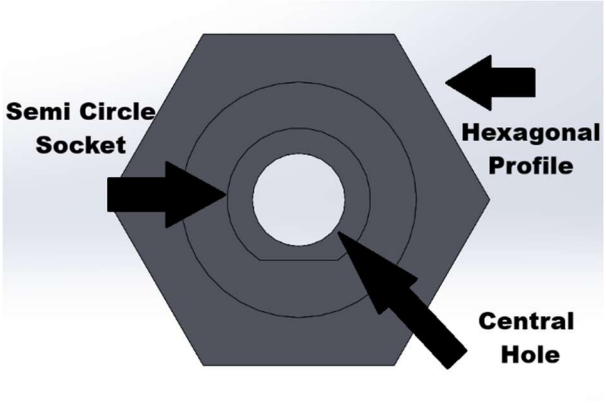


Figure 23: Extender Hole

The extender is designed to transfer rotational motion from the motor to the wheel. The motor shaft cannot directly reach the wheel, so the extender acts as a mechanical link between both elements. One end of the extender features a hexagonal profile that fits precisely into the corresponding hexagonal cavity of the wheel. This allows torque to be delivered effectively so that the wheel and extender rotate together without any slippage. The opposite end is shaped as a circle with a semi-circle socket to match the profile of the motor shaft, so that it is coupled securely and further supports transmission of rotational motion. In addition, a central hole is included in the design to accommodate a fastening screw firmly connecting all three components as a single rotating assembly.

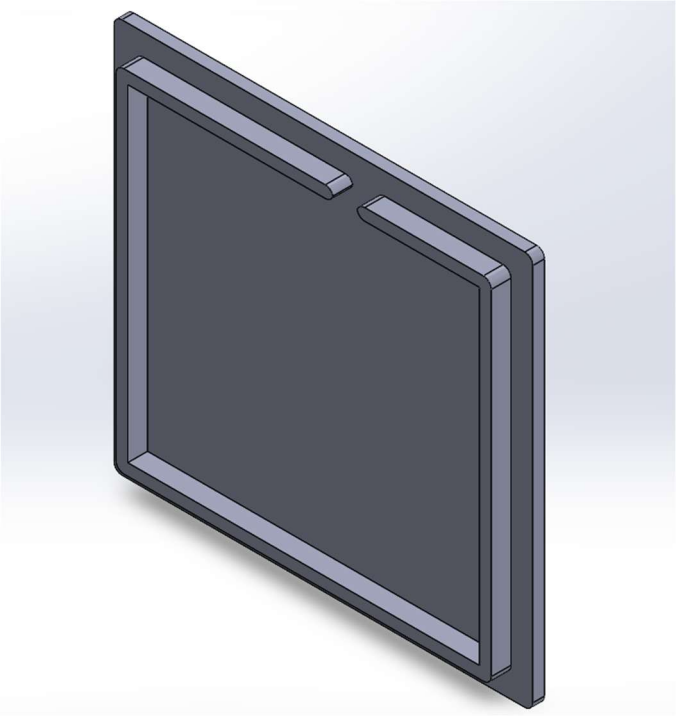


Figure 24: Component Box Lid

The component box lid is used to securely close the box once all the components have been stored inside. Its primary function is to provide an additional layer of protection for the components, while also improving the overall presentation of the setup by concealing any exposed cables, resulting in a cleaner and more organized appearance.

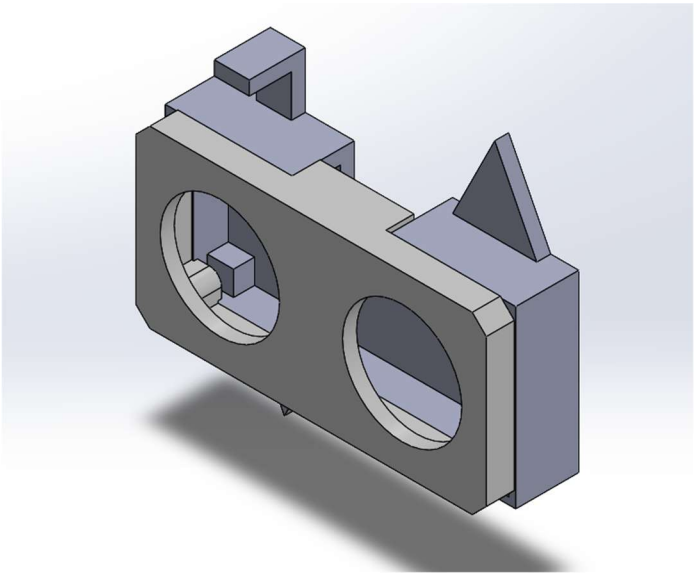
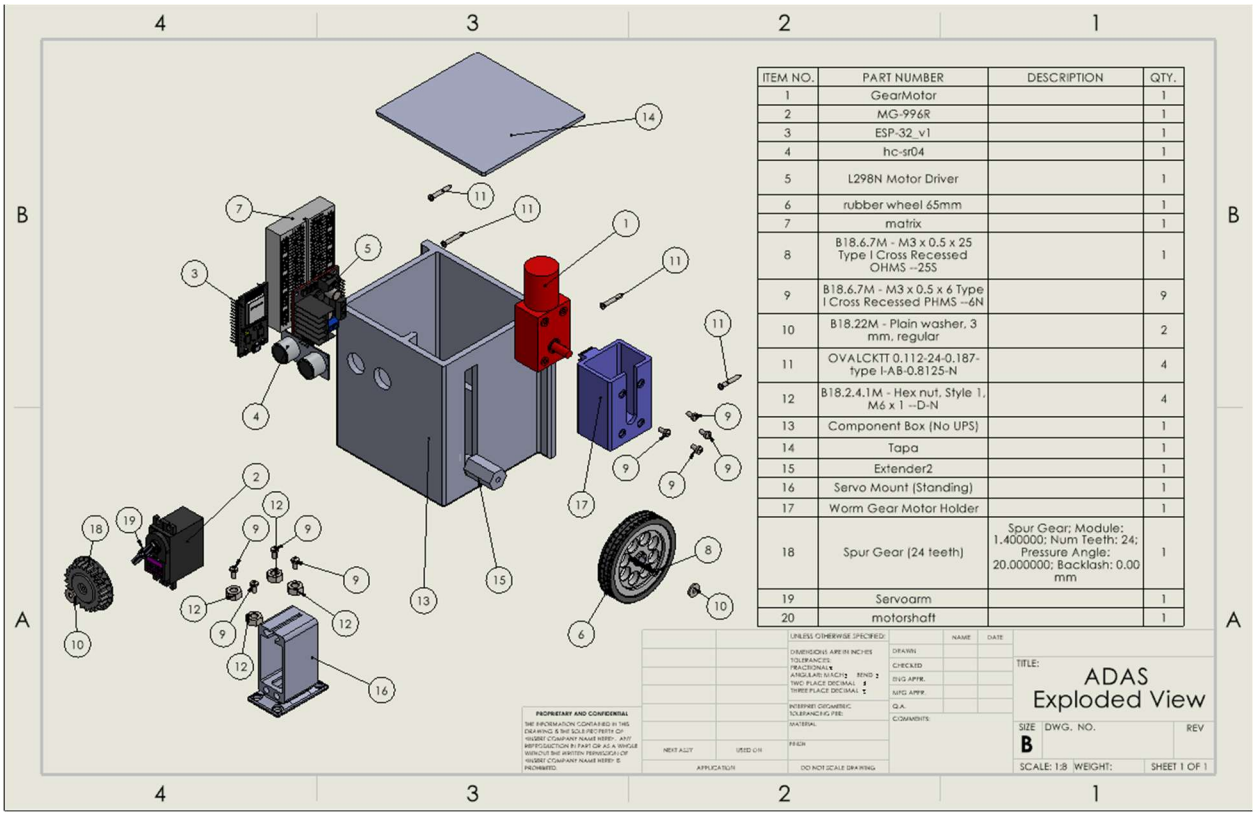


Figure 25: Ultrasonic Sensor Mount

Finally, a mount is constructed to safeguard the external ultrasonic sensor that will be positioned on the opposite side of the door. This sensor is used to identify any obstacle that may interfere with the door's operation and is designed to offer protection against environmental factors, while allowing the sensor to maintain its full operational capacity.



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	GearMotor		1
2	MG-996R		1
3	ESP-32_v1		1
4	hc-sr04		1
5	L298N Motor Driver		1
6	rubber wheel 65mm		1
7	matrix		1
8	B18.6.7M - M3 x 0.5 x 25 Type I Cross Recessed OHMS --25S		1
9	B18.6.7M - M3 x 0.5 x 6 Type I Cross Recessed PHMS --6N		9
10	B18.22M - Plain washer, 3 mm, regular		2
11	OVALCKTT 0.112-24-0.187-type I-AB-0.8125-N		4
12	B18.2.4.1M - Hex nut, Style 1, M6 x 1 --D-N		4
13	Component Box (No UPS)		1
14	Tapa		1
15	Extender2		1
16	Servo Mount (Standing)		1
17	Worm Gear Motor Holder		1
18	Spur Gear (24 teeth)	Spur Gear; Module: 1.400000; Num Teeth: 24; Pressure Angle: 20.000000; Backlash: 0.00 mm	1
19	Servoarm		1
20	motorshaft		1

Figure 26: Exploded View

In the previous figure, an exploded view of the complete product is presented, illustrating the arrangement and relationship of all individual components within the assembly. This view provides a clear representation of how the parts fit together and identifies each component through its corresponding part number, facilitating component recognition, assembly procedures, maintenance activities, and reference to the bill of materials.

5.4 Electrical Schematic

The electrical design was assembled with a perforated board and jumper wires; see the electrical schematic to ensure that components are properly connected.

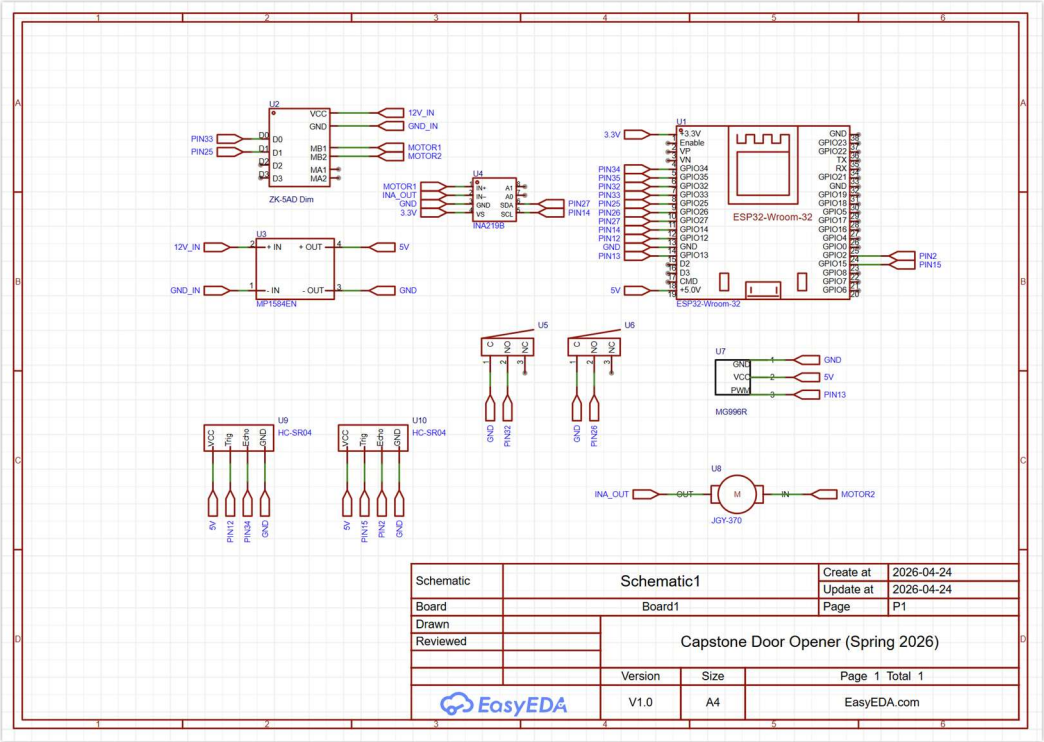


Figure 27: Electrical Schematic

Legend for Components:
U1 = ESP32

U2 = H-Bridge Motor Driver (ZK-5AD)

U3 = Voltage Regulator (MP-1584EN)

U4 = Current Sensor (INA219B)

U5 = Limit Switch

U6 = Limit Switch

U7 = Servo Motor (MG996R)

U8 = DC Motor (JGY-370)

U9 = Ultrasonic Sensor (HC-SR04)

U10 = Ultrasonic Sensor (HC-SR04)

5.5 Software Code and Flowchart

The software for this project is organized into two main components within the GitHub repository (github.com/yosefsantiago/door_opener/):

1. A **Progressive Web App (PWA)** for user interaction
2. Embedded **ESP32 firmware** for system control

These components are separated into folders within the repository to maintain modularity and clarity.

We have licensed this project under the General Public License Version 3 (GPLv3) to help future students iterate on our work without worrying about Intellectual Property infringement. This license ensures that if the project is revisited, future students can use our code as a reference or a starting point for their own versions, as long as your project also remains open and free for others to use (Intellectual Property Seminar).

The user interface is implemented as a PWA and is hosted using GitHub Pages. This allows the application to be accessed directly from a browser without requiring installation, while still supporting app-like behavior such as offline access and mobile compatibility. For iOS devices the Bluetooth feature is not supported via PWAs, so an app was included to allow access to the feature.

The repository includes a workflow configuration file (GitHub Actions) that automates deployment. This workflow extracts the contents of the site folder and publishes them to GitHub Pages. As a result, any updates made to the website folder are automatically reflected in the live application after each commit.

The site folder contains all files required for the frontend interface, including:

- HTML structure for the user interface
- JavaScript for handling user interactions and Bluetooth communication
- Service worker files to enable offline functionality and caching (PWA behavior)

The interface is designed to be simple and functional, typically including controls such as:

- Connect: find the device via Bluetooth.
- Open door: control motors to open the door.
- Close door: control motors to close the door.
- Setup: allow the user to set up the DC motor position properly.

This minimal design ensures reliable interaction with the hardware while maintaining ease of use.

The esp32 folder contains the embedded programs intended to be uploaded to the ESP32 microcontroller. These programs are responsible for handling all real-time control, sensing, and communication tasks.

Within this folder, there are additional subfolders used for:

- Testing different subsystems (e.g., motor control, BLE communication, sensors)
- Iterating on features before integrating them into the main control program

This structure allows for isolated testing and debugging before deploying a fully integrated system.

The current main firmware implements the complete control logic of the system. Its core functionalities include:

Communication

- Uses **Bluetooth Low Energy (BLE)** to communicate with the PWA
- Receives single-character commands (e.g., open, close, stop)
- Sends feedback and warning messages back to the user

Actuation

- Controls a **DC motor via an H-bridge** to open and close the door
- Uses a **servo motor** to engage or disengage the mechanical linkage

Sensing and Safety

- Limit switches detect fully open/closed positions
- A current sensor (INA219) monitors motor load
- Optional ultrasonic sensors detect nearby obstacles

Control Logic

- Implements dedicated routines for opening and closing the door
- Continuously checks for:
    - Obstructions (via current spikes or sensors)
    - Timeout conditions

- Invalid limit switch states
- Emergency stop commands

If any fault is detected, the system safely stops the motor and notifies the user. The following flowchart represents the ESP32 control logic. To explain the programming logic, the Bluefy App section is what the user sees on their device. First, they access the Web App via this link: yosefsantiago.github.io/door_opener/. The first time they access the app, it uses Wifi to cache or “save” the app to the device. If there is cached information on the device, then it uses that information to load the app. Then it displays the user interface (UI) with the commands available to send to the ESP32. Second, on the ESP32, there is a Setup section that runs once; here, the devices to be controlled are initialized. In the software, the Sensors and Limit Switches are placed as optional; there is a variable to enable the devices when uploading the code to the ESP32. Third, the Loop section is the one that runs constantly on the ESP32. If the sensors are enabled, the obstruction is checked to see whether someone is going to open the door manually; if not, the wheel can be disengaged via the app, then the ESP32 checks the Bluetooth connection, and if the device is connected, then it checks for commands. The open and close commands first make sure the motor is engaged and then turn on the motor in the appropriate direction. Second, they check if the door has reached the open or closed position; if the limit switches are not enabled, then it uses a timer to stop the door. Third, as long as the door is moving, it checks for obstructions using the enabled sensors, and if one of the sensors is tripped, then it disengages the wheel and stops the motor. For the setup wheel command allows the gear to disengage completely from the rack on the first tap and on the second tap it should re-engage appropriately, so the wheel touches the floor properly and it uses the full servo motor range. The manual command disengages the wheel, allows someone to manually operate the door and reengages by the end of a timer.

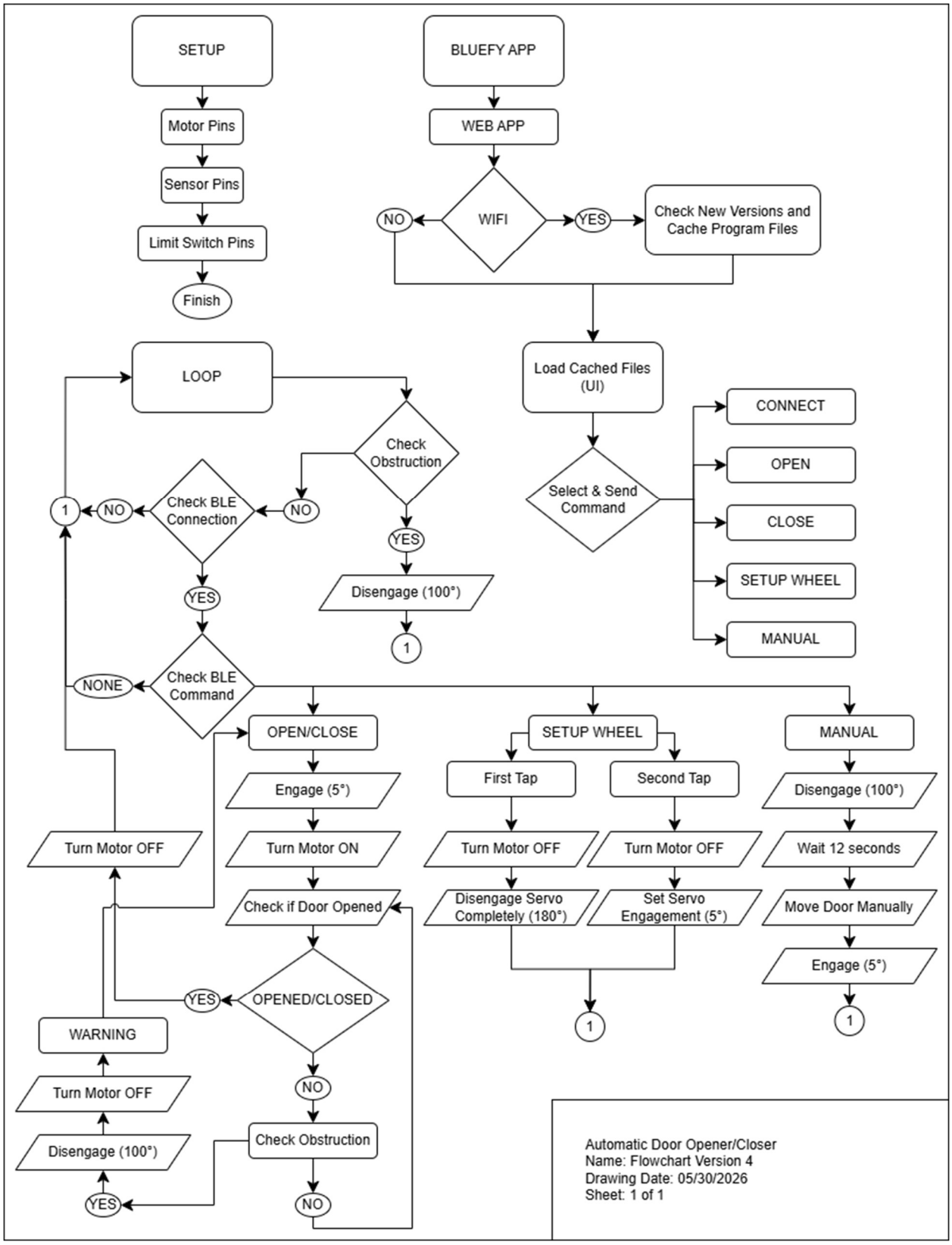


Figure 28: Flowchart

5.6 Conclusion

This section outlines all the key design details needed to build and understand the automatic door system. The Bill of Materials (B.O.M.) lists all required components, including both purchased and custom-made parts, providing a clear overview of what is needed for the system. The manufactured components section shows how custom parts were designed to support the mechanical structure and ensure proper operation.

The electrical schematic explains how all electronic components are connected, while the software section describes how the system is controlled and operated. The flowchart helps visualize how the system responds to user inputs and different conditions during operation. Additionally, concepts discussed in the Artificial Intelligence seminar were applied to support tasks such as code development, problem-solving, and documentation efficiency throughout the project.

A detailed Manufacturing and Assembly Manual was also developed to provide detailed instructions for the fabrication, assembly, and integration of the system components. This manual serves as a practical guide for reproducing the project, ensuring consistency in construction, simplifying maintenance procedures, and facilitating future modifications or improvements.

Overall, this section ensures that the design is clear, organized, and easy to replicate. It provides all the necessary information to build the system while maintaining flexibility for future improvements.

6 Conclusions and Discussion

6.1 Conclusion

The Product Design Specification (PDS) was followed closely throughout the development process, and the final system meets its requirements while successfully demonstrating its intended functionality as an interior door opener/closer. The device provides a reliable

mechanism to open and close a door, supports remote operation via Bluetooth without reliance on Wi-Fi, and incorporates basic safety features such as controlled motion and obstruction detection, while remaining compact, cost-effective, and built from readily available components. A structured design methodology was applied, beginning with 52 initial concepts that were reduced to 15 using the plus-minus method, then narrowed to 5 through a weighted rating matrix, and finally refined to 3 using a second, more detailed weighted evaluation on preliminary designs; these final concepts were developed into CAD models and brought to life through multiple prototypes, which were iteratively tested and improved to enhance performance and reliability. A complete control system was implemented, including custom-written code and fully developed circuitry integrating all electronic components, and detailed engineering calculations were carried out for energy consumption, motor selection based on torque requirements, battery capacity to ensure sufficient runtime, and servo power to guarantee effective actuation. Additionally, the disengaging mechanism was modified so that the system automatically disengages whenever the user selects the manual operation mode through the application interface. This allows the door to be operated manually when desired, providing greater user control without requiring the system to disengage after every operating cycle. This mechanism was necessary because the door could unintentionally open or close on its own due to air pressure and forces acting on it. Despite these successes, not all PDS criteria were fully achieved: the system is not water resistant due to enclosure gaps that expose internal components, and it cannot currently operate without mains power because the originally specified UPS did not arrive and had to be replaced with an alternative solution, meaning only partial backup capability is available through an external unit. Furthermore, the obstacle detection functionality was not successfully implemented, as the current-sensing module used for obstacle detection suffered a hardware failure during testing and became inoperable. In addition, the ultrasonic sensors exhibited unreliable performance and malfunctions, preventing the development of a dependable obstacle detection system. Additionally, PLA was used for 3D-printed components instead of PETG due to printer limitations, although PETG would be more suitable for future iterations because of its higher impact resistance, improved flexibility, greater durability, and superior temperature

resistance. Several improvements are recommended, including redesigning the enclosure with rubber gaskets, silicone seals, and cable glands to enhance water resistance, applying conformal coating to protect electronics, replacing the servo motor with a stepper motor for more precise and continuous motion control, and redesigning the mounting system to clamp onto the door rather than requiring permanent screws. Future developments should also include the design of a dedicated PCB with improved connectors for better electrical reliability, integration of a compact internal backup power supply to fully meet the PDS, and further refinement of materials and enclosure design to improve durability and environmental protection. Overall, the project represents a well-structured and evidence-based engineering process that delivers a functional proof of concept, with clear pathways for improving robustness, usability, and real-world applicability.

6.2 Relevance of Engineering Factors

FACTOR	RELEVANT TO THE PROJECT	NOR RELEVANT
PUBLIC HEALTH	X	
SAFETY	X	
WELFARE	X	
GLOBAL ASPECTS		X
CULTURAL ASPECTS		X
SOCIAL ASPECTS	X	
ENVIRONMENTAL ASPECTS	X	
ECONOMIC ASPECTS	X	

6.3 Discussion of Relevant Factors

The proposed automatic door system was developed with consideration of several engineering factors, including public health, safety, welfare, social, environmental, and economic aspects.

Public health and welfare were considered by reducing the need for direct contact with door handles, which can help minimize the spread of contaminants in shared environments. This is particularly beneficial in high-traffic or public spaces where hygiene is important.

Safety was a primary consideration throughout the design process. The system incorporates limit switches to prevent overtravel of the door, as well as current monitoring and optional ultrasonic sensors to detect obstructions. These features allow the system to stop operation under unsafe conditions, reducing the risk of injury or mechanical damage.

Social aspects were addressed by improving accessibility and ease of use. The system enables automated door operation, which can assist individuals with limited mobility and enhance overall user convenience, aligning with accessibility principles.

Environmental considerations were included by optimizing energy consumption and material usage. The system operates with low energy per cycle and uses a rechargeable power source, while the enclosure design minimizes material waste through efficient geometry and sizing.

Economic factors influenced the selection of cost-effective and widely available components, such as the ESP32 microcontroller and standard sensors. Additionally, the use of 3D printing for custom components reduces manufacturing costs and enables rapid prototyping and iteration.

6.4 Performance Evaluation

The performance of the system was evaluated through analytical calculations and prototype testing. Calculations were used to determine motor torque requirements, power consumption, and battery life to ensure that the selected components meet system requirements. Structural analyses, including drop and impact testing, were conducted to verify that the enclosure can withstand realistic loading conditions.

Prototype testing demonstrated that the system is capable of opening and closing the door within the desired time frame while maintaining stable operation. The integration of sensors and control logic was also validated to ensure proper response to user inputs and environmental conditions.

6.5 Learning and Knowledge Application

The development of this project required the application of knowledge from multiple engineering areas, including mechanical design, electronics, and embedded systems.

Concepts such as torque analysis, power calculations, and structural evaluation were applied to ensure proper system functionality.

Additionally, new knowledge was acquired in areas such as 3D modeling, circuit integration, and microcontroller programming. Capstone seminars contributed to this learning process by introducing concepts such as agile (Scrum-based) development, engineering innovation, and failure-based design, which were applied throughout the project. Artificial intelligence tools were also utilized to support calculations, improve design decisions, and enhance documentation efficiency.

6.6 Engineering Standards

The design and analysis of the system were guided by established engineering standards and best practices. Drop and impact analyses were based on recognized testing methodologies such as ASTM and IEC standards to ensure a realistic evaluation of structural performance. Geometric Dimensioning and Tolerancing (GD&T) principles, following ASME standards, were applied in the technical drawings to ensure proper manufacturing and assembly.

Additionally, accessibility considerations were informed by ADA guidelines to improve usability for a wide range of users. These standards contributed to ensuring that the system is safe, reliable, and aligned with professional engineering practices.

6.7 Ethical Considerations

Ethical considerations in this project include ensuring user safety, reliability, and transparency in reporting results. The system was designed with safety features to prevent hazardous conditions, and all analyses were conducted using realistic assumptions to avoid misleading conclusions.

Furthermore, the design promotes accessibility and usability, reflecting a commitment to inclusive engineering practices. Proper acknowledgment of external sources and adherence to academic integrity were also maintained throughout the project.

6.8 Intellectual Property and Legal Considerations

The project primarily uses commercially available components and open-source tools, minimizing the risk of intellectual property conflicts. Custom-designed components developed for this project are considered original work.

All external references, standards, and resources used in the development of the system have been properly cited. While no known patent conflicts were identified, further evaluation would be required if the system were to be commercialized to ensure compliance with applicable regulations and intellectual property laws.

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8 Appendix

8.1 Appendix A

The Plus-Minus method (observed in table 11) is a simple way to compare multiple ideas against a chosen reference point, or datum. In this case, the datum selected is 1, which serves as the baseline for evaluation. Each idea is compared to this datum: if an idea performs worse than the datum, it receives a score of -1; if it performs similarly, it receives a 0; and if it performs better, it receives a +1. The table presented here illustrates this method, showing a few example comparisons for clarity. For example, Idea #2 performed worse than the datum in the cost criterion, so it receives a -1. For the compact size criterion, it performed similarly, earning a 0, and for the response level criterion, it performed better than the datum, receiving a +1.

Table 11: Plus/Minus method Example

Datum #1	Criteria/Ideas	# 1	# 2	# 3	# 4	# 5	# 6	# 7	# 8	# 9	# 10
1	Cost	0	-1	1	0	1	0	0	-1	1	1
2	Compact Size	0	0	0	0	1	1	-1	-1	1	1
3	Battery	0	0	1	1	0	0	1	1	1	1
4	Safety	0	0	0	0	-1	-1	-1	-1	-1	-1
5	Lifetime	0	0	1	1	0	0	-1	-1	1	1
6	Serviceability	0	0	0	0	-1	-1	-1	-1	-1	-1
7	Time to Open	0	0	0	0	1	1	1	1	1	1
8	Response Level	0	1	0	1	0	1	0	1	0	1
9	Positional Accuracy	0	1	0	1	0	1	0	1	0	1
10	Motion Smoothness	0	0	0	0	1	1	1	1	1	1
	SUM	0	1	3	4	2	3	-1	0	4	6

Note: It is important to note that the table only represents a fraction of all the calculations carried out.

A Weighted Rating method was employed to reduce the number of concepts from 15 to 5 by systematically comparing each idea against a set of evaluation criteria. Each criterion was assigned a percentage weight according to its importance in the design process, meaning that criteria considered more important had a greater influence on the final score. The concepts were then evaluated using a scale from 1 to 10, where the scale represents how well each individual idea performs for a specific criterion. Higher scale values indicate better performance, while lower values indicate weaker performance. The scale value for each idea was multiplied by the corresponding criterion weight to obtain a weighted score.

For example, the criterion of manufacturing cost was assigned a weight of 10%. For this criterion, Idea 2 received a scale value of 10, while Idea 1 received a scale value of 8, indicating that Idea 2 performed better in terms of manufacturing cost. When multiplied by 10% weight, Idea 2 obtained a weighted score of 1, while Idea 1 received 0.8. In contrast, for the criterion of disengaging simplicity, Idea 1 performed better than Idea 2, obtaining a weighted score of 0.7 compared to 0.49 for Idea 2. However, although Idea 1 performed better in disengaging simplicity, the lower importance (weight) assigned to this criterion

could not outweigh the higher importance of manufacturing cost, which carried the larger 10% weight. By combining both the performance ratings and the importance of each criterion, the Weighted Rating method clearly identifies which concepts perform best overall and allows the strongest ideas to be selected for further development.

Table 12: Weighted Rated Method (Top 5):

Criterion	Weight (%)	# 1(28)	# 2(22)	# 3(16)	# 4(32)	# 5(48)	# 6(36)	# 7(52)	# 8(4)	# 9(15)	# 10(23)	# 11(31)	# 12(46)	# 13(50)	# 14(121)	# 15(30)
Manufacturing Cost	10.00 %	0.8	1	0.9	0.7	0.5	0.7	0.5	0.5	0.8	0.8	0.7	0.5	0.5	0.9	0.7
Maintenance Cost	5.00 %	0.4	0.4	0.4	0.25	0.25	0.35	0.25	0.25	0.4	0.4	0.25	0.25	0.25	0.4	0.25
Volume	5.00 %	0.5	0.5	0.5	0.25	0.4	0.25	0.4	0.4	0.5	0.5	0.25	0.4	0.4	0.5	0.25
Weight	5.00 %	0.5	0.5	0.5	0.25	0.4	0.25	0.4	0.4	0.5	0.5	0.25	0.4	0.4	0.5	0.25
Battery Life	5.00 %	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.25	0.5	0.25	0.25	0.25	0.25
Rechargeability	15.00 %	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	1.5	0.75	1.5	0.75	0.75	0.75	0.75
IP Rating	5.00 %	0.35	0.35	0.35	0.35	0.4	0.4	0.4	0.35	0.35	0.35	0.35	0.4	0.4	0.35	0.35
Ergonomic/Safe Design	3.00 %	0.21	0.15	0.15	0.21	0.24	0.24	0.24	0.24	0.15	0.21	0.21	0.24	0.24	0.15	0.21
Lifetime	8.00 %	0.56	0.56	0.56	0.56	0.48	0.56	0.48	0.56	0.56	0.56	0.56	0.48	0.48	0.56	0.56
Serviceability	15.00 %	1.05	1.05	1.05	1.05	0.75	1.2	0.75	1.2	0.75	1.05	0.75	0.75	0.75	0.75	0.75
Time to Open	7.00 %	0.7	0.7	0.7	0.56	0.49	0.56	0.49	0.56	0.7	0.7	0.56	0.49	0.49	0.7	0.56
Response Level	7.00 %	0.7	0.7	0.7	0.7	0.21	0.7	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.7	0.7
Positional Accuracy	3.00 %	0.24	0.24	0.24	0.21	0.03	0.21	0.03	0.21	0.09	0.24	0.21	0.03	0.03	0.24	0.21
Disengaging Simplicity	7.00 %	0.7	0.49	0.49	0.49	0.21	0.49	0.35	0.35	0.49	0.7	0.49	0.21	0.35	0.49	0.49
SUM	100%	8.71	8.64	8.54	7.58	6.36	7.91	6.5	7.23	7.5	7.22	6.79	5.36	5.5	7.24	6.28

8.2 Appendix B: Geometric Dimensioning and Tolerancing (GD and T) of Custom Designed Components (In-House Fabrication)

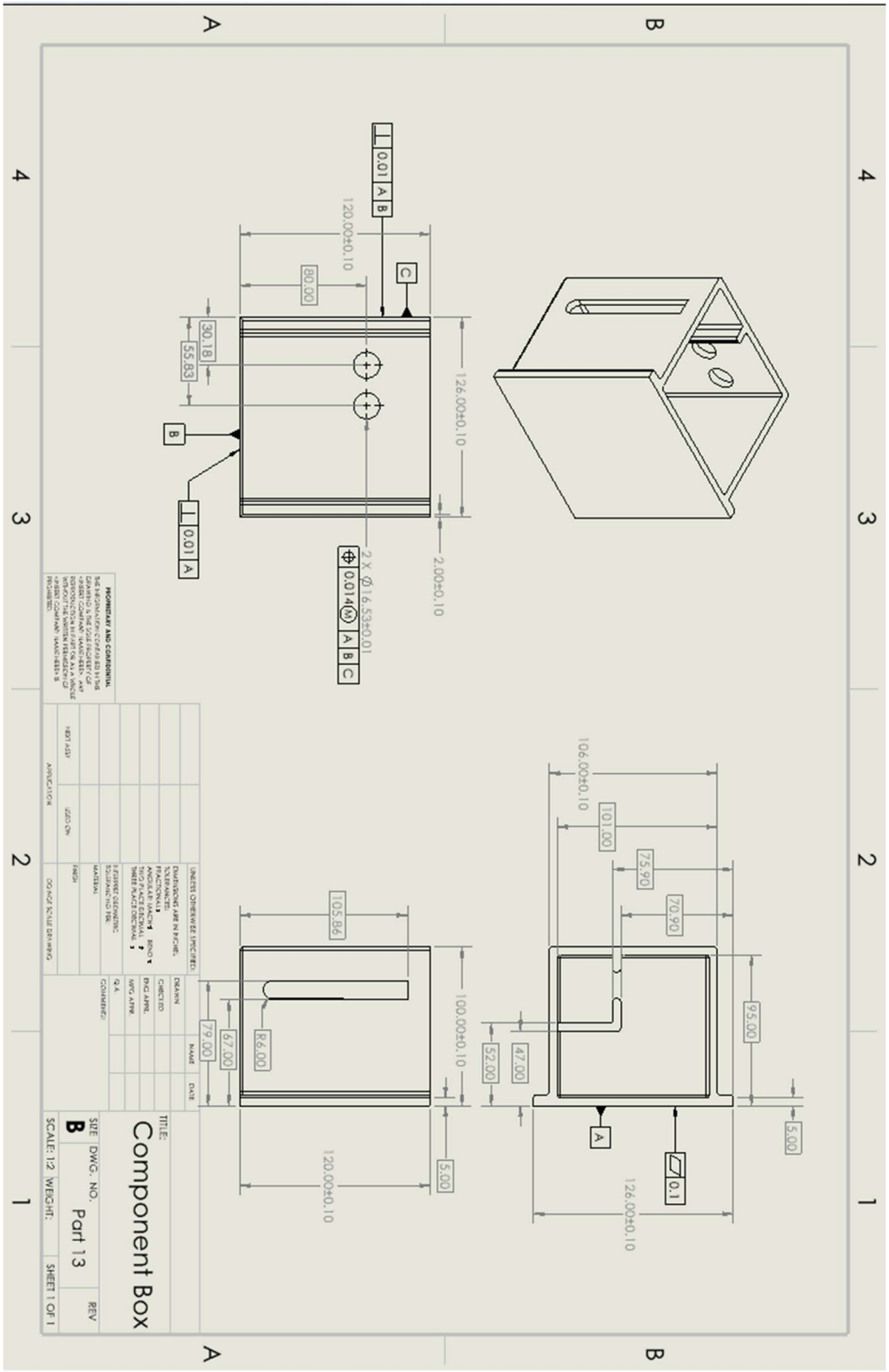


Figure 29: Component Box (Technical Drawing)

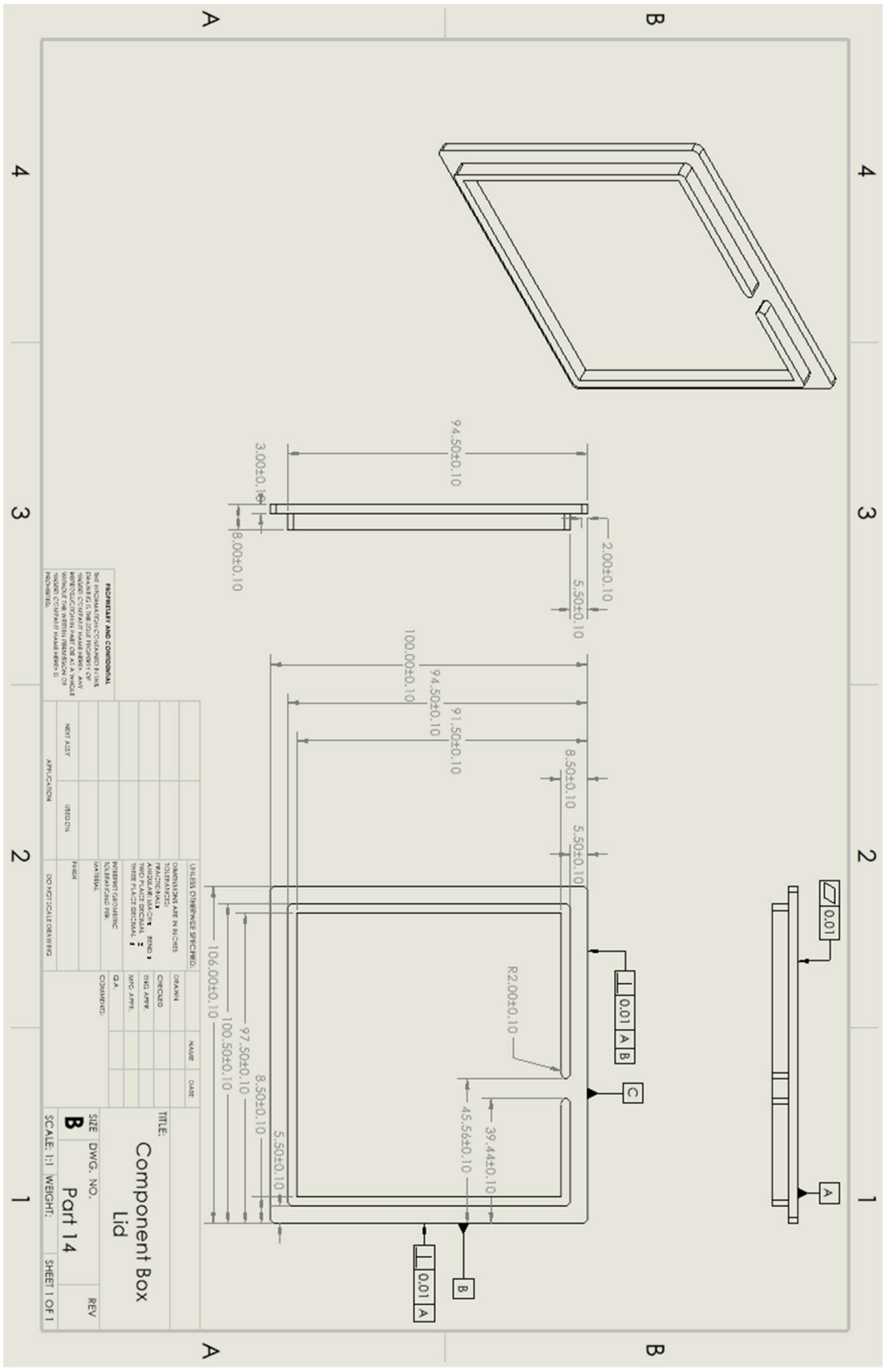


Figure 30: Component Box Lid (Technical Drawing)

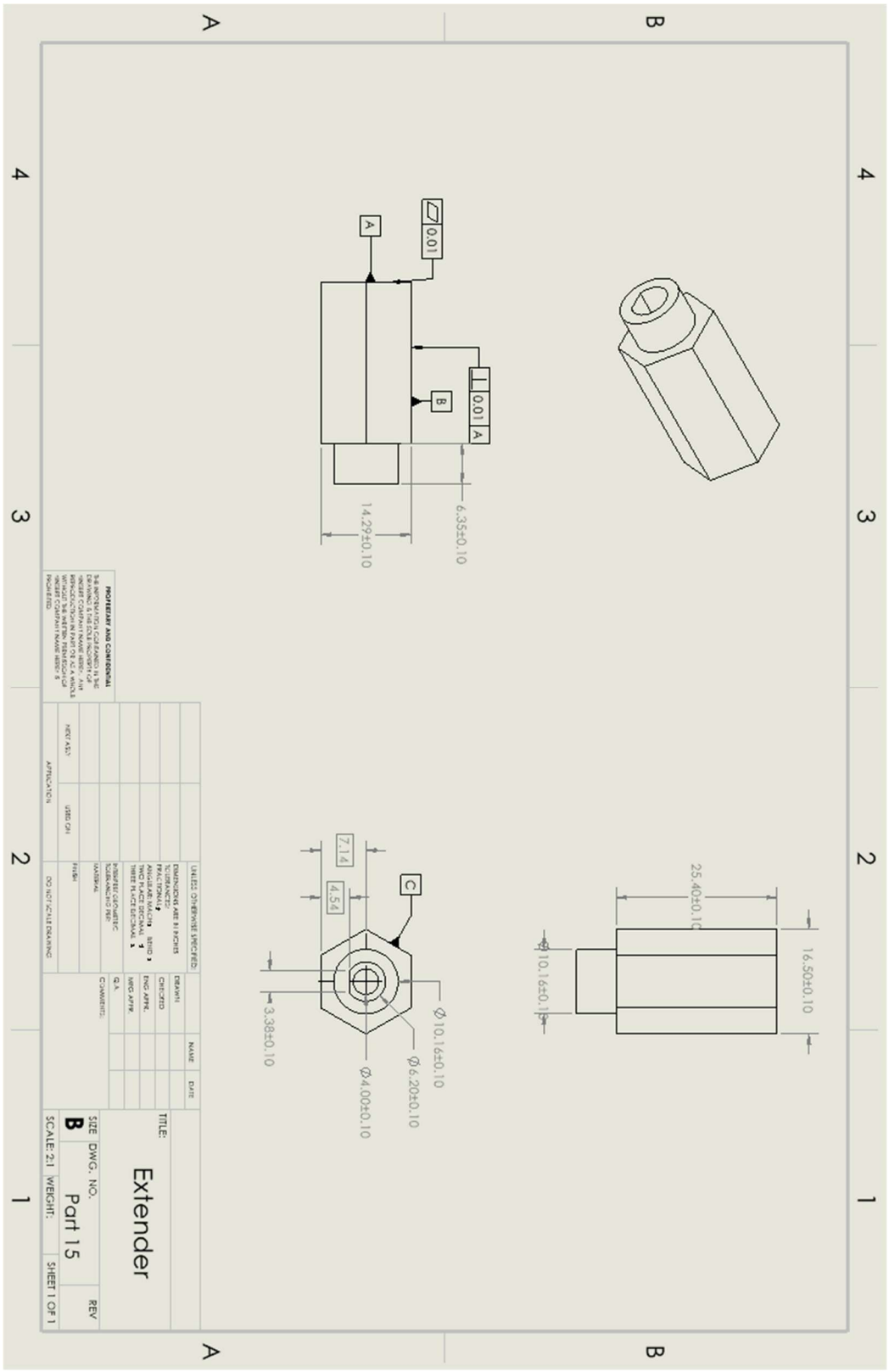


Figure 31: Extender (Technical Drawing)

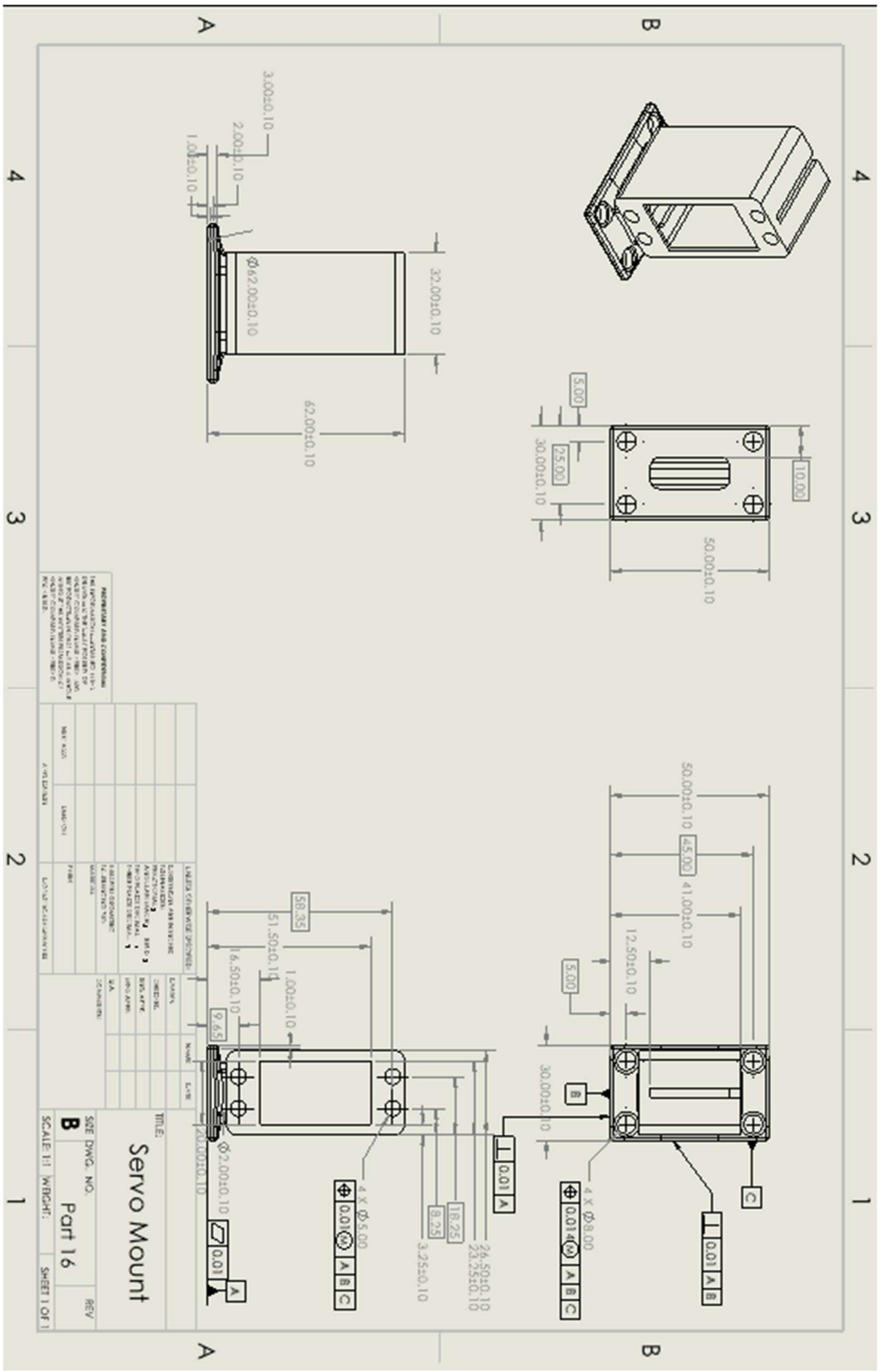


Figure 32: Servo Motor Mount (Technical Drawing)

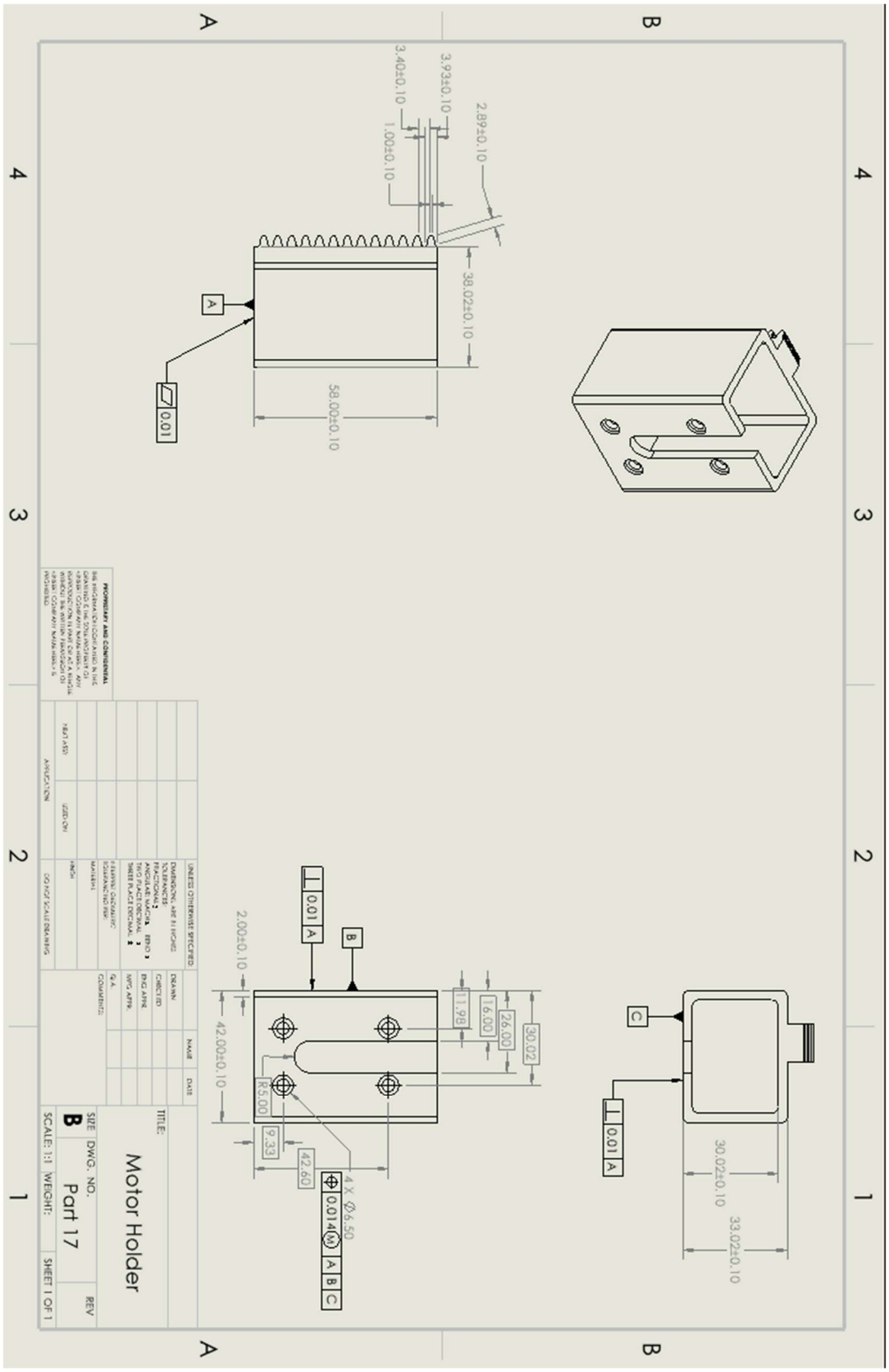


Figure 33: Motor Holder (Technical Drawing)

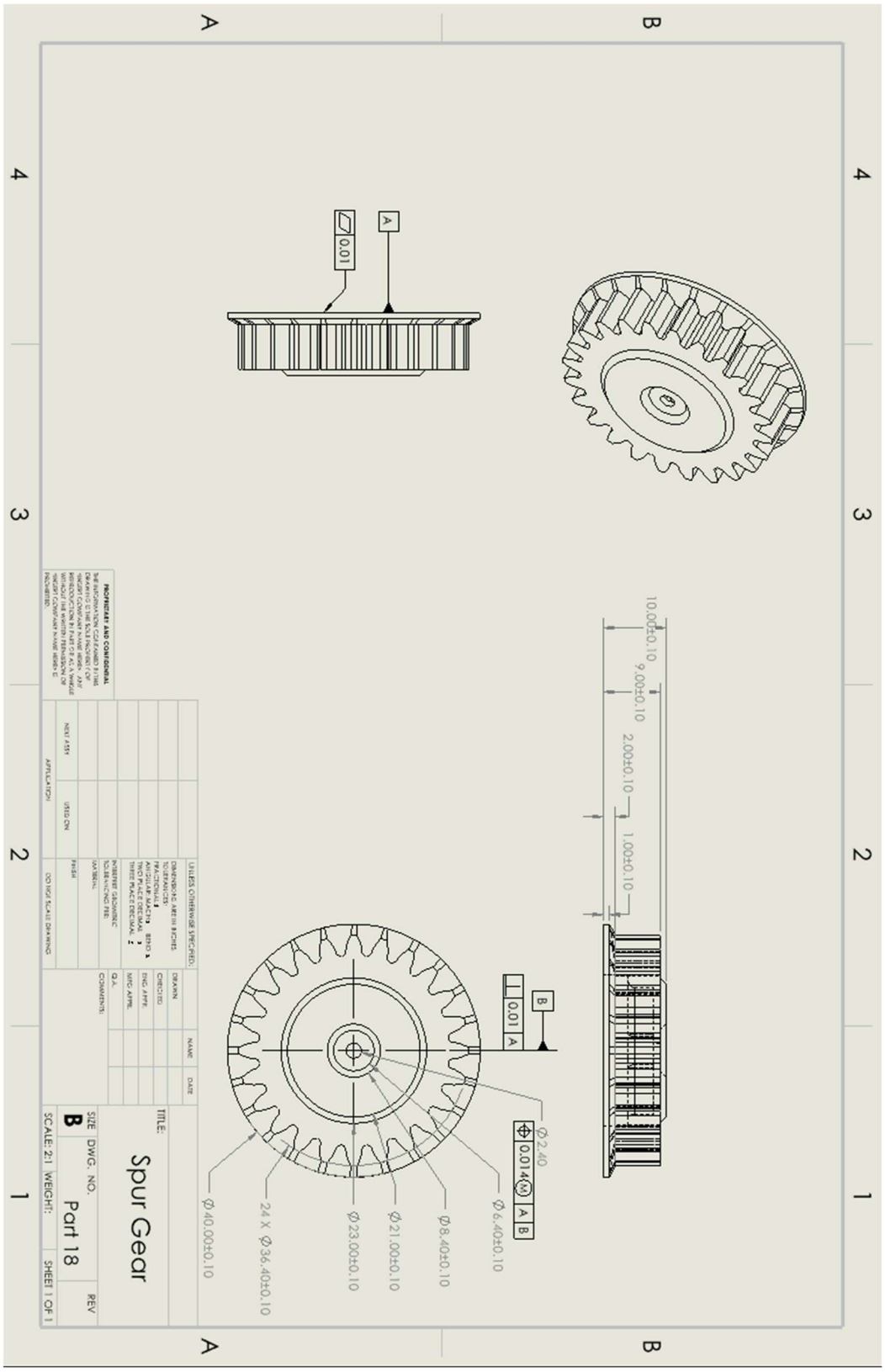


Figure 34: Spur Gear (Technical Drawing)

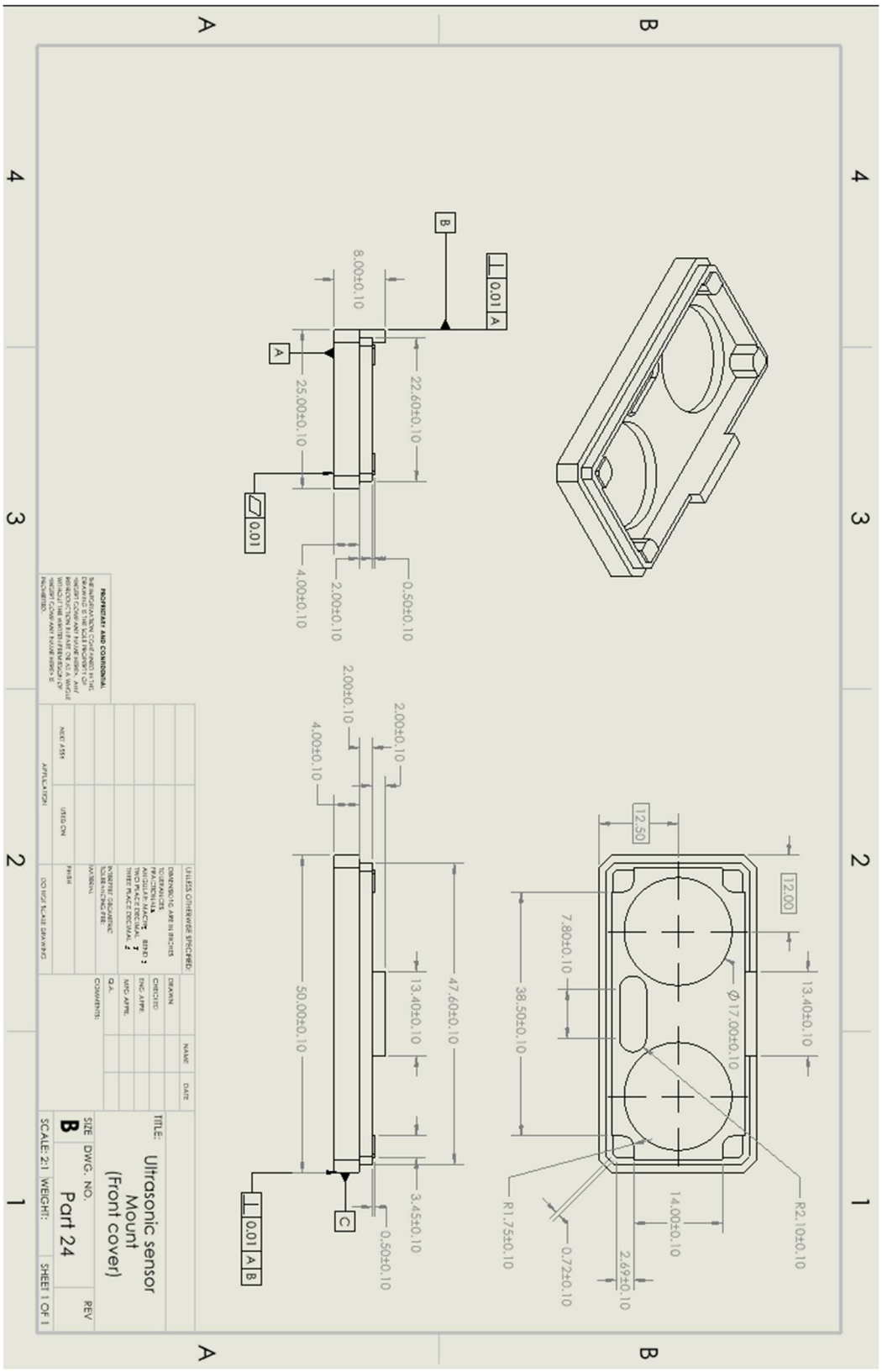


Figure 35: Ultrasonic Sensor Mount (Front Cover)

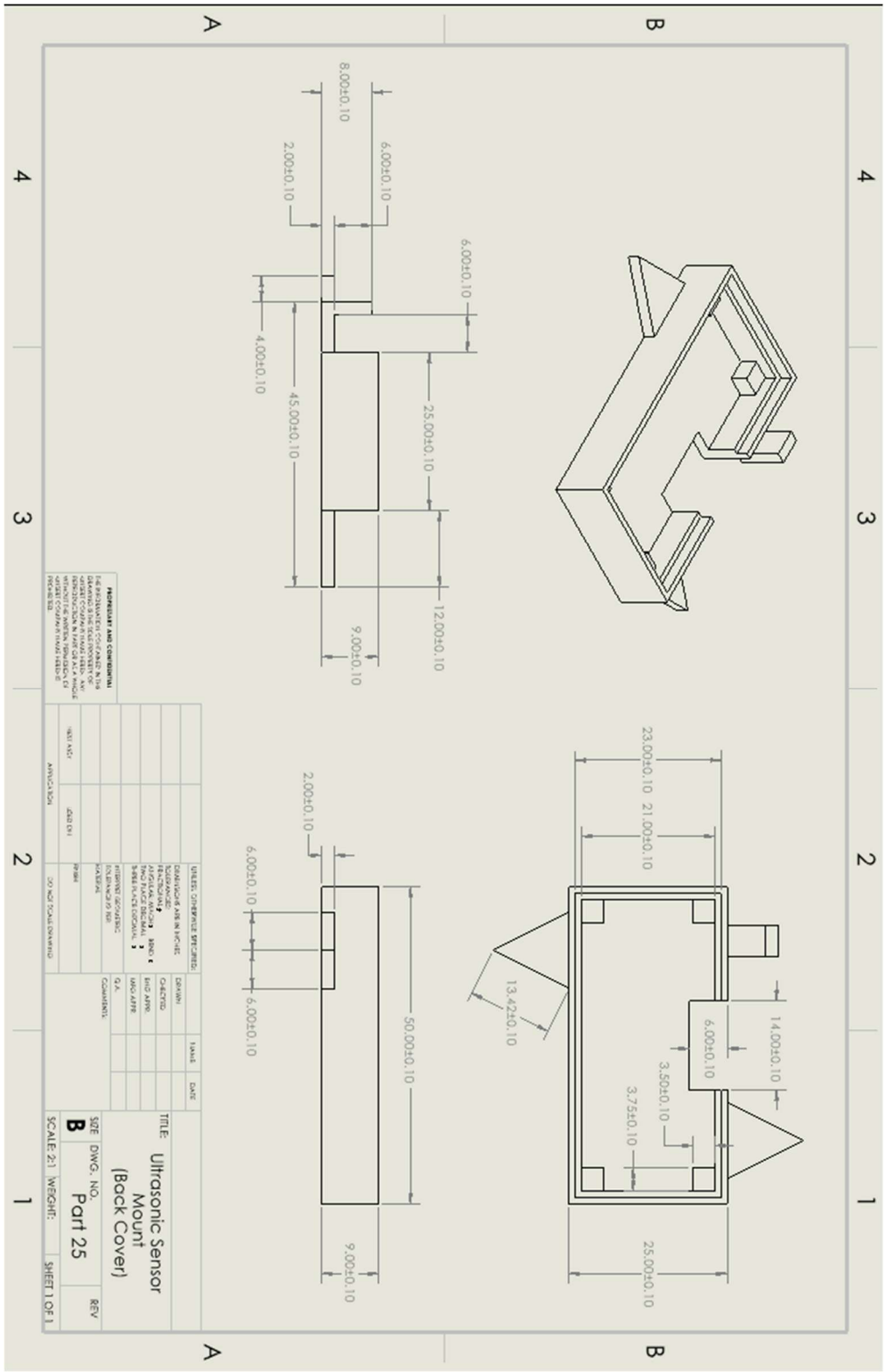


Figure 36: Ultrasonic Sensor Mount (Back Cover)

8.3 Appendix C: Manufacturing and Assembly Manual

8.3.1 Overview



Figure 1: ADAS System

This Manufacturing and Assembly Manual provides comprehensive instructions for the fabrication, assembly, and verification of a low-cost automatic door opener/closer designed for individuals with severe mobility impairments, including quadriplegic patients. The system is intended to enable independent operation of a standard interior hinged door through a compact, reliable, and user-friendly mechanism.

The device integrates mechanical, electrical, and control subsystems to achieve consistent door actuation while maintaining safety, durability, and ease of installation in residential environments. Emphasis has been placed on accessibility, ensuring that the system can be operated remotely without requiring physical interaction with the door.

This manual outlines all required components, materials, tools, and manufacturing processes necessary to build the system. It includes a detailed Bill of Materials (BOM), part specifications, engineering drawings with dimensions and tolerances following ASME standards, and exploded assembly views to guide the integration of components. Step-by-step assembly procedures are provided to ensure correct construction, along with quality control checks to verify proper functionality.

Additionally, the manual documents all custom-designed parts and fabrication files, including STL files for additive manufacturing where applicable. The goal is to ensure that the system can be reliably reproduced, assembled, and validated by engineers or technicians while maintaining consistency with the design requirements defined in the Product Design Specification (PDS).

8.3.2 B.O.M.

Part Num.	Part Name and Number	Description	Qty	Price	Supplier	Notes	Link
1	BRINGSMART DC Motor	12V Worm Gear DC Motor (30 RPM, 6 kg·cm) – High torque, low speed, self-locking motor for door actuation	1	$ 15.87	Amazon		DC motor
2	MG 996R Servo	MG996R Servo Motor – High-torque (≈11 kg·cm), metal gear, PWM-controlled positioning actuator	1	$ 4.25	Amazon	Sold in packs ($16.99 for 4)	Servomotor
3	ESP32	ESP32 Dev Board – Dual-core 240 MHz microcontroller with Wi-Fi/Bluetooth for system control	1	$ 8.98	Amazon	Sold in packs ($17.95 for 2)	Esp32
4	Ultrasonic Sensor	HC-SR04 Ultrasonic Sensor – Non-contact distance measurement (2–500 cm) using 40 kHz sound waves	2	$ 2.00	Amazon	Sold in packs ($9.99 for 5)	URsensor
5	H Bridge	Dual H-Bridge Motor Driver (3–14V, 5A) – Bidirectional DC motor control with	1	$ 4.94	Amazon	Sold in packs ($9.88 for 2)	HBridge

		PWM speed regulation					
6	Wheel	65mm rubber wheel	1	$ 3.24	Control Hobby		ControlHobby
7	Perfboard	ELEGOO 32 Pcs Double sided PCB Board Prototype Kit	1	$ 9.99	Amazon		PerfBoard
8	M3 Screws (35mm)	M3-0.5x35mm Zinc Pan Head Phillips Drive Machine Screw	1	$ 1.95	Home Depot	Were adjusted to size as needed	M3 Screws (35 mm)
9	M3 Screws (6mm)	M3-0.5 x 6 mm Zinc Pan Head Phillips Drive Machine Screw	5	$ 1.95	Home Depot	Sold in packs	M3 Screws (6mm)
10	M3 Washer	M3 Zinc-Plated Steel Flat Washers	2	$ 1.06	Home Depot	Sold in packs ($2.67 for 5)	M3 Washer
11	Screws (6-32 X 1/2)	Round Machine Screws (6-32 X 1/2)	4	$0.32	National	Sold Individually	National
12	Hex Nuts (6-32 ZP)	Hex Nuts (6-32 ZP)	4	$0.24	National	Sold Individually	National
13	Component Box (Part 13)	Structural enclosure designed to contain and protect electronic components, providing mounting features and cable management for the system.	1	$ 4.53	In-House Fabrication	Printed in PLA for prototype purposes. Final product should be printed in PETG or similar/stronger material	
14	Lid (Part 14)	Removable cover designed to secure and protect internal components while allowing	1	$ 0.83	In-House Fabrication	Printed in PLA for prototype purposes. Final product should be printed in PETG or	

		access for maintenance and assembly.				similar/stronger material	
15	Extender (Part 15)	Custom extension used to secure the wheel to the motor and transfer torque effectively.	1	$0.05	In-House Fabrication	Printed in PLA for prototype purposes. Final product should be printed in PETG or similar/stronger material	
16	Servo Holder (Part 16)	Custom bracket designed to securely mount the servo motor, ensuring proper alignment and stable operation during actuation.	1	$ 0.70	In-House Fabrication	Printed in PLA for prototype purposes. Final product should be printed in PETG or similar/stronger material	
17	Motor Holder (Part 17)	Support structure designed to hold and align the DC worm gear motor, providing stability and proper positioning for torque transmission.	1	$ 1.90	In-House Fabrication	Printed in PLA for prototype purposes. Final product should be printed in PETG or similar/stronger material	
18	Gear (Part 18)	Custom-designed gear used to transfer rotational motion from the motor to the door mechanism,	1	$ 0.13	In-House Fabrication	Printed in PLA for prototype purposes. Final product should be printed in PETG or similar/stronger material	

		enabling controlled movement and torque transmission.					
19	Power Supply	12VDC Switching Adaptor 2A	1	$3.50	Alpha3D Lab		Alpha3D
20	Female Jack	Female jack for power supply	1	$0.50	Alpha3D Lab		Alpha3D
21	UPS	Talentcell Rechargeable 12 V 6000mAh/5V	1	$32.99	Amazon		UPS
22	Jumper Wires	ELEGOO 120 pcs Multicolored Dupont Wire (Male to male, male to female, female to female)	1	$6.98	Amazon	Multiple Wires are included	JumperWires
23	Screws (6 X 1-1/4)	Flat Wood screws 6 X 1-1/4	4	$0.40	National	Sold Individually	National

8.3.3 Tools for Manufacturing and Assembly

8.3.3.1 Manufacturing Tools

- FDM 3D Printer
- Filament Material
- Slicing Software
- Caliper

8.3.3.2 Post Processing Tools

- Pliers (for support removal)
- Sandpaper (various grits for surface finish)
- Deburring tool or precision knife

8.3.3.3 Assembly Tools

- Philips Screwdriver (for M3 screws)
- Wire strippers

- Wire cutter
- Soldering iron and solder
- Heat shrink tubing or electrical tape
- Multimeter
- Drill (for mounting screws)
- Pliers
- Double Sided Mounting Tape

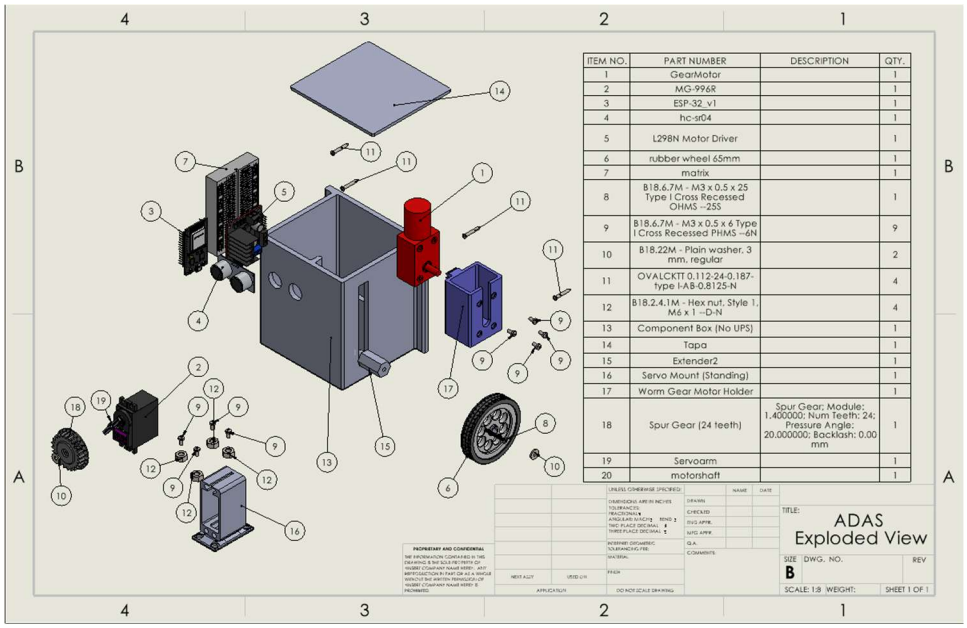
8.3.4 Manufacturing Information

The components of the automatic door opener system were manufactured primarily using fused deposition modeling (FDM) 3D printing, allowing for rapid prototyping, design iteration, and cost-effective production of custom parts. All prototype components were fabricated using polylactic acid (PLA) filament due to its ease of printing, dimensional stability, and suitability for initial testing and validation. For final production, it is recommended to use polyethylene terephthalate glycol (PETG) or a similar higher-strength thermoplastic material to improve mechanical durability, impact resistance, and long-term performance under load. Printing parameters such as layer height, infill density, and print orientation were selected to balance strength and material usage, with load-bearing components oriented to maximize structural integrity. Post-processing steps, including support removal, light sanding, and dimensional verification, were performed to ensure proper fit and function. All parts were designed with standard tolerances compatible with FDM printing, and assembly features such as holes, slots, and mounting interfaces were incorporated to facilitate reliable and repeatable assembly.

8.3.5 Quality Control and Tolerances

Quality control procedures were implemented to ensure that all manufactured components met the required dimensional and functional specifications. Critical features, such as mounting holes, shaft interfaces, and alignment surfaces, were designed and verified with a tolerance of ±0.2 mm to ensure proper fit and reliable assembly with mechanical and electrical components. These dimensions were measured and inspected after printing to confirm compliance. For all non-critical features, standard FDM 3D printing tolerances were applied, accounting for typical variations in layer adhesion, thermal expansion, and printer accuracy. Visual inspection was also conducted to identify defects such as warping, stringing, or incomplete layers that could affect performance. Any parts that did not meet dimensional or structural requirements were reprinted. This approach ensures a balance between manufacturing efficiency and the precision necessary for consistent system operation.

8.3.6 Components and Part Numbers



ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	GearMotor		1
2	MG-996R		1
3	ESP-32_v1		1
4	hc-sr04		1
5	L298N Motor Driver		1
6	rubber wheel 65mm		1
7	matrix		1
8	B18.6.7M - M3 x 0.5 x 25 Type I Cross Recessed OHMS --25S		1
9	B18.6.7M - M3 x 0.5 x 6 Type I Cross Recessed PHMS --6N		9
10	B18.22M - Plain washer, 3 mm, regular		2
11	OVALCKTT.0.112-24-0.187-type I-AB-0.8125-N		4
12	B18.2.4.1M - Hex nut, Style 1, M6 x 1 --D-N		4
13	Component Box (No UPS)		1
14	Tapa		1
15	Extender2		1
16	Servo Mount (Standing)		1
17	Worm Gear Motor Holder		1
18	Spur Gear (24 teeth)	Spur Gear; Module: 1.400000; Num Teeth: 24; Pressure Angle: 20.000000; Backlash: 0.00 mm	1
19	Servoarm		1
20	motorshaft		1

ADAS Exploded View

Figure 2: Exploded View

8.3.7 Electrical Schematic

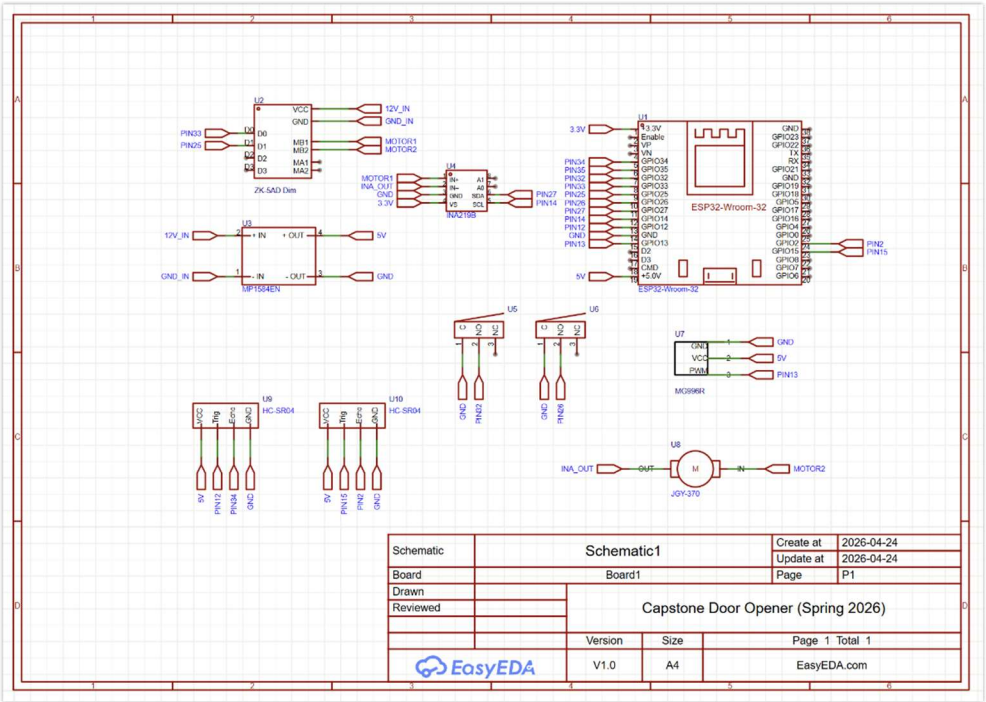


Figure 3: Electrical Schematic

8.3.8 Manufacturing/Assembly Instructions

1. **Fabricate all 3D-printed components** using the specified material and printing parameters. Remove supports and clean all parts as required to ensure proper fit and finish.

   List:

   a) Component Box
      - Part Number: 13
      - STL File Name: Component Box (No UPS)

   b) Component Box Lid
      - Part Number: 14
      - STL File Name: Tapa

   c) Extender
      - Part Number: 15
      - STL File Name: Extender2

   d) Servomotor Mount
      - Part Number: 16
      - STL File Name: Servo Mount (Standing)

   e) Motor Holder
      - Part Number: 17
      - STL File Name: Worm Gear Motor Holder2

   f) Spur Gear
      - Part Number: 18
      - STL File Name: Spur Gear (24 Teeth)

2. **Prepare the DC motor (Part #1)** by soldering two wires (Part #22) to its terminals, as shown in Figure 1. Use a red wire for the positive terminal and a black (or brown/green) wire for the negative terminal. Connect the opposite ends of the wires to the H-bridge motor driver (Part #5).

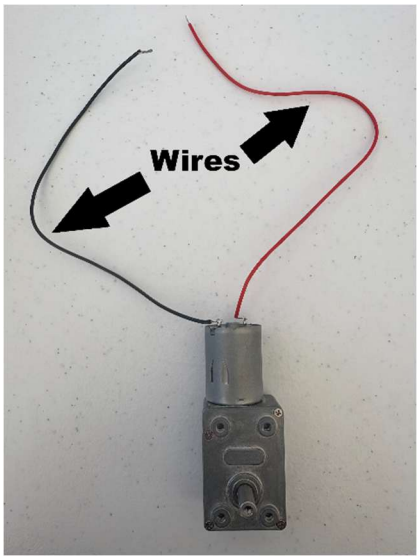


Figure 4: Prepare DC Motor

3. **Connect the power supply (Part #19)** by attaching a 12V DC source to the female power jack(Part #20). Then connect the positive and negative outputs from the jack to the corresponding terminals on the H-bridge (Part #5), as shown in Figure 2.

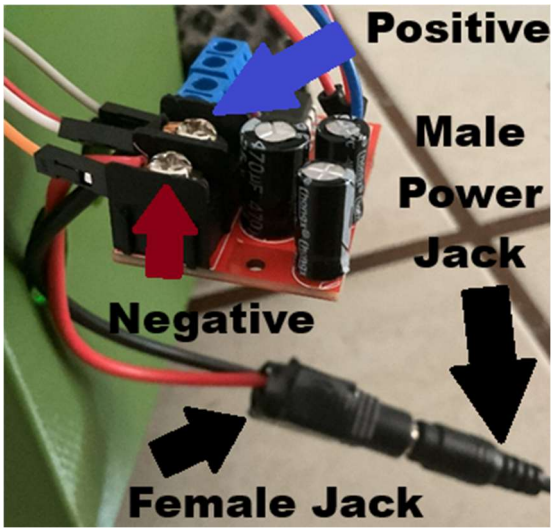


Figure 5: Connect the Power Supply

4. **Install the DC motor (Part #1)** into the motor holder (Part #17) and secure it using M3 × 6 mm screws (Part #9) with a Phillips screwdriver. Place the assembled motor holder into its designated position inside the component box (Part #13).

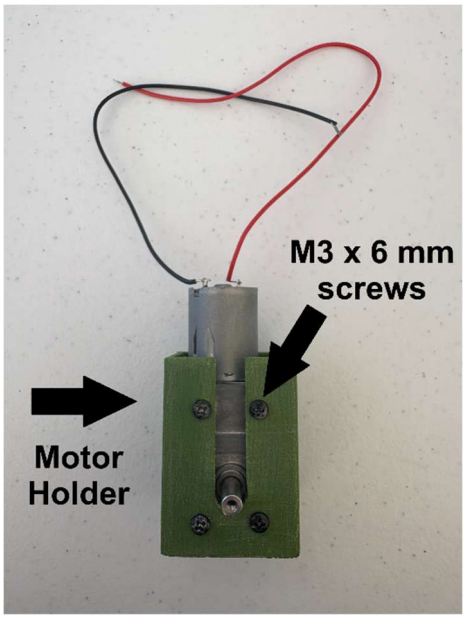


Figure 6: DC motor inside Motor Holder

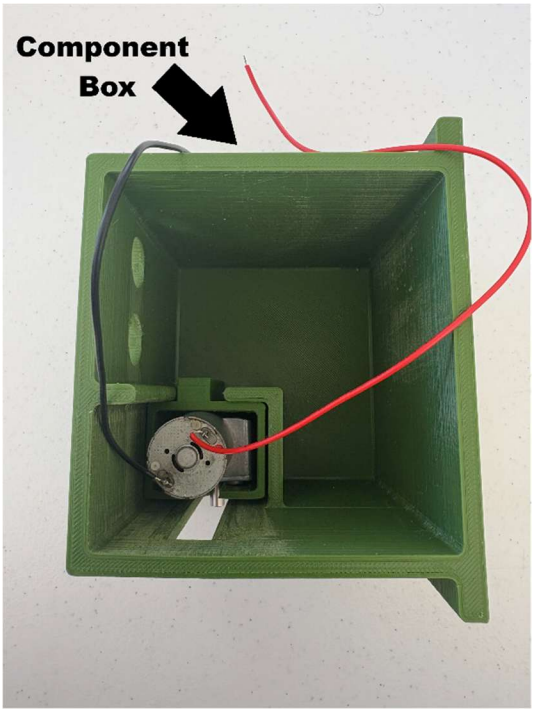


Figure 7: Insert Motor Holder in Component box

5. **Attach the wheel assembly** by connecting the wheel to the motor shaft using the extender, an M3 × 35 mm screw (Part #8), and a M3 washer (Part #10). Tighten securely using a Phillips screwdriver.

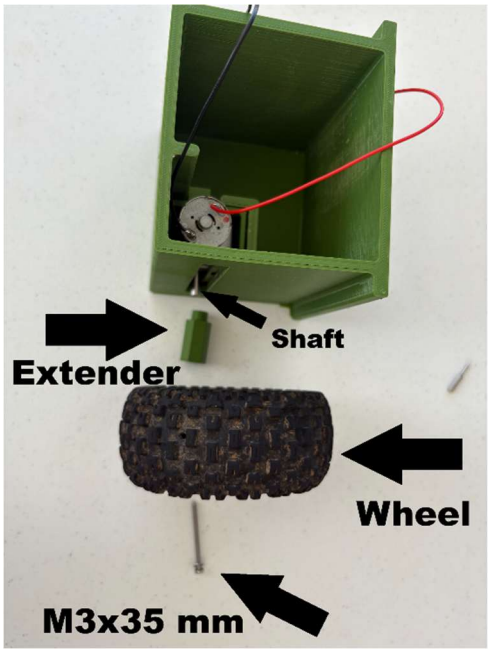


Figure 8: Wheel Assembly (1)

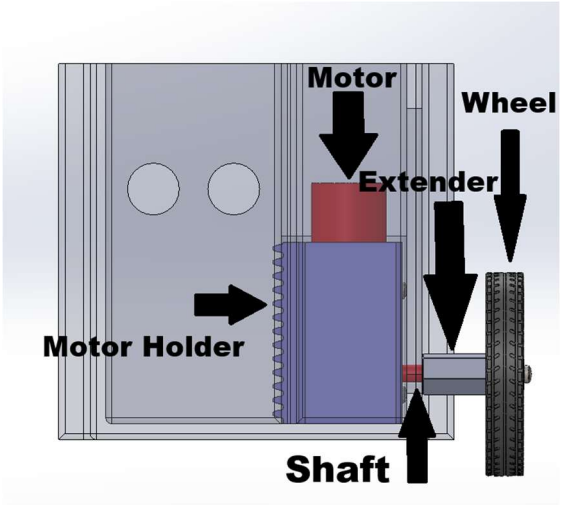


Figure 9: Wheel Assembly (2)

6. **Install the servomotor** by placing the MG996R servo (Part #2) into the servo holder and securing it in place. Attach the gear to the servo shaft using an M3 × 6 mm (Part #9) screw and M3 washer (Part #10).

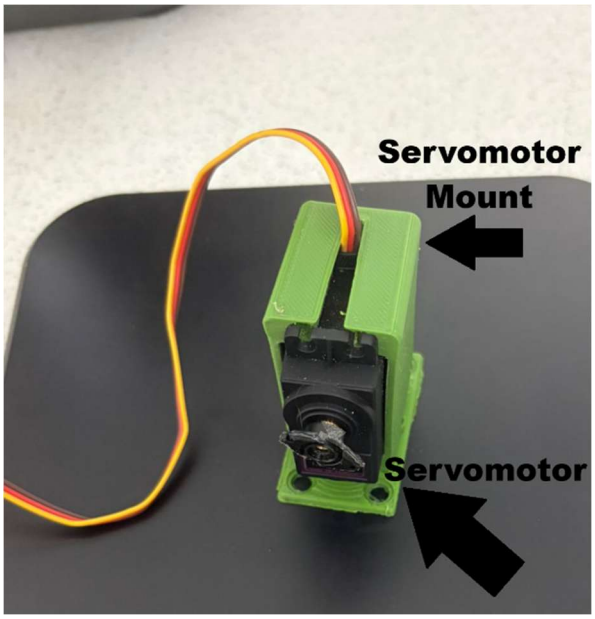


Figure 10: Servomotor inside of Servomotor Mount

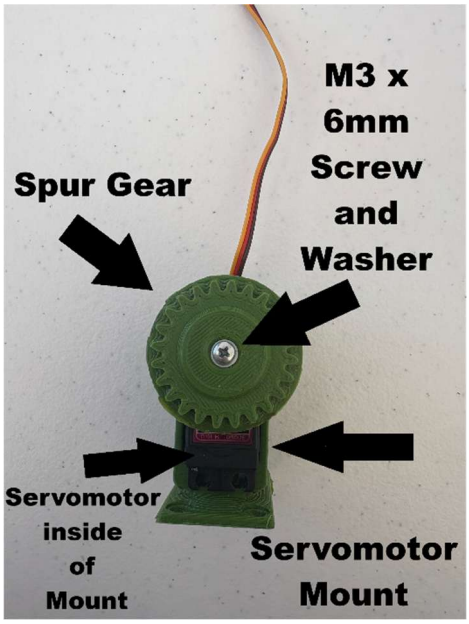


Figure 11: Servomotor Installed

7. **Align the rack and pinion mechanism** by ensuring proper meshing between the gear (Part #18) and the motor holder's (Part #17) rack. Secure the servo holder (Part #16) inside the component box (Part #13) using mounting screws (Part #11) and nuts (Part #12) or high-strength adhesive tape.

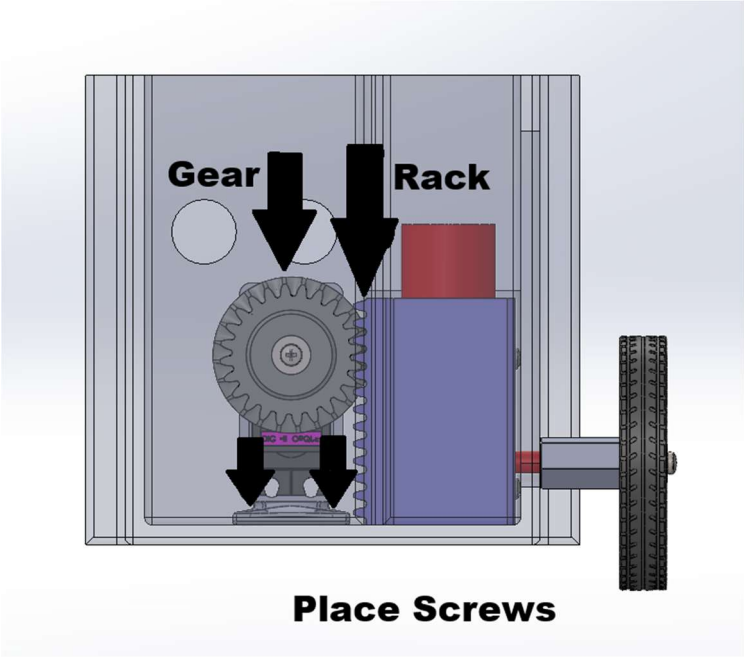


Figure 12: Align the Rack and Pinion

8. **Complete the electrical connections** by wiring all components to the breadboard or Perfboard (Part #7) according to the electrical schematic (Figure 3). Extend wires (Part #22) if necessary and ensure all connections are secure and properly insulated.

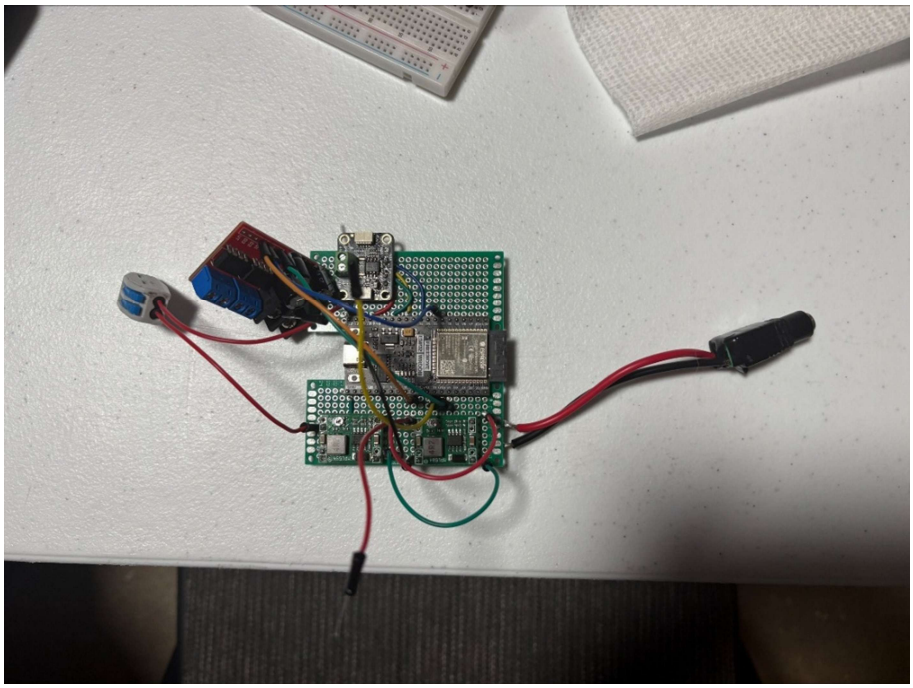


Figure 13: Electrical Connections

9. **Secure the component box to the door** using four wood screws (Part #23). Adjust the vertical position of the box to ensure proper contact between the drive wheel (Part #6) and the floor for effective operation.



Figure 14: Installed ADAS System

8.4 Appendix D: User's Manual

8.4.1 Introduction



Figure 1: ADAS System

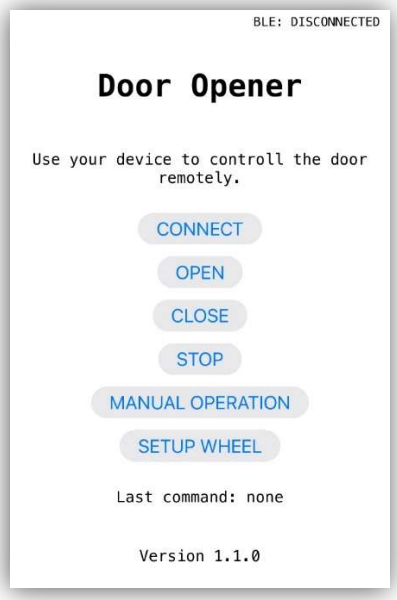


Figure 2: ADAS Interface

This product is an **automatic door opening and closing system** designed to assist individuals with limited mobility, including quadriplegic patients.

The system enables remote or automatic operation of an interior door, reducing the need for physical effort and increasing user independence in residential environments.

8.4.1.1 System Features

- Automatic door opening and closing system
- High-torque motor for reliable operation
- Electronic control using a microcontroller (ESP32)
- Ultrasonic sensor for presence detection
- Limit switches to control opening and closing limits
- Compact design adaptable to standard doors
- Quiet and safe operation

8.4.2 Warnings ⚠

8.4.2.1 Electrical Safety

- Use only the specified 12V DC power supply.
- Do not operate the system with damaged wires or loose connections.
- Disconnect power before performing any maintenance or adjustments.
- When changing the direction of rotation of the motor, please be sure to wait for the motor to completely stop, otherwise the driver will be damaged due to excessive current.
- Reverse connection protection is not supported.

8.4.2.2 Mechanical Safety

- Keep hands, fingers, and loose clothing away from **moving parts** (wheel, gear, and motor).
- Do not place objects in the path of the door while the system is operating.
- Ensure all components are properly secured before use.

8.4.2.3 Installation Safety

- Verify that the device is **firmly mounted** to the door before operation.
- Ensure the wheel maintains proper contact with the floor to prevent slipping or unexpected movement.

- Confirm that cables are routed safely and do not interfere with door movement.

8.4.2.4 Operational Safety

- Do not use the system if unusual noise, vibration, or malfunction is detected.
- Always maintain a clear path for the door during operation.
- Supervise operation when used around children or pets.

8.4.2.5 Environmental Conditions

- This system is intended for **indoor use only**.
- Do not expose the system to water, moisture, or extreme temperatures.

8.4.2.6 Modification Warning

- Do not modify the electrical or mechanical components unless properly qualified.
- Unauthorized modifications may result in malfunction or injury.

8.4.2.7 Operation without Sensors

- Make sure there are no obstructions while operating the device
- Stop the door via the app when it reaches fully opened or closed positions (there are timers set as a redundance as well)
- Press the MANUAL OPERATION button to disengage the wheel for 10 seconds and move the door freely

Failure to follow these instructions may result in damage to the device or personal injury.

8.4.3 Installation Instructions

1. Position the system (component box) at the bottom of the door. Making sure it is at the opposite side of the door hinges.



Figure 3: Positioned Box

Note: Measure and adjust the height of the component box to ensure the wheel will have appropriate contact with the floor.

2. Secure device to door using 4 wood screws.



Figure 4: Add screws at side sections of Component Box

3. Connect the 12V power supply to the system.

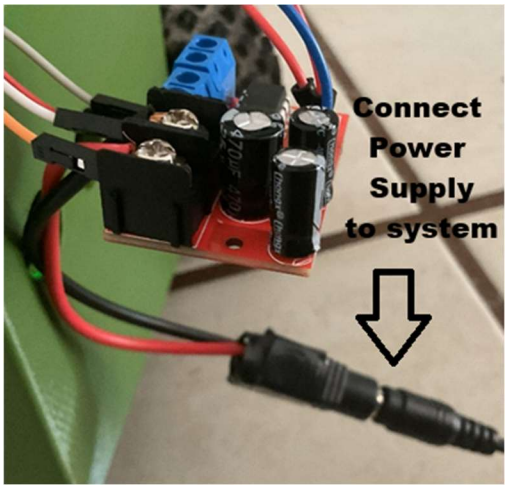


Figure 5: Connect Power Supply to System

4. Connect the 12V power supply to the outlet.



Figure 6: Connect Power Supply to Wall Outlet

8.4.4 Operating Instructions

8.4.4.1 1. Install the Mobile Application

Download and install the **Bluefy – Web BLE Browser** application from the App Store or Google Play Store.

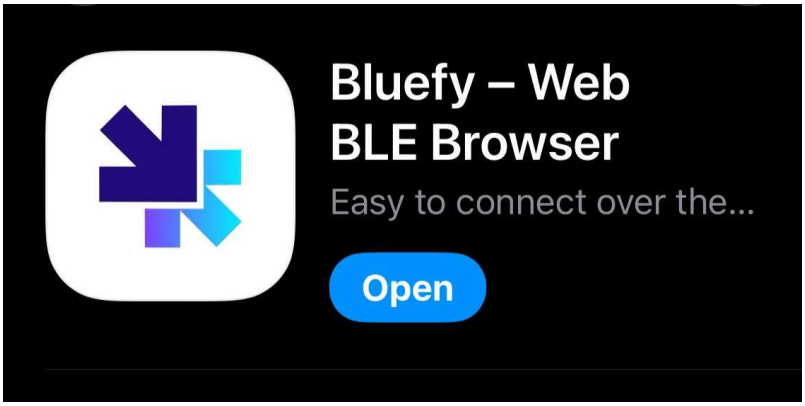


Figure 7: Bluefy-Web BLE Browser

8.4.4.2 2. Access the Control Interface

Once the application is installed, open it and enter the following link in the search bar:

https://yosefsantiago.github.io/door_opener/

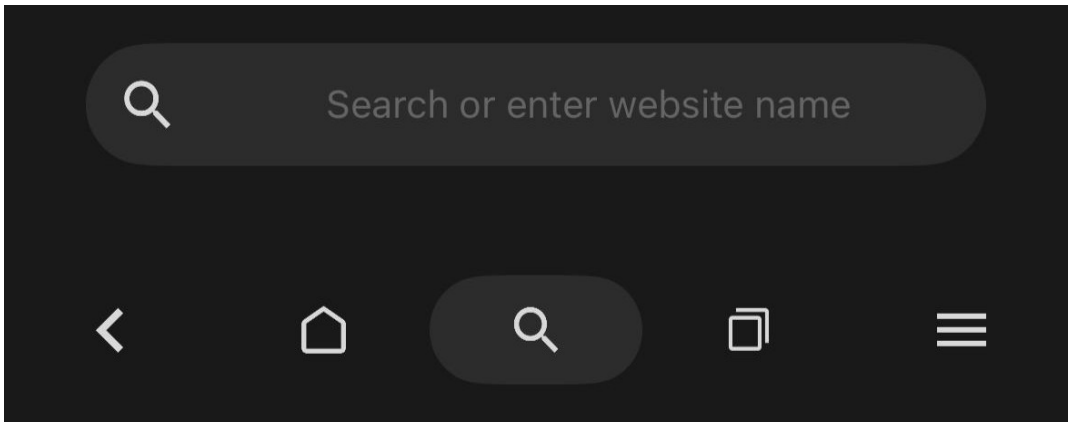


Figure 8: Search Bar

8.4.4.3 3. Load the Interface

After accessing the link, the control interface shown below should appear on your device.

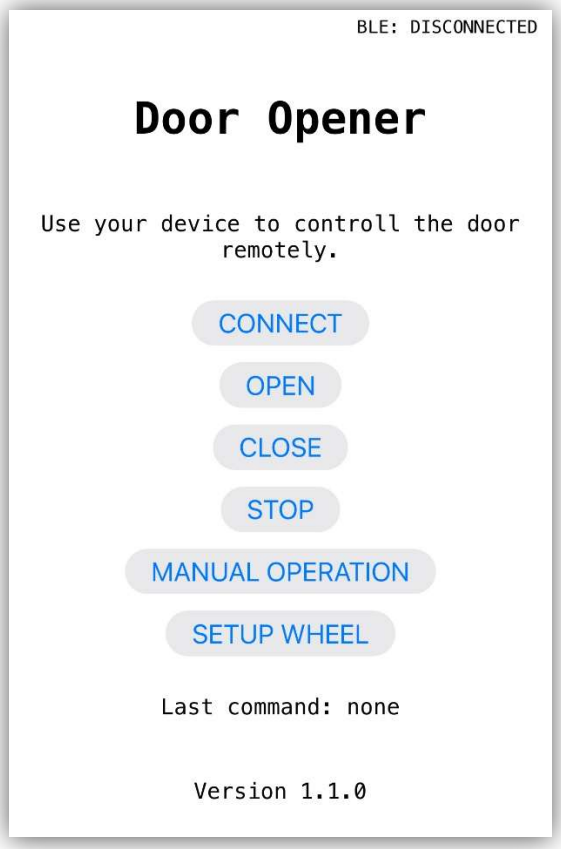


Figure 9: User Interface

8.4.4.4 4. Connect to the Device

Ensure that the ESP32 is powered on (a flashing indicator light should be visible).

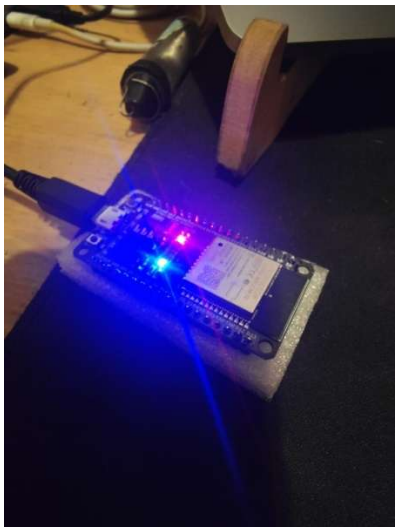


Figure 10: ESP32 is on

Press the **“CONNECT”** button on the interface. A device selection menu will appear—select **“ESP32”** from the list.

Note: Ensure *Bluetooth* is enabled on your device before attempting to connect.

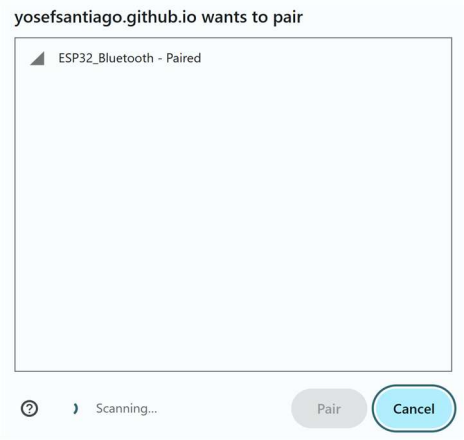


Figure 11: ESP32 Paired

8.4.4.5 5. Verify Connection

Once connected, the status indicator located at the top-right corner of the interface should display **“Connected”**.

BLE: CONNECTED (ESP32_Bluetooth)

Figure 12: Bluetooth Connected

8.4.4.6 6. Operate the System

Use the interface buttons to control the door:

- **OPEN** – Opens the door
- **CLOSE** – Closes the door
- **STOP** – Stops movement immediately
- **MANUAL OPERATION** – Disengages the wheel for 10 seconds, allowing free manual movement of the door
- **SETUP WHEEL** – Set the wheel properly with respect to the servo angle

The system will respond in real time to the selected command.

8.4.4.7 7. Manual Operation Option

1. Press the “Manual Operation” button in the user interface
2. The system will automatically release the wheel so the door can move freely.

3. Open the door manually and pass through normally.
4. Close the door after passing through.
5. After a few seconds, the system will automatically turn back on and continue normal operation.

8.4.5 Maintenance

Regular maintenance is required to ensure reliable operation and extend the lifespan of the automatic door opener system.

8.4.5.1 Tools/Products for Maintenance

- Microfiber cloth – For removing dust and debris from the system surfaces.
- Phillip's screwdriver – For tightening screws and fasteners.
- Digital multimeter – For checking electrical connections and power supply operation.

8.4.5.2 Cleaning

- Clean the system periodically using a dry cloth to remove dust and debris.
- Do not use water or liquid cleaners, as they may damage electrical components.
- Ensure the ultrasonic sensor is free of dirt or obstructions for proper operation.

8.4.5.3 Inspection

- Check that all screws and fasteners are securely tightened.
- Verify that all components remain properly aligned, especially the motor, gear, and rack mechanism.
- Inspect wiring for any signs of wear, damage, or loose connections.
- Ensure smooth movement of the wheel, gear, and door mechanism.
- Check that the wheel maintains proper contact with the floor to prevent slipping.
- Inspect 3D printed parts for cracks, deformation, or wear, especially load-bearing components.

8.4.6 Electrical System

- Verify that the power supply connections are secure and functioning correctly.
- Ensure the ESP32 and control system operate as expected.
- Replace or repair any damaged cables immediately.

8.4.6.1 Functional Testing

- Periodically test the system by performing open, close, and stop operations.
- Confirm that limit switches are functioning correctly and stopping the door at the intended positions.
- Verify that the ultrasonic sensor detects presence accurately.

8.4.6.2 Storage and Environment

- Keep the system in a dry indoor environment.
- Avoid exposure to excessive heat, humidity, or dust.

8.4.7 Recommended Maintenance Interval

- Inspect and clean the system every 2–4 weeks depending on usage.

8.4.8 Troubleshooting

Step 1: Check Power Supply Voltage in Motor Driver (12V)

Step 2: Check Buck Convert voltage (5V)

Step 3: Verify Component functionality

- Test 2: Bluetooth Connection
- Test 3: Ultrasonic Detection
- Test 4: DC Motor Movement
- Test 5: Servo Motor Movement

Step 4: Check Individual Pin Burnouts

- Test 0: Place selected pin High and check voltage of certain pins.

Step 5: Check Limit Switch Position

- Make sure they make contact to the at the wall and door frame.

Step 6: Check Wheel Screw

- Verify the 3M screw has not loosened.

Step 7: Check Gear and Rack Alignment

- Make sure the servo is properly positioned within the box.

The tests are available in the esp32 folder of the github repository: https://github.com/yosefsantiago/door_opener

IMPORTANT: Remote operation of the device requires the door latch to remain in the disengaged position. If the latch is engaged, remote operation cannot be performed, as the device is designed to operate only when the latch is disengaged.

As a potential enhancement for future revisions, consideration should be given to the incorporation of a mechanism that enables remote engagement and disengagement of the door latch.

8.5 Appendix E: Datasheets

8.5.1 ESP 32

https://cdn.sparkfun.com/datasheets/IoT/esp32_datasheet_en.pdf

1. Overview

ESP32 is a single chip 2.4 GHz Wi-Fi and Bluetooth combo chip designed with TSMC ultra low power 40 nm technology. It is designed and optimized for the best power performance, RF performance, robustness, versatility, features and reliability, for a wide variety of applications, and different power profiles.

1.1 Featured Solutions

1.1.1 Ultra Low Power Solution

ESP32 is designed for mobile, wearable electronics, and Internet of Things (IoT) applications. It has many features of the state-of-the-art low power chips, including fine resolution clock gating, power modes, and dynamic power scaling.

For instance, in a low-power IoT sensor hub application scenario, ESP32 is woken up periodically and only when a specified condition is detected; low duty cycle is used to minimize the amount of energy that the chip expends. The output power of the power amplifier is also adjustable to achieve an optimal trade off between communication range, data rate and power consumption.

Note:
For more information, refer to Section 3.7 RTC and Low-Power Management.

1.1.2 Complete Integration Solution

ESP32 is the most integrated solution for Wi-Fi + Bluetooth applications in the industry with less than 10 external components. ESP32 integrates the antenna switch, RF balun, power amplifier, low noise receive amplifier, filters, and power management modules. As such, the entire solution occupies minimal Printed Circuit Board (PCB) area.

ESP32 uses CMOS for single-chip fully-integrated radio and baseband, and also integrates advanced calibration circuitries that allow the solution to dynamically adjust itself to remove external circuit imperfections or adjust to changes in external conditions.

As such, the mass production of ESP32 solutions does not require expensive and specialized Wi-Fi test equipment.

1.2 Basic Protocols

1.2.1 Wi-Fi

- 802.11 b/g/n/e/i
- 802.11 n (2.4 GHz), up to 150 Mbps
- 802.11 e: QoS for wireless multimedia technology
- WMM-PS, UAPSD
- A-MPDU and A-MSDU aggregation
- Block ACK

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- Fragmentation and defragmentation
- Automatic Beacon monitoring/scanning
- 802.11 i security features: pre-authentication and TSN
- Wi-Fi Protected Access (WPA)/WPA2/WPA2-Enterprise/Wi-Fi Protected Setup (WPS)
- Infrastructure BSS Station mode/SoftAP mode
- Wi-Fi Direct (P2P), P2P Discovery, P2P Group Owner mode and P2P Power Management
- UMA compliant and certified
- Antenna diversity and selection

Note:
For more information, refer to Section 3.5 Wi-Fi.

1.2.2 Bluetooth

- Compliant with Bluetooth v4.2 BR/EDR and BLE specification
- Class-1, class-2 and class-3 transmitter without external power amplifier
- Enhanced power control
- +10 dBm transmitting power
- NZIF receiver with -98 dBm sensitivity
- Adaptive Frequency Hopping (AFH)
- Standard HCI based on SDIO/SPI/UART
- High speed UART HCI, up to 4 Mbps
- BT 4.2 controller and host stack
- Service Discover Protocol (SDP)
- General Access Profile (GAP)
- Security Manage Protocol (SMP)
- Bluetooth Low Energy (BLE)
- ATT/GATT
- HID
- All GATT-based profile supported
- SPP-Like GATT-based profile
- BLE Beacon
- A2DP/AVRCP/SPP, HSP/HFP, RFCOMM
- CVSD and SBC for audio codec
- Bluetooth Piconet and Scatternet

2. Pin Definitions

2.1 Pin Layout

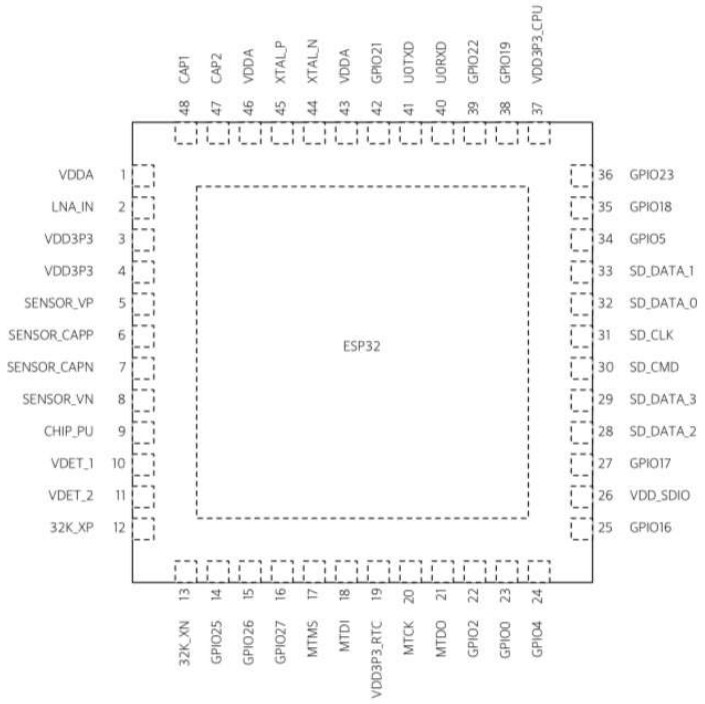


Figure 2: ESP32 Pin Layout

2.2 Pin Description

Table 1: Pin Description

Name	No.	Type	Function
Analog			
VDDA	1	P	Analog power supply (2.3V ~ 3.6V)
LNA_IN	2	I/O	RF input and output
VDD3P3	3	P	Amplifier power supply (2.3V ~ 3.6V)
VDD3P3	4	P	Amplifier power supply (2.3V ~ 3.6V)
VDD3P3_RTC			
SENSOR_VP	5	I	GPIO36, ADC_PRE_AMP, ADC1_CH0, RTC_GPIO0 Note: Connects 270 pF capacitor from SENSOR_VP to SENSOR_CAPP when used as ADC_PRE_AMP.

Name	No.	Type	Function
SENSOR_CAPP	6	I	GPIO37, ADC_PRE_AMP, ADC1_CH1, RTC_GPIO1 Note: Connects 270 pF capacitor from SENSOR_VP to SENSOR_CAPP when used as ADC_PRE_AMP.
SENSOR_CAPN	7	I	GPIO38, ADC1_CH2, ADC_PRE_AMP, RTC_GPIO2 Note: Connects 270 pF capacitor from SENSOR_VN to SENSOR_CAPN when used as ADC_PRE_AMP.
SENSOR_VN	8	I	GPIO39, ADC1_CH3, ADC_PRE_AMP, RTC_GPIO3 Note: Connects 270 pF capacitor from SENSOR_VN to SENSOR_CAPN when used as ADC_PRE_AMP.
CHIP_PU	9	I	Chip Enable (Active High) High: On, chip works properly Low: Off, chip works at the minimum power Note: Do not leave CHIP_PU pin floating
VDET_1	10	I	GPIO34, ADC1_CH6, RTC_GPIO4
VDET_2	11	I	GPIO35, ADC1_CH7, RTC_GPIO5
32K_XP	12	I/O	GPIO32, 32K_XP (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9
32K_XN	13	I/O	GPIO33, 32K_XN (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8
GPIO25	14	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
GPIO26	15	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
GPIO27	16	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
MTMS	17	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
MTDI	18	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
VDD3P3_RTC	19	P	RTC IO power supply input (1.8V - 3.3V)
MTCK	20	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
MTDO	21	I/O	GPIO15, ADC2_CH3, TOUCH3, RTC_GPIO13, MTDO, HSPICS0, HS2_CMD, SD_CMD, EMAC_RXD3
GPIO2	22	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPIWP, HS2_DATA0, SD_DATA0
GPIO0	23	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
GPIO4	24	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPIHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER
VDD_SDIO			
GPIO16	25	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
VDD_SDIO	26	P	1.8V or 3.3V power supply output
GPIO17	27	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
SD_DATA_2	28	I/O	GPIO9, SD_DATA2, SPIHD, HS1_DATA2, U1RXD
SD_DATA_3	29	I/O	GPIO10, SD_DATA3, SPIWP, HS1_DATA3, U1TXD
SD_CMD	30	I/O	GPIO11, SD_CMD, SPICS0, HS1_CMD, U1RTS
SD_CLK	31	I/O	GPIO6, SD_CLK, SPICLK, HS1_CLK, U1CTS

8.5.2 Bringsmart 12V DC Motor

https://media.digikey.com/pdf/data%20sheets/seeed%20technology/108990007_web.pdf



(images/product/ASLONG JGY 370 12V motor.jpg)

ASLONG JGY-370 12V DC Worm Gear Motor

SKU: 108990007

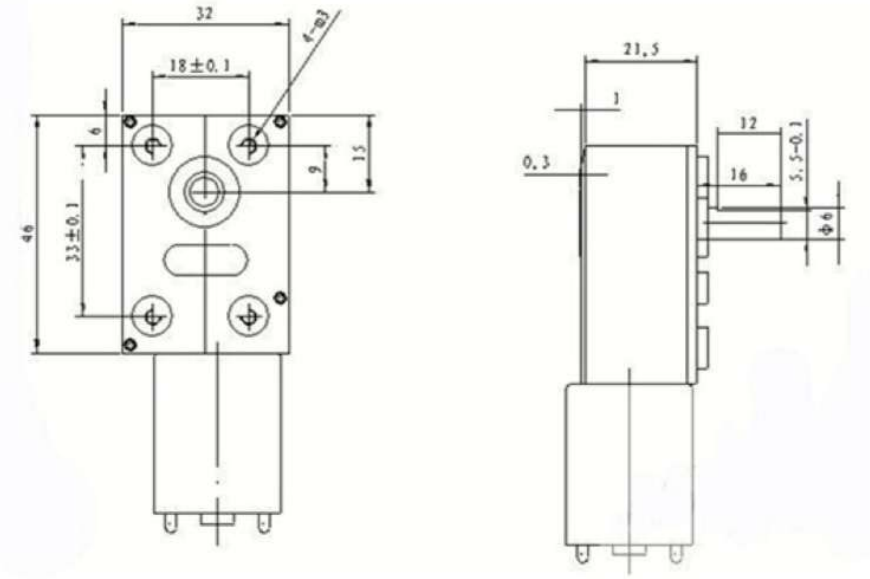
WIO LINK **is LIVE at Kickstarter!**

(https://www.kickstarter.com/projects/seeed/wio-link-3-steps-5-minutes-build-your-iot-applicat/description?ref=banner_depot)

Description

Mainly used in robot platform and car provide power, Good quality and long lifetime, High torgue and Low nosie. It's running at 160RPM.

Specifications



•Rated Voltage: DC 12V
•No load speed: 160rpm
•No Load Current: 35mA
•load torque: 0.55kg.cm
•load Speed: 128rpm
•load Current: 180mA
•load out put: 1.1w
•Stall torque: 2.2kg.cm
•Stall Current: 1A
•Ratio: 37.3
•Weight:165g

For any technical support or suggestion, please kindly go to our forum (http://www.seeedstudio.com/forum/viewforum.php?f=57).

Overview

(http://www.seeedstudio.com/depot/Shenzhen-2U-t-11.html?ref=pinfo) Designer: Others Other Products From This Designer (http://www.seeedstudio.com/depot/Others-m-24.html?ref=pinfo)
Weight: 156 g

8.5.3 MG 966R Servo Motor

https://www.handsontec.com/dataspecs/motor_fan/MG996R.pdf

HT **Handson Technology**

User Guide

MG996R Metal Gear Servo Motor

This High-Torque MG996R Digital Servo features metal gearing resulting in extra high 10kg stalling torque in a tiny package. It features upgraded shock-proofing and a redesigned PCB and IC control system that make it much more accurate than its predecessor MG995. The gearing and motor have also been upgraded to improve dead bandwidth and centering. This high-torque standard servo can rotate approximately 120º (60º in each direction). The MG996R Metal Gear Servo also comes with a selection of arms and hardware to get you set up nice and fast servo control projects!



SKU: EMH-1056

Brief Data:

- Stall torque: 9.4 kgf·cm (4.8V), 11 kgf·cm (6 V)
- Operating speed: 0.17 s/60º (4.8 V), 0.14 s/60º (6 V)
- Operating voltage: 4.8V a 7.2V
- Running Current: 500mA.
- Stall Current: 2.5A (6V).
- Dead band width: 5µs
- Stable and shock proof double ball bearing design.
- Weight: 55g.
- Dimension: 40.7 x 19.7 x 42.9 mm approx.

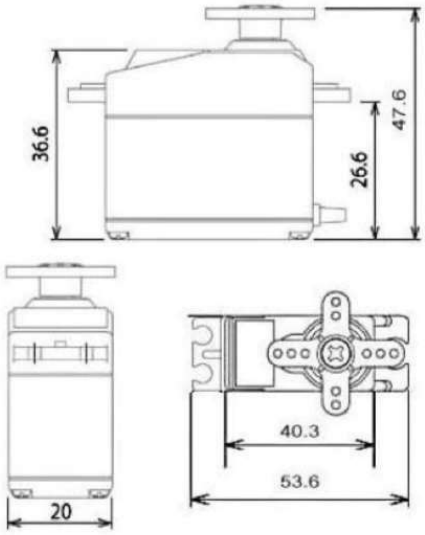
1 www.handsontec.com

Package Included:

- 1x MG996R Servo Motor.
- 4x Arms.
- 4x Fixing Screws and rubber grommet.
- 1x M4 arms fixing screw.

Mechanical Dimensions:

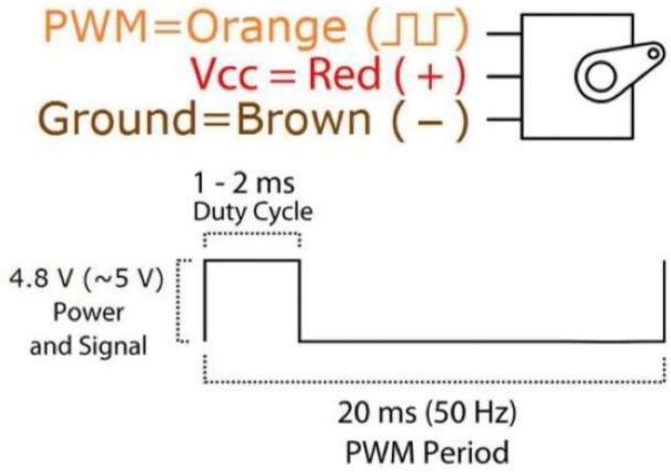
Unit: mm



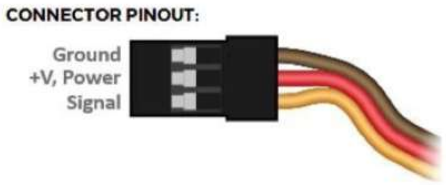
Metal Gear Construction:



Electrical Control Signal:



Position "0" (1.5ms pulse) is middle, "90" (~2ms pulse) is middle, is all the way to the right, "-90" (~1ms pulse) is all the way to the left.



Application With Arduino:

Circuit:

Servo motors have three wires: power, ground, and signal. The power wire is typically red, and should be connected to the 5V pin on the Arduino board. The ground wire is typically black or brown and should be connected to a ground pin on the board. The signal pin is typically yellow, orange or white and should be connected to pin 9 on the board.

8.5.4 HC-SR01 Ultrasonic Sensor

https://www.mouser.com/datasheet/2/813/HCSR04-1022824.pdf?srsltid=AfmBOoom9PtPkIZuxsgaxZVo7TKL9LKgAnQHT0MFGxyLUIQy3nst-tOZ

ELEC Freaks Tech Support: services@elecfreaks.com

Ultrasonic Ranging Module HC - SR04

Product features:

Ultrasonic ranging module HC - SR04 provides 2cm - 400cm non-contact measurement function, the ranging accuracy can reach to 3mm. The modules includes ultrasonic transmitters, receiver and control circuit. The basic principle of work:
(1) Using IO trigger for at least 10us high level signal,
(2) The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back.
(3) IF the signal back, through high level , time of high output IO duration is the time from sending ultrasonic to returning.
Test distance = (high level time×velocity of sound (340M/S) / 2,

Wire connecting direct as following:

- 5V Supply
- Trigger Pulse Input
- Echo Pulse Output
- 0V Ground

Electric Parameter

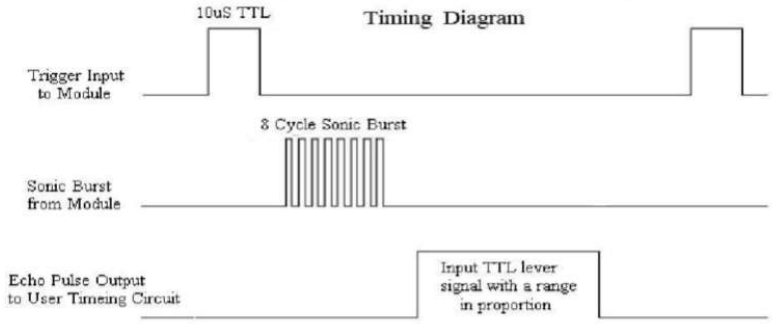
Working Voltage	**DC 5 V**
Working Current	**15mA**
Working Frequency	**40Hz**
Max Range	**4m**
Min Range	**2cm**
MeasuringAngle	**15 degree**
Trigger Input Signal	**10uS TTL pulse**
Echo Output Signal	**Input TTL lever signal and the range in proportion**
Dimension	**45*20*15mm**



Vcc Trig Echo GND

Timing diagram

The Timing diagram is shown below. You only need to supply a short 10uS pulse to the trigger input to start the ranging, and then the module will send out an 8 cycle burst of ultrasound at 40 kHz and raise its echo. The Echo is a distance object that is pulse width and the range in proportion .You can calculate the range through the time interval between sending trigger signal and receiving echo signal. Formula: uS / 58 = centimeters or uS / 148 =inch; or: the range = high level time * velocity (340M/S) / 2; we suggest to use over 60ms measurement cycle, in order to prevent trigger signal to the echo signal.



Attention:

- The module is not suggested to connect directly to electric, if connected electric, the GND terminal should be connected the module first, otherwise, it will affect the normal work of the module.
- When tested objects, the range of area is not less than 0.5 square meters and the plane requests as smooth as possible, otherwise ,it will affect the results of measuring.

www.Elecfreaks.com

8.5.5 ZK-5AD Motor Driver

https://a.co/d/07QM6wlO

Specifications:

- Model: ZK-5AD
- Channels: 2 (independent control)
- Operating Voltage: DC 3.0V to 14V
- Input Signal Voltage: DC 2.2V to 6.0V
- Max Drive Current: 5A per channel
- Standby Current: 10μA
- Control Pins: D0/D1 (Motor 1), D2/D3 (Motor 2)
- Signal Logic Levels:
  - High: 2.2V – 6.0V
  - Low: 0.0V – 0.7V
- Operating Temperature: -20°C to 85°C
- Humidity: 0% to 95% RH
- Dimensions: 38mm x 27mm x 18mm
- Mounting Hole Diameter: 2mm

Features:

- Dual 5A H-Bridge motor driver
- Independent speed and direction control for two motors
- Supports PWM control
- Compact and energy-efficient design
- Supports remote control with 433MHz receiver module
- Easy to interface with microcontrollers (e.g., Arduino)

8.6 Appendix F: ESP32 Firmware Code

```
/*************************************************************************
  ESP32 Door Opener/Closer Controller
*************************************************************************/
#include <BLEDevice.h>
#include <BLEServer.h>
#include <BLEUtils.h>
#include <BLE2902.h>
#include <ESP32Servo.h>
#include <Wire.h>
#include <Adafruit_INA219.h>

char bleCommand = '\0';
bool deviceConnected = false;
bool restartAdvertising = false;
bool disengaged = false;
bool setupModeDisengaged = false;
bool motorRunning = false;
unsigned long overCurrentStart = 0;                    // log when over
current was detected
unsigned long idleOverCurrentStart = 0;
const bool ultrasonicEnabled = false;
const bool currentSpikeEnabled = false;
const bool limitSwitchEnabled = false;
```

```
const int INA_SDA = 27;                                    // Serial Data pin
for I2C
const int INA_SCL = 14;                                    // Serial Clock pin
for I2C
const int openedPin = 13;                                  // limit switch
(normally open)
const int closedPin = 21;                                  // limit switch
(normally open)
const int pinX = 16;                                       // left H-bridge
control (DO)
const int pinY = 17;                                       // right H-bridge
control (D1)
const int servoPin = 19;
const int echoPin1 = 33;                                   // opening
const int trigPin1 = 25;                                   // opening
const int echoPin2 = 22;                                   // closing
const int trigPin2 = 23;                                   // closing
const int engagedAngle = 5;
const int disengagedAngle = 70;
const float minDistance = 5.0;                             // ultrasonic
detection range (cm)
const float peakCurrent_obst = 400.0;                      // tuned expected
load (mA)
const float peakCurrent_norm = 0.10;                       // tuned expected
load (mA)
const float motorCurrentLimit_obst = peakCurrent_obst * 1.5; // obstruction
threshold (mA)
const float motorCurrentLimit_norm = peakCurrent_norm * 1.5; // normal door
operation threshold (mA)
const unsigned long maxOpenTime = 12000;                   // timeout safety
const unsigned long maxCloseTime = 12000;                  // timeout safety
const unsigned long currentTripTime = 100;                 // time above limit
before fault (ms)
const unsigned long motorStartupIgnoreTime = 300;          // ignore startup
spike
const unsigned long manualOverrideTime = 10000;            // time servo stays
disengaged for manual use

Servo engagingServo;
Adafruit_INA219 ina219;

#define SERVICE_UUID           "6E400001-B5A3-F393-E0A9-E50E24DCCA9E"
#define CHARACTERISTIC_UUID_RX "6E400002-B5A3-F393-E0A9-E50E24DCCA9E"
#define CHARACTERISTIC_UUID_TX "6E400003-B5A3-F393-E0A9-E50E24DCCA9E"

BLECharacteristic* pTxCharacteristic = nullptr;
```

```
//=====================================================================
// BLUETOOTH LOW ENERGY (BLE)

/**
 * @brief listens for when a phone connects or disconnects from BLE.
 */
class MyServerCallbacks : public BLEServerCallbacks {
  void onConnect(BLEServer* pServer) override {
    deviceConnected = true;
    Serial.println("BLE client connected");
  }

  void onDisconnect(BLEServer* pServer) override {
    deviceConnected = false;
    restartAdvertising = true;
    Serial.println("BLE client disconnected");
    Serial.println("Restart advertising requested");
  }
};

/**
 * @brief when phone sends a command read it and store it.
 */
class MyCallbacks : public BLECharacteristicCallbacks {
  void onWrite(BLECharacteristic* pCharacteristic) override {
    String value = pCharacteristic->getValue();

    if (value.length() > 0) {
      bleCommand = value[0];   // take first letter only
      Serial.print("BLE command received: ");
      Serial.println(bleCommand);
    }
  }
};

/**
 * @brief setup BLE functionality
 * 1. initialize bluetooth server
 * 2. define comunication characteristics between esp32 and phone
 * 3. make the esp32 discoverable to other devices
 */
void setupBLE(const char* bleName) {
  BLEDevice::init(bleName);

  BLEServer* pServer = BLEDevice::createServer();
```

```
  pServer->setCallbacks(new MyServerCallbacks());

  BLEService* pService = pServer->createService(SERVICE_UUID);

  pTxCharacteristic = pService->createCharacteristic(
    CHARACTERISTIC_UUID_TX,
    BLECharacteristic::PROPERTY_NOTIFY
  );
  pTxCharacteristic->addDescriptor(new BLE2902());

  BLECharacteristic* pRxCharacteristic = pService->createCharacteristic(
    CHARACTERISTIC_UUID_RX,
    BLECharacteristic::PROPERTY_WRITE
  );
  pRxCharacteristic->setCallbacks(new MyCallbacks());

  pService->start();

  BLEAdvertising *pAdvertising = BLEDevice::getAdvertising();
  pAdvertising->addServiceUUID(SERVICE_UUID);
  pAdvertising->setScanResponse(true);
  pAdvertising->setMinPreferred(0x06);
  pAdvertising->setMinPreferred(0x12);

  BLEDevice::startAdvertising();

  Serial.println("BLE ready");
  Serial.println("Advertising started");
  Serial.println("Waiting for connection...");
}

/**
 * @brief verify if phone is connected, otherwise advertise esp32
 */
bool checkConnection() {
  if (restartAdvertising && !deviceConnected) {
    restartAdvertising = false;
    delay(200);
    BLEDevice::startAdvertising();
    Serial.println("Advertising restarted");
  }
  if (deviceConnected) {
    return true;
  }
  return false;
}
```

```
/**
 * @brief allows to send phone messages from the esp32
 */
void sendPhoneMessage(const char* msg) {
  if (deviceConnected && pTxCharacteristic != nullptr) {
    pTxCharacteristic->setValue(msg);
    pTxCharacteristic->notify();

    Serial.print("Message sent: ");
    Serial.println(msg);
  }
}

//=====================================================================
// LIMIT SWITCHES

/**
 * @brief read opened state limit switch
 */
bool doorIsOpen() {
  if (!limitSwitchEnabled) {
    return false;
  }

  return digitalRead(openedPin) == LOW;
}

/**
 * @brief read closed state limit switch
 */
bool doorIsClosed() {
  if (!limitSwitchEnabled) {
    return false;
  }

  return digitalRead(closedPin) == LOW;
}

/**
 * @brief check if both limitswitches are pressed; should not happen
 */
bool invalidLimitState() {
  if (!limitSwitchEnabled) {
    return false;
  }
```

```
  return doorIsOpen() && doorIsClosed();
}

//=====================================================================
// DC MOTOR CONTROL

/**
 * @brief H-bridge control functions to set the DC motor's spin
 */
void forward() { digitalWrite(pinX,HIGH); digitalWrite(pinY,LOW); motorRunning =
true; }
void reverse() { digitalWrite(pinX,LOW); digitalWrite(pinY,HIGH); motorRunning =
true; }
void motorstop() { digitalWrite(pinX,LOW); digitalWrite(pinY,LOW); motorRunning =
false; }
void motorbrake() { digitalWrite(pinX,HIGH); digitalWrite(pinY,HIGH);
motorRunning = false; }

/**
 * @brief verify if emergency stop was detected
 */
bool emergencyStopRequested() { return bleCommand == 'S'; }

//=====================================================================
// SERVO MOTOR CONTROL

/**
 * @brief set servo angle to 100 degrees
 */
void engage() {
  engagingServo.write(engagedAngle);
  Serial.print("Servo engaged at angle: ");
  Serial.println(engagedAngle);
  disengaged = false;
}

/**
 * @brief set servo angle to 5 degrees
 */
void disengage() {
  engagingServo.write(disengagedAngle);
  Serial.print("Servo disengaged at angle: ");
  Serial.println(disengagedAngle);
  disengaged = true;
}
```

```
//=====================================================================
//  OBSTRUCTION DETECTION

/**
 * @brief check ultrasonic sensor
 */
bool checkUltrasonic(const int trigPin, const int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin,HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin,LOW);
  float duration = pulseIn(echoPin,HIGH, 50000); // time (us); wait 50 ms for a
response
  if (duration == 0) {
    Serial.println("No echo detected");
    // change to true for fail-safe behavior: stop if the sensor isn't read
    return false; // assume no obstruction
  }
  float distance = duration * 0.034 / 2; // converts echo time to distance (cm)
  return distance <= minDistance;
}

/**
 * @brief read current from INA219
 */
float readMotorCurrent_mA() {
  float current_mA = ina219.getCurrent_mA();

  // sometimes sensor noise can give tiny negative values
  if (current_mA < 0) {
    current_mA = -current_mA;
  }

  return current_mA;
}

/**
 * @brief control the rate a which the currents are printed in serial monitor;
 * for debug and tuning purposes
 */
void printCurrent(const char* label, float current) {
  static unsigned long lastPrintTime = 0;
  const unsigned long printInterval = 1000;
```

```
  if (millis() - lastPrintTime >= printInterval) {
    lastPrintTime = millis();

    Serial.print(label);
    Serial.print(": ");
    Serial.println(current);
  }
}

/**
 * @brief check if overcurrent conditions have been met
 * 1. don't account for startup spikes
 * 2. read current
 * 3. check if current exceeds limit
 * 4. check if the current spike is sustained
 */
bool overCurrentDetected(unsigned long motionStartTime) {
  if (!currentSpikeEnabled) {
    return false;
  }

  // ignore normal motor startup spike
  if (millis() - motionStartTime < motorStartupIgnoreTime) {
    overCurrentStart = 0;
    return false;
  }

  float current_mA = readMotorCurrent_mA();

  printCurrent("Motor current mA", current_mA);

  if (current_mA > motorCurrentLimit_obst) {
    if (overCurrentStart == 0) {
      overCurrentStart = millis();
    }

    if (millis() - overCurrentStart >= currentTripTime) {
      return true;
    }
  }
  else {
    overCurrentStart = 0;
  }

  return false;
}
```

```
/**
 * @brief check if someone is operating the door normally (i.e. pushing the door)
 */
bool idleLoadDetected() {
  if (!currentSpikeEnabled) {
    return false;
  }

  float current_mA = readMotorCurrent_mA();

  printCurrent("Idle current mA", current_mA);

  if (!motorRunning && !disengaged) {
    if (current_mA > motorCurrentLimit_norm) {
      if (idleOverCurrentStart == 0) {
        idleOverCurrentStart = millis();
      }
      if (millis() - idleOverCurrentStart > 100) {
        return true;
      }
    }
    else {
      idleOverCurrentStart = 0;
    }
  }
  else {
    idleOverCurrentStart = 0;
  }

  return false;
}

/**
 * @brief detect if a person is near the door while the system is idle
 *
 * If the motor is not running and the servo is engaged, either ultrasonic
 * sensor can trigger a manual override. This disengages the servo so the
 * person can operate the door manually.
 */
bool idleUltrasonicDetected() {
  if (!ultrasonicEnabled) {
    return false;
  }

  if (motorRunning || disengaged) {
```

```
    return false;
  }

  bool openingSideDetected = checkUltrasonic(trigPin1, echoPin1);
  bool closingSideDetected = checkUltrasonic(trigPin2, echoPin2);

  return openingSideDetected || closingSideDetected;
}

/**
 * @brief temporarily disengage servo to allow manual door operation
 */
void allowManualOperation() {
  Serial.println("Manual operation detected → disengaging servo");
  disengage();

  sendPhoneMessage("MANUAL: Door movement detected, servo disengaged");

  delay(manualOverrideTime);

  Serial.println("Manual override time ended → engaging servo");
  engage();

  sendPhoneMessage("MANUAL: Servo re-engaged");
}

/**
 * @brief check all obstruction sensors
 * 1. check current sensor
 * 2. determine if we are opening or closing to check the appropriate sensor
 * 3. check ultrasonic1 if mode is 'O' and ultrasonic2 if mode is 'C', and they
are enabled
 */
bool obstructionDetected(char mode, unsigned long motionStartTime) {
  if (currentSpikeEnabled && overCurrentDetected(motionStartTime)) {
    return true;
  }
  if (ultrasonicEnabled) {
      if (mode == 'O') {
      return checkUltrasonic(trigPin1,echoPin1);
    }
    else if (mode == 'C') {
      return checkUltrasonic(trigPin2,echoPin2);
    }
  }
  return false;
```

```
}

//=====================================================================
// CONTROL LOOPS

/**
 * @brief send warning to phone
 * 1. 'O': Obstruction detected, command stopped
 * 2. 'T': Timeout fault, check limit switches
 * 3. 'U': Ultrasonic sensor failure
 * 4. 'L': Both limit switches were detected simultaneously
 */
void warning(char w) {
  if (w == 'O') {
    // WARNING: Obstruction detected
    sendPhoneMessage("WARNING: Obstruction detected, motor stopped");
  }
  else if (w == 'T') {
    // WARNING: Timeout fault, check limit switches
    sendPhoneMessage("WARNING: Switch was not detected, check limit switch
position");
  }
  else if (w == 'U') {
    // WARNING: Ultrasonic sensor failure
    sendPhoneMessage("WARNING: Ultrasonic sensor failed to read");
  }
  else if (w == 'L') {
    // WARNING: Both limit switches active
    sendPhoneMessage("WARNING: Both limit switches are pressed at the same
time");
  }
  else {
    sendPhoneMessage("WARNING: Unknown fault");
  }
}

/**
 * @brief Opening door control loop
 * 1. check if door is open
 * 2. engage servo motor
 * 3. turn on DC motor
 * 4. check emergency stop, obstructions, if the door has been opened,
 *    if the limit switches have invalid behavior, and if the door went
 *    past what was expectected (i.e. missed limit switch)
 */
void openDoor() {
```

```
Serial.println("Starting openDoor()");

// clear start command so only a new 'S' matters during motion
bleCommand = '\0';

if (doorIsOpen()) {
  Serial.println("Door already open");
  return;
}

Serial.println("Opening door...");
forward();

unsigned long startTime = millis();

while (true) {
  if (emergencyStopRequested()) {
    Serial.println("Emergency stop during opening");
    motorstop();
    return;
  }
  if (obstructionDetected('O', startTime)) {
    Serial.println("Obstruction detected during opening");
    motorbrake();
    delay(50);
    motorstop();
    disengage();
    warning('O');
    return;
  }
  if (invalidLimitState()) {
    Serial.println("Invalid limit switch state");
    warning('L');
    motorstop();
    return;
  }
  if (doorIsOpen()) {
    Serial.println("Door reached open position");
    motorstop();
    return;
  }
  if (millis() - startTime > maxOpenTime) {
    Serial.println("Open timeout reached, check limit switch position");
    motorstop();
    warning('T');
    return;
```

```
    }
    delay(10);
  }
}

/**
 * @brief Closing door control loop
 * 1. check if door is closed
 * 2. engage servo motor
 * 3. turn on DC motor
 * 4. check emergency stop, obstructions, if the door has been closed,
 *    if the limit switches have invalid behavior, and if the door went
 *    past what was expectected (i.e. missed limit switch)
 */
void closeDoor() {
  Serial.println("Starting closeDoor()");

  // clear start command so only a new 'S' matters during motion
  bleCommand = '\0';

  if (doorIsClosed()) {
    Serial.println("Door already closed");
    return;
  }

  Serial.println("Closing door...");
  reverse();

  unsigned long startTime = millis();

  while (true) {
    if (emergencyStopRequested()) {
      Serial.println("Emergency stop during closing");
      motorstop();
      return;
    }
    if (obstructionDetected('C', startTime)) {
      Serial.println("Obstruction detected during closing");
      motorbrake();
      delay(50);
      motorstop();
      disengage();
      warning('O');
      return;
    }
    if (invalidLimitState()) {
```

```
      Serial.println("Invalid limit switch state");
      warning('L');
      motorstop();
      return;
    }
    if (doorIsClosed()) {
      Serial.println("Door reached closed position");
      motorstop();
      return;
    }
    if (millis() - startTime > maxCloseTime) {
      Serial.println("Close timeout reached, check limit switch position");
      motorstop();
      warning('T');
      return;
    }
    delay(10);
  }
}

void setupWheelEngagement() {
  motorstop();

  if (!setupModeDisengaged) {
    Serial.println("Wheel setup step 1: disengaging servo.");
    Serial.println("Press motor/wheel down so gear locks in, then press SETUP
again.");

    engagingServo.write(180);
    Serial.print("Servo disengaged at angle: ");
    Serial.println(180);
    disengaged = true;
    setupModeDisengaged = true;

    sendPhoneMessage("SETUP: Press wheel down, then press SETUP again");
  }
  else {
    Serial.println("Wheel setup step 2: engaging servo.");

    engage();
    setupModeDisengaged = false;

    sendPhoneMessage("SETUP: Wheel engaged");
  }
}
```

```
//=====================================================================
// MAIN

void setup() {
  Serial.begin(115200);

  setupBLE("ESP32_Bluetooth");

  if (limitSwitchEnabled) {
    pinMode(openedPin, INPUT_PULLUP);
    pinMode(closedPin, INPUT_PULLUP);
  }

  if (ultrasonicEnabled) {
    pinMode(echoPin1, INPUT);
    pinMode(echoPin2, INPUT);
    pinMode(trigPin1, OUTPUT);
    pinMode(trigPin2, OUTPUT);
    digitalWrite(trigPin1, LOW);
    digitalWrite(trigPin2, LOW);
  }

  pinMode(pinX,OUTPUT);
  pinMode(pinY,OUTPUT);
  digitalWrite(pinX,LOW);
  digitalWrite(pinY,LOW);

  pinMode(servoPin,OUTPUT);
  engagingServo.setPeriodHertz(50);          // standard servo frequency
  engagingServo.attach(servoPin, 500, 2500); // pulse range in microseconds

  if (currentSpikeEnabled) {
    Wire.begin(INA_SDA, INA_SCL);

    if (!ina219.begin()) {
      Serial.println("Failed to find INA219 current sensor");
      sendPhoneMessage("WARNING: INA219 not detected");
    }
    else {
      Serial.println("INA219 current sensor ready");
    }
  }

  engage();
  disengaged = false;
```

```
  delay(1000);
}

void loop() {
  if (idleLoadDetected()) {
    Serial.println("External force detected → disengaging");
    disengage();
    idleOverCurrentStart = 0;
    sendPhoneMessage("WARNING: External force detected");
  }
  if (idleUltrasonicDetected()) {
    allowManualOperation();
  }
  if (checkConnection()) {
    if (bleCommand == 'O') {
      Serial.println("Processed command: O");
      engage();
      openDoor();
      bleCommand = '\0';
    }
    else if (bleCommand == 'C') {
      Serial.println("Processed command: C");
      engage();
      closeDoor();
      bleCommand = '\0';
    }
    else if (bleCommand == 'S') {
      Serial.println("Processed command: S");
      motorstop();
      bleCommand = '\0';
    }
    else if (bleCommand == 'W') {
      Serial.println("Processed command: W");
      setupWheelEngagement();
      bleCommand = '\0';
    }
    else if (bleCommand == 'M') {
      Serial.println("Processed command: M");
      allowManualOperation();
      bleCommand = '\0';
    }
    else if (bleCommand != '\0') {
      Serial.println("Unknown command processed");
      bleCommand = '\0';
    }
  }
```

```
}
```